Generalized Disjunctive Programming with Derivative-Free Configuration Search

Albert Joon Lee [1], Woosik Kim [2], and David E. Bernal Neira [1,*]

[1]Davidson School of Chemical Engineering, Purdue University, West Lafayette, IN 47907, USA

[2]Elmore Family School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN 47907, USA

*Email: dbernaln@purdue.edu

Abstract

Generalized disjunctive programming (GDP) describes process design alternatives with Boolean variables, disjunctions, and logical propositions, and fixing one configuration leaves a subproblem in the continuous variables. Conventional GDP methods reformulate the model as a mixed-integer nonlinear program (MINLP) or decompose it with logic-based algorithms; both need the equations of every alternative, which a process simulator does not expose, and MINLP reformulations of nonconvex process models can be intractable. The external-variable reformulation turns configuration choice into an optimization over an integer lattice whose objective is the subproblem value, which derivative-free optimization (DFO) can search while the GDP model continues describing the alternatives. We pair GDP models with surrogate DFO methods and the direct-search method `NOMAD` and compare them with discrete steepest descent (`D-SDA`), logic-based and MINLP solvers, and enumeration on nine process models, 21 test functions, a pharmaceutical flowsheet simulated in `PharmaPy`, and a catalytic distillation column with 100,947 configurations. On the largest reactor network, the surrogate methods `MISO` and `RBFOpt` reach the reference in nine or ten of ten runs with about 30 subproblem solves, whereas logic-based methods solve nearly one subproblem for each admissible configuration and most `D-SDA` descents end at one local minimum. In the flowsheet example, where GDP solvers do not apply, `RBFOpt` and `MISO` find the best of 36 simulated configurations in every run with 20 evaluations. For the catalytic column, `D-SDA` and `MISO` reach the best known design with 100 evaluations, whereas MINLP solvers find no design within 15% of it and global solvers return contradictory certificates. A GDP model can therefore be solved by a derivative-free configuration search whenever its subproblems can be evaluated, including by simulation. This search pays off when the configuration space is too large to enumerate, has deceptive local minima, or has MINLP reformulations that fail, and enumeration or a direct reformulation is simpler for small algebraic models.

Keywords: Generalized disjunctive programming; Black-box optimization; Derivative-free optimization; Surrogate modeling; Process systems engineering; Computational benchmarking.

1 Introduction

In chemical process design, evaluating a single candidate configuration can require costly optimization or simulation, making the choice of which configuration to explore next a central computational challenge. Process design involves discrete decisions, such as equipment selection, the number of processing stages, and operating modes, along with continuous variables such as flow rates, temperatures, and equipment sizes. Generalized disjunctive programming (GDP) provides a framework to represent these decisions through Boolean variables, logical propositions, and conditional constraints (Raman and Grossmann, 1994). In logic-based GDP solution methods, fixing the Boolean variables associated with a selected configuration determines which constraints remain active and yields a reduced optimization subproblem. When all discrete decisions

are fixed, this subproblem can be formulated as a nonlinear programming (NLP) problem. Exploring multiple configurations may therefore require repeatedly solving NLP subproblems, which can be computationally expensive. Consequently, identifying high-quality configurations with fewer evaluations of NLP subproblems is an important objective in solving GDP problems.

Conventional GDP solution methods generally rely on mixed-integer nonlinear programming (MINLP) reformulations or logic-based decomposition. The Big-M and Hull reformulations transform disjunctive constraints into algebraic formulations (Grossmann and Lee, 2003; Raman and Grossmann, 1994), while logic-based approaches, such as logic-based outer approximation (LOA) and the relaxation with integer cuts (RIC) of `GDPopt`, select discrete configurations and solve the corresponding reduced subproblems (Bergamini et al., 2005; Pyomo Development Team, n.d.; Türkay and Grossmann, 1996). In logic-based approaches, the strategy used to select discrete configurations can significantly affect the number of subproblem evaluations required. Both families need the algebraic equations of every disjunct. When a unit is described by a process simulator, these equations are not available, and when the model is nonconvex, a reformulation keeps the equations of units that a configuration does not contain, whose terms can be poorly scaled or undefined at the values that absent units take.

The Discrete-Steepest Descent Algorithm (`D-SDA`) was previously applied to process superstructure optimization by Liñán et al. (2020a). Building on this approach, our previous work introduced the Logic-Based Discrete-Steepest Descent Algorithm (`LD-SDA`), which represents ordered Boolean decisions using external integer variables (Ovalle et al., 2025). With this external-variable reformulation, the GDP becomes an optimization over an integer lattice whose objective is the value of the subproblem of each configuration. The GDP model still describes the alternatives, their logic, and the subproblem of each configuration, but the method that searches the lattice needs only the value of each configuration. `LD-SDA` explores the integer space through neighborhood search and discrete line search, evaluating selected configurations by solving their corresponding reduced optimization subproblems. However, as a local search method, `LD-SDA` may converge to different locally optimal solutions depending on the starting point and the selected neighborhood (Peng et al., 2024).

Because the lattice search needs only configuration values, any derivative-free optimization (DFO) method that handles integer variables can perform it, and the GDP model can be solved even when its subproblems are simulations or its MINLP reformulations fail. We pair GDP models in their external integer representation with DFO methods that search the configurations. Figure 1 summarizes this approach and the conventional GDP methods with which we compare it. Surrogate-based DFO uses accumulated objective observations to guide new evaluations (Larson et al., 2019). `RBFOpt` builds surrogate models from radial basis functions and balances predicted objective values with exploration (Costa and Nannicini, 2018), and `MISO` combines radial-basis-function surrogate models with sampling strategies designed for integer variables (Müller, 2016). The comparison also includes Branch-and-Model (`BAM`) (Ma et al., 2023) and a variant of the factorization machine with quantum annealing (`FMQA`) framework (Kitai et al., 2020), in which classical simulated annealing replaces quantum annealing; we denote this variant `FMQA-SA`. `FMQA` represents its surrogate as a quadratic unconstrained binary optimization problem, the input format of quantum annealers, which makes it a candidate for a possibly quantum-assisted configuration search. We also include the mesh adaptive direct-search solver `NOMAD` (Audet et al., 2022), which, with `MISO`, performed best on mixed-integer problems in the comparison of Ploskas and Sahinidis (2022). We ask three questions. First, do these methods find high-quality configurations with fewer NLP subproblem solves than `D-SDA` and conventional GDP methods, and when does this advantage translate into shorter computing time? Second, does the same search solve GDP models whose subproblems are evaluated by a simulator? Third, does it solve GDP models whose MINLP reformulations defeat conventional solvers?

Candidate selection has its own cost, and evaluation costs vary between configurations (Moré and Wild, 2009). We therefore report three measures for every method: how often it reaches the reference objective, how many NLP subproblems it solves, and its estimated completion time. The complete enumeration of each configuration domain and uniform sampling serve as references, and all methods share initial configurations, evaluation budgets, and the responses of each configuration. The contributions are as follows:

- A use of the external-variable reformulation as the interface between a GDP model and a derivative-free configuration search, extended to subproblems defined by a process simulator and demonstrated on a lomustine manufacturing flowsheet simulated in `PharmaPy` and written as a GDP.

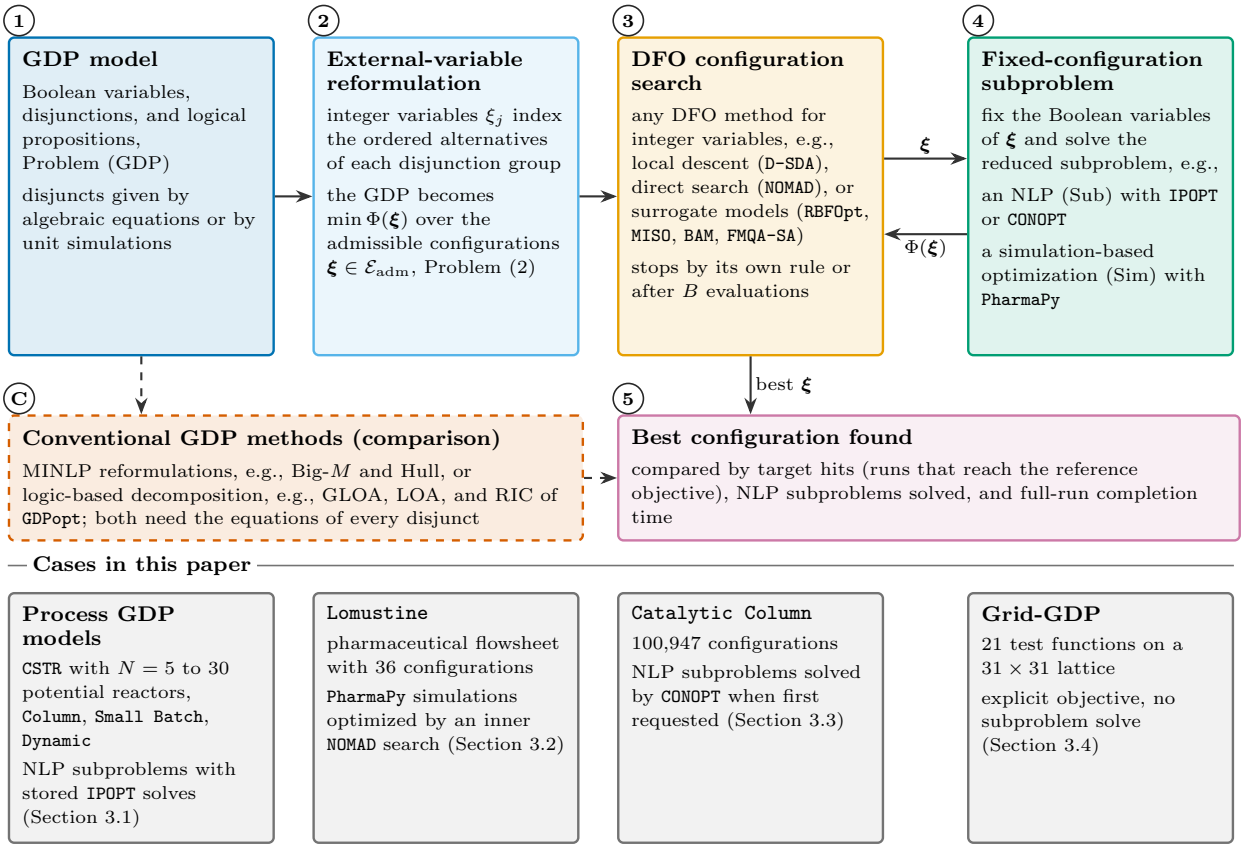


Figure 1: Methodology of this study. Top, the general method: a GDP model (1) is rewritten with external integer variables $\boldsymbol{\xi}$ that index the alternatives of its disjunctions (2), so that solving it becomes a search over the admissible configurations whose objective is the subproblem value $\Phi(\boldsymbol{\xi})$. A DFO method (3) proposes a configuration, which is evaluated by fixing its Boolean variables and solving the reduced subproblem (4), and the returned value updates the search until the method stops or the evaluation budget B is spent; the best configuration found (5) is the result. The dashed path (C) shows the conventional GDP methods, MINLP reformulations or logic-based methods such as global logic-based outer approximation (GLOA), logic-based outer approximation (LOA), and relaxation with integer cuts (RIC), which solve the same models when the equations of every disjunct are available. Bottom: the four cases of this paper and the subproblem evaluation each one uses.

- A comparison of surrogate-based and direct-search configuration methods with local descent, logic-based and direct GDP methods, and full enumeration on four process GDP families and 21 Grid test functions, measured by target attainment, NLP subproblems solved, and completion time.
- Evidence from execution records that the logic-based methods RIC and LOA solve one subproblem for each admissible configuration of most of these models, whereas surrogate search reaches near-optimal configurations with a small fraction of those subproblems.
- A benchmark with live NLP evaluations on a catalytic distillation column whose 100,947 configurations are too many to enumerate, together with valid Big-M and Hull reformulations of its GDP model, on which conventional MINLP solvers fail, and global solvers return contradictory certificates.
- An analysis of landscape features that determines when the surrogate search outperforms local descent, with recommendations for choosing between enumeration, direct reformulation, local descent, and surrogate search.

2 Background

2.1 Generalized Disjunctive Programming and Fixed-Configuration Evaluation

GDP represents operational and design alternatives using Boolean variables and associated algebraic constraints (Raman and Grossmann, 1994). The original continuous variables have domain $\mathcal{X} = [\underline{\mathbf{x}}, \overline{\mathbf{x}}] \subseteq \mathbb{R}^{n_x}$. The original integer variables have domain $\mathcal{Z} = [\underline{\mathbf{z}}, \overline{\mathbf{z}}] \cap \mathbb{Z}^{n_z}$. Let K index the disjunctions and D_k index the disjuncts in disjunction $k \in K$. Each disjunct $i \in D_k$ has a Boolean variable Y_{ik}. The GDP formulation is given by

$$
\begin{aligned}
\min_{\mathbf{x},\mathbf{z},\mathbf{Y}} \quad & f(\mathbf{x},\mathbf{z}) \\
\text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{z}) \le \mathbf{0}, \\
& \bigvee_{i \in D_k} \begin{bmatrix} Y_{ik} \\ \mathbf{h}_{ik}(\mathbf{x},\mathbf{z}) \le \mathbf{0} \end{bmatrix}, \qquad k \in K, \\
& \Omega(\mathbf{Y}) = \text{True}, \\
& \mathbf{x} \in \mathcal{X}, \quad \mathbf{z} \in \mathcal{Z}, \quad \mathbf{Y} \in \{\text{False}, \text{True}\}^{n_y}.
\end{aligned}
\tag{GDP}
$$

Here, f is the objective function, $\mathbf{g}$ represents the global constraints, and $\mathbf{h}_{ik}$ represents the constraints associated with each disjunct. The operator $\veebar$ requires that exactly one disjunct be selected within each disjunction. Logical propositions $\Omega(\mathbf{Y})$ impose additional relationships between Boolean variables.

Following the external-variable reformulation of Ovalle et al. (2025), each ordered group of m_j Boolean alternatives is represented by an external integer variable $\xi_j \in \{1, \ldots, m_j\}$, which identifies the selected alternative. Ordered alternatives, such as the position of a feed along a column, are one of the conditions that Ovalle et al. (2025) use because they make neighboring values of ξ_j physically similar configurations. The representation itself does not require this order: unordered alternatives, such as a choice among equipment types, can be indexed in any fixed order, as in Section 3.2, although the order then shapes the neighborhoods and surrogate models of the search (Section 4.2). Let n_E denote the number of external integer variables, and $\mathcal{Y}_{\text{logic}}$ the set of Boolean assignments that satisfy the conditions `exactly-one`, and $\Omega(\mathbf{Y})$. The external-variable lattice and its logically admissible subset are

$$
\begin{aligned}
\mathcal{L}_E &= \prod_{j=1}^{n_E} \{1, \ldots, m_j\}, \\
\mathcal{E}_{\text{adm}} &= \{\boldsymbol{\xi} \in \mathcal{L}_E : \mathbf{Y}(\boldsymbol{\xi}) \in \mathcal{Y}_{\text{logic}}\}.
\end{aligned}
\tag{1}
$$

For each logically admissible configuration $\boldsymbol{\xi}$, the corresponding Boolean variables are fixed, yielding the following reduced optimization subproblem:

$$
\begin{aligned}
\Phi(\boldsymbol{\xi}) = \inf_{\mathbf{x},\mathbf{z}} \quad & f(\mathbf{x},\mathbf{z}) \\
\text{s.t.} \quad & \mathbf{g}(\mathbf{x},\mathbf{z}) \le \mathbf{0}, \\
& \mathbf{h}_{ik}(\mathbf{x},\mathbf{z}) \le \mathbf{0} \quad \text{for all } (i,k) \text{ with } Y_{ik}(\boldsymbol{\xi}) = \text{True}, \\
& \mathbf{x} \in \mathcal{X}, \quad \mathbf{z} \in \mathcal{Z}.
\end{aligned} \tag{Sub}
$$

Here, $\Phi(\boldsymbol{\xi})$ represents the optimal objective value of the reduced subproblem. The reduced subproblem is an NLP when all remaining decision variables are continuous. If the reduced subproblem is infeasible, we define $\Phi(\boldsymbol{\xi}) = +\infty$.

When the external-variable mapping represents all logically admissible Boolean configurations, the original GDP can be expressed as the following upper-level optimization problem (Ovalle et al., 2025):

$$
\min_{\boldsymbol{\xi} \in \mathcal{E}_{\text{adm}}} \Phi(\boldsymbol{\xi}). \tag{2}
$$

A logically admissible configuration may still yield an infeasible reduced subproblem. Problem (2) separates the two tasks of solving a GDP: the model defines the admissible configurations and the subproblem of each, and the method that solves Problem (2) decides which configurations to evaluate.

Problem (2) uses only the value $\Phi(\boldsymbol{\xi})$ of each configuration, so it also applies when a configuration is evaluated by a process simulator. Fixing $\boldsymbol{\xi}$ then selects the unit models of a flowsheet, and the simulator $\mathcal{S}_{\boldsymbol{\xi}}$ computes its states $\mathbf{s}$, such as stream compositions and crystal size distributions, from operating inputs $\mathbf{u} \in \mathcal{U}_{\boldsymbol{\xi}}$. The reduced subproblem becomes

$$
\Phi(\boldsymbol{\xi}) = \min_{\mathbf{u} \in \mathcal{U}_{\boldsymbol{\xi}}} \big\{ f(\mathbf{u},\mathbf{s}) : \ \mathbf{s} = \mathcal{S}_{\boldsymbol{\xi}}(\mathbf{u}), \ \mathbf{g}(\mathbf{u},\mathbf{s}) \le \mathbf{0} \big\}, \tag{Sim}
$$

whose objective and constraints are known only through simulation outputs. In `PharmaPy`, for example, each unit is described by differential-algebraic equations (DAEs), and the simulator integrates them unit by unit in the order of the flowsheet. A DAE model can instead be discretized by orthogonal collocation on finite elements into algebraic equations, which turns its subproblem into an NLP; the `Dynamic` benchmark of Section 3.1 is solved this way with `pyomo.dae` (Nicholson et al., 2018). A derivative-free method in $\mathbf{u}$ solves Problem (Sim), so $\Phi(\boldsymbol{\xi})$ is approximated by the best value that this inner search finds within its simulation budget. Logic-based and direct GDP methods require the equations of every disjunct and do not apply to Problem (Sim). The configuration-search methods compared here use it through the same evaluation interface as an NLP subproblem, which Section 3.2 illustrates with a pharmaceutical flowsheet simulated in `PharmaPy` (Casas-Orozco et al., 2021).

2.2 Conventional GDP Solution Approaches

Big-M and Hull reformulations transform GDP models into MINLP formulations by associating Boolean variables Y_{ik} with binary variables y_{ik} and translating logical propositions into algebraic constraints (Grossmann and Lee, 2003; Raman and Grossmann, 1994). In the Big-M reformulation, the disjunctive constraints are expressed as $\mathbf{h}_{ik}(\mathbf{x},\mathbf{z}) \le \mathbf{M}_{ik}(1 - y_{ik})$, where the inequality is applied componentwise. The Big-M values must be sufficiently large to render the corresponding constraints redundant when $y_{ik} = 0$, but excessively large values can weaken the continuous relaxation and cause numerical difficulties.

Hull reformulations introduce disaggregated variables, linking constraints, and bounds scaled by binary indicators (Grossmann and Lee, 2003). Disjunctive constraints are imposed on the disaggregated variables when $y_{ik} = 1$, while $y_{ik} = 0$ forces these variables to zero, making the corresponding disjunct inactive. For convex disjunctive sets, this formulation can provide a stronger continuous relaxation than Big-M, although it introduces additional variables and constraints. For nonconvex GDP problems, the Hull reformulation does not guarantee a convex reformulated model or global optimality (Chen et al., 2022).

Logic-based methods address GDP problems directly by exploiting their logical and disjunctive structure to select discrete configurations and construct the corresponding optimization subproblems. LOA alternates between a mixed-integer master problem and fixed-configuration NLP subproblems (Türkay and Grossmann, 1996). Under suitable convexity and regularity assumptions, the master problem uses linearization cuts to obtain lower bounds, while feasible solutions to the NLP subproblem provide upper bounds. For nonconvex

GDP problems, these linearizations may not provide valid global lower bounds, motivating global logic-based outer approximation (GLOA), which requires globally valid relaxations and subproblem bounds. Bergamini et al. (2005) developed a GLOA approach for process networks using piecewise under- and overestimators.

`GDPopt` provides implementations of logic-based methods within `Pyomo.GDP` (Chen et al., 2022; Pyomo Development Team, n.d.). Its RIC method uses a mixed-integer linear programming (MILP) relaxation to select configurations and adds no-good cuts to exclude evaluated discrete assignments, without generating the successive outer-approximation cuts used by LOA (Pyomo Development Team, n.d.).

`LD-SDA` is an existing alternative that searches ordered external configurations by improving neighborhood moves and discrete line searches (Ovalle et al., 2025). It selects the best improving neighbor, continues along the resulting direction while improvement persists, and resumes neighborhood exploration. The present work uses this discrete-search structure as a reference and compares other configuration-selection strategies and their computational costs.

2.3 Derivative-Free Optimization Solvers

DFO uses function observations without requiring derivatives of the objective with respect to the search variables (Larson et al., 2019). Black-box configuration search selects configurations using their evaluation responses without accessing the subproblem equations. For GDP configuration search, the outer method can be derivative-free, even if its evaluation routine uses a derivative-based nonlinear solver. `D-SDA` also uses this derivative-free interface, although it does not construct a surrogate model. The upper-level search operates over external integer variables that represent the logical decisions in the GDP. Some logical restrictions are embedded in the external-variable reformulation, while additional constraints can be imposed on the configuration space. One approach to DFO uses surrogate models to guide the search. Surrogate methods seek to reduce expensive evaluations by fitting models to observed data, with computational cost arising from both objective evaluations and model-based candidate selection.

Several DFO methods handle integer or categorical variables directly, and Ploskas and Sahinidis (2022) compare many of them on mixed-integer test problems. Mesh adaptive direct search, implemented in `NOMAD`, extends pattern search to mixed variables through user-defined discrete neighborhoods (Abramson et al., 2009; Audet et al., 2022), and `MISO` combines radial basis function surrogates with sampling strategies for integer variables (Müller, 2016). Bayesian optimization has also been adapted to combinatorial spaces, for example by fitting sparse second-order models of binary variables in `BOCS` (Baptista and Poloczek, 2018), which are closely related to the factorization-machine surrogate used here, or by placing kernels on graphs of discrete configurations in `COMBO` (Oh et al., 2019). We compare four surrogate-based methods that represent different modeling choices: `RBFOpt` and `MISO` interpolate all observations with global radial basis function models but select candidates differently, `BAM` partitions the domain and fits local algebraic models, and `FMQA-SA` fits a quadratic model over a binary encoding. `NOMAD` adds a direct-search method that polls trial configurations on an adaptive mesh and keeps no global surrogate.

2.3.1 Branch-and-Model (`BAM`)

`BAM` combines domain partitioning with local surrogate construction (Ma et al., 2023). The evaluated points define boxes, and a potentially optimal selection procedure identifies regions for further modeling and search. Within selected boxes, Automated Learning of Algebraic Models (`ALAMO`) constructs sparse algebraic models from nearby observations (Cozad et al., 2014). `BARON` optimizes these local models to propose candidates, while box-size exploration maintains broader coverage (Ma et al., 2023; Sahinidis, 1996; The Optimization Firm, 2026). Within `BAM`, `BARON` optimizes these learned models and never receives the GDP itself. Its support for integer restrictions allows for the configuration domains considered here (The Optimization Firm, 2026).

2.3.2 Factorization-Machine-Based Optimization

A second-order factorization machine (FM) models individual feature effects and pairwise interactions through shared latent parameters (Rendle, 2010). For a binary encoding $\mathbf{b} \in \{0,1\}^m$ of an external in-

teger configuration $\boldsymbol{\xi}$, the FM fit surrogate is

$$\widehat{\Phi}_{\mathrm{FM}}(\mathbf{b}) = \beta_0 + \sum_{i=1}^{m} \beta_i b_i + \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} \langle \mathbf{a}_i, \mathbf{a}_j \rangle b_i b_j, \tag{3}$$

where $\widehat{\Phi}_{\mathrm{FM}}(\mathbf{b})$ denotes the predicted training response, m is the number of binary features, β_0 is the intercept, β_i is the linear coefficient of b_i, and $\mathbf{a}_i \in \mathbb{R}^r$ is its latent vector of dimension r. The inner product $\langle \mathbf{a}_i, \mathbf{a}_j \rangle$ determines the pairwise interaction coefficient. The model parameters are learned from the evaluated configurations and their corresponding training responses.

For binary inputs, $b_i^2 = b_i$, the fitted model is a binary quadratic model. Excluding the constant term β_0 gives the following quadratic unconstrained binary optimization (QUBO) problem:

$$\min_{\mathbf{b} \in \{0,1\}^m} \mathbf{b}^{\mathsf{T}} \mathbf{Q} \mathbf{b}, \qquad Q_{ij} = \begin{cases} \beta_i, & i = j, \\ \langle \mathbf{a}_i, \mathbf{a}_j \rangle, & i < j, \\ 0, & i > j. \end{cases} \tag{4}$$

The QUBO models the learned surrogate, and a binary string may decode to a configuration outside the domain or one that violates the logical constraints.

The `FMQA` framework combines FM surrogate fitting with annealing-based candidate generation (Kitai et al., 2020). Based on observed evaluations, the method fits the FM, samples binary candidates from the resulting quadratic model, and evaluates a selected decoded configuration through the objective-evaluation routine before updating the training data.

`FMQA-SA` uses classical simulated annealing to sample binary candidates from the learned QUBO. Temperature-dependent acceptance of uphill moves supports exploration, while gradual cooling favors lower-energy solutions (Kirkpatrick et al., 1983). In this study, no quantum hardware was used.

2.3.3 `RBFOpt`

`RBFOpt` is an open-source surrogate optimizer for bounded black-box problems, with support for continuous, integer, and categorical variables (Costa and Nannicini, 2018; Nannicini, 2021). Given N observed objective values in external integer configurations $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(N)}$, `RBFOpt` constructs a radial basis function (RBF) surrogate model of the form

$$\widehat{\Phi}_N(\boldsymbol{\xi}) = \sum_{r=1}^{N} \lambda_r \varphi\Big(\|\boldsymbol{\xi} - \boldsymbol{\xi}^{(r)}\|_2 \Big) + p(\boldsymbol{\xi}), \tag{5}$$

where $\widehat{\Phi}_N(\boldsymbol{\xi})$ approximates the objective value for a candidate configuration $\boldsymbol{\xi}$, λ_r are interpolation coefficients determined from the observed values, φ is the RBF kernel evaluated at the Euclidean distance between configurations, and $p(\boldsymbol{\xi})$ is a polynomial tail depending on the selected kernel.

Candidate selection balances low predicted values against exploration away from evaluated points. The metric stochastic response surface method (MSRSM) combines scaled surrogate predictions and distances to existing samples, changing their relative weights during the search (Regis and Shoemaker, 2007). New objective evaluations update the model.

2.3.4 `MISO`

`MISO` uses the radial basis function surrogate of Equation (5) but selects candidates with strategies designed for integer and mixed-integer variables (Müller, 2016). Its coordinate perturbation and target value (CPTV) strategy first perturbs randomly chosen coordinates of the best configuration found and scores the candidates by surrogate value and distance to the evaluated points, as MSRSM does. When this stage stops improving, it switches to a target value step, which proposes the configuration at which the surrogate could attain a prescribed target value with the least added bumpiness.

2.3.5 NOMAD

NOMAD implements mesh adaptive direct search (MADS) (Audet et al., 2022). Each poll evaluates trial points on a mesh around the incumbent, and the mesh is refined after unsuccessful polls and coarsened after successful ones; for integer variables, it cannot become finer than the integer spacing (Abramson et al., 2009). By default, NOMAD 4 complements the poll with search steps based on local quadratic models, but it does not retain a global surrogate of the objective.

3 Experimental Methods

3.1 Process GDP benchmark problems

The process benchmarks include a CSTR network (Liñán et al., 2020a), a 17-stage distillation column (Ghouse et al., 2018), and a small batch plant (Kocis and Grossmann, 1988). We use their GDP formulations reported by Ovalle et al. (2025), and the nine-stage nonlinear mode-switching model of Peng et al. (2024). Their objectives minimize the total reactor volume (`CSTR`), condenser and reboiler duties, plus the active tray cost (`Column`), the investment cost (`Small Batch`), and the negative integral of the squared state (`Dynamic`). The external variables represent the reactor/feed cutoff and recycle location (`CSTR`), reflux and boilup positions (`Column`), parallel unit counts (`Small Batch`), and final stages of modes 1 and 2 (`Dynamic`). For these process benchmarks, fixing the external choices leaves only continuous variables, so the resulting subproblems are NLPs. The `CSTR` superstructure contains N potential reactor positions, where $N \in \{5, 10, 15, 20, 25, 30\}$. We denote each `CSTR` instance by its size parameter N, such as `CSTR30` for $N = 30$. The `Dynamic` model is a DAE model that `pyomo.dae` discretizes by orthogonal collocation with 30 finite elements per stage and three Radau collocation points (Nicholson et al., 2018), so each of its fixed-configuration subproblems is an NLP. The reference configurations in external variable coordinates are (N, N) for `CSTR`, $(13, 4)$ for `Column` (physical trays $(14, 5)$), $(2, 2, 1)$ for `Small Batch`, and $(1, 4)$ for `Dynamic`. The full GDP formulations of the `CSTR`, `Column`, and `Small Batch` benchmarks are provided in Appendices A.1-A.3 of Ovalle et al. (2025), respectively, while the `Dynamic` formulation is described in Section IV.C of Peng et al. (2024). External-variable mappings for the three design cases are provided in Appendices B.1, B.2, and B.4 of Ovalle et al. (2025). The `CSTR` volume is reported in liters and the `Small Batch` investment in the source model's dollars. The `Column` uses the source model's weighted cost proxy based on duties and active trays. The model sizes and unit conventions are given in the Supplementary Section S4 (Table S3). `Dynamic` is an abstract benchmark with no calibrated physical units for state or time. Minimizing its negative squared-state integral favors greater integrated state magnitude and tests mode sequencing coupled to dynamic equations.

Table 1 summarizes the original GDP and DFO candidate domains, the number of configurations whose objective values the interior-point NLP solver `IPOPT` (Wächter and Biegler, 2006) returns with an accepted status, the number of configurations with measured evaluation times, and the evaluation budgets. The process searches use frozen objective tables from fixed-configuration `IPOPT` evaluations; only accepted finite objectives are used as valid lookup objectives. The objective, acceptance status, and point cost are taken from the same recorded evaluation attempt for the same model instance and configuration. Measured coverage includes all configurations in each DFO candidate box, including unsuccessful and structural-rejection evaluations. The evaluation budget B limits the total number of objective evaluations, including initialization, repeated calls, and unsuccessful evaluations. The full GDP box precedes the logical and algebraic feasibility restrictions. `Column` searches $\{8, \ldots, 15\} \times \{1, \ldots, 8\}$ (64 coordinates), the logical domain of its original $\{1, \ldots, 15\}^2$ box. Other cases use their full boxes. The `Column` bounds encode the reflux and boilup placement restrictions; algebraic constraints are still checked during configuration evaluation. All DFO methods use this box, and the `Column` conclusions apply only to it. For `CSTR`, recycle/feed ordering admits $N(N + 1)/2$ logical candidates, but not all yield objective values satisfying the solver acceptance criteria defined in Section 3.6. The `CSTR` comparisons do not isolate scalability: evaluation budgets change with size. `Dynamic` admits only configurations with $\xi_1 < \xi_2$ or $(\xi_1, \xi_2) = (9, 9)$, which leaves 37 logically admissible configurations. `Small Batch` uses all 27 configurations in $\{1, 2, 3\}^3$, whose decisions count parallel mixer, reactor, and centrifuge units. Of these configurations, 15 are rejected during feasibility-based bound tightening (FBBT) preprocessing.

Table 1: Process GDP models: configuration domains, evaluation budgets, reference objectives, and the cost of full enumeration.

Model	N	Full GDP box	Full box size	DFO domain size	Accepted	B	f_{ref}	Unit	Enumeration time (s)
`CSTR`	5	$\{1,\dots,N\}^2$	25	25	15	8	3.06201	L	30.27
	10		100	100	54	15	2.88954	L	122.06
	15		225	225	120	22	2.82939	L	298.42
	20		400	400	210	29	2.79941	L	553.38
	25		625	625	325	36	2.78148	L	899.39
	30		900	900	464	43	2.76957	L	1,719.00
`Column`	–	$\{1,\dots,15\}^2$	225	64	29	13	19,346.1	none	137.24
`Small Batch`	–	$\{1,2,3\}^3$	27	27	12	10	167,428	$	10.18
`Dynamic`	–	$\{1,\dots,9\}$ $\times\{2,\dots,9\}$	72	72	37	12,15,20, 30,37,50	-212.009	none	47.25

N: number of potential reactors of `CSTR`; –: not applicable. Full GDP box: external-variable values before logical restrictions. DFO domain: configurations that the DFO methods search; it equals the full GDP box except for `Column`, whose DFO domain is its logical domain $\{8,\dots,15\}\times\{1,\dots,8\}$, and every configuration in it has a measured evaluation cost. Accepted: configurations of the DFO domain whose fixed-configuration `IPOPT` solve returns a finite objective, violates no constraint or bound by more than 10^{-5}, and ends with an accepted solver status (Section 3.6). B: evaluation budget. f_{ref}: reference objective, the lowest accepted objective of the DFO domain. Unit: L, reactor volume in liters; $, investment in the dollars of the source model; none, no physical unit (the weighted cost of duties and active trays of the `Column` source model and the negative squared-state integral of `Dynamic`). Enumeration time: summed measured cost of evaluating every configuration of the DFO domain once. The 980 primary process runs (`D-SDA`, `NOMAD`, `RBFOpt`, `MISO`, `BAM`, `FMQA-SA`, and uniform sampling; 14 model-budget conditions and ten seeds per condition) evaluate 2,106 distinct configurations over all models and sizes.

3.2 Simulation-based benchmark: lomustine manufacturing

The `Lomustine` benchmark selects the flowsheet of a manufacturing process for the anticancer drug lomustine, whose model is a `PharmaPy` simulation (Casas-Orozco et al., 2021, 2023). The superstructure was introduced by Laky and Casas-Orozco (2022), related flowsheet synthesis frameworks are described by Barhate et al. (2025), and Park and Bernal Neira (2026) studied its discrete landscape with Ising-based solvers; we use the corrected-model data of SECQUOIA (2026b). The superstructure contains two reactors in series, each a batch reactor, a CSTR, or a PFR, followed by a batch vaporizer and a crystallizer that is either an unseeded batch cooling crystallizer or a train of one, two, or three mixed-suspension, mixed-product-removal (MSMPR) crystallizers. The external variables ξ_1 and ξ_2 select the two reactors in the order (batch, CSTR, PFR), and ξ_3 selects the crystallizer in the order (batch, one, two, three MSMPR stages), which gives $3\times3\times4=36$ configurations, all logically admissible. These choices are categorical, so this case does not meet the ordering condition of the external-variable reformulation; we index the alternatives in the order of the source enumeration, and the neighborhoods of `D-SDA` and `NOMAD` and the surrogate models act on this order.

Written as a GDP, the flowsheet is

$$
\begin{aligned}
\min_{\mathbf{u},\mathbf{s},\mathbf{Y}} \quad & f(\mathbf{u},\mathbf{s}) \\
\text{s.t.} \quad & \bigvee_{a\in\mathcal{R}} \begin{bmatrix} Y_a^1 \\ \mathbf{s}_1 = \mathcal{S}_a^{\mathrm{R}}(\mathbf{u},\mathbf{s}_0) \end{bmatrix}, \qquad \bigvee_{a\in\mathcal{R}} \begin{bmatrix} Y_a^2 \\ \mathbf{s}_2 = \mathcal{S}_a^{\mathrm{R}}(\mathbf{u},\mathbf{s}_1) \end{bmatrix}, \\
& \mathbf{s}_3 = \mathcal{S}^{\mathrm{V}}(\mathbf{u},\mathbf{s}_2), \qquad \bigvee_{c\in\mathcal{C}} \begin{bmatrix} Y_c^{\mathrm{C}} \\ \mathbf{s}_4 = \mathcal{S}_c^{\mathrm{C}}(\mathbf{u},\mathbf{s}_3) \end{bmatrix}, \\
& \mathbf{g}(\mathbf{u},\mathbf{s}) \le \mathbf{0}, \qquad \mathbf{u}\in\mathcal{U}, \qquad \mathbf{Y}\in\{\text{False},\text{True}\}^{10},
\end{aligned}
\tag{Lom}
$$

where $\mathcal{R}=\{\text{batch},\text{CSTR},\text{PFR}\}$ and $\mathcal{C}=\{\text{batch},\text{one},\text{two},\text{three MSMPR stages}\}$. Each disjunct contains the `PharmaPy` model of one unit, which maps the operating inputs $\mathbf{u}$ and the state $\mathbf{s}_{k-1}$ of the incoming stream to the outlet state $\mathbf{s}_k$, and $\mathbf{s}_0$ is the reagent feed. The vaporizer $\mathcal{S}^{\mathrm{V}}$ is present in every configuration,

f is the objective defined below, and $\mathbf{g}$ is the solubility path constraint. The external variables ξ_1, ξ_2, and ξ_3 index the three disjunctions, and fixing them selects one unit model in each, which gives Problem (Sim). The unit models are available only as simulations, so Problem (Lom) is a GDP whose disjuncts have no algebraic form for the conventional GDP methods.

For each configuration, `NOMAD` optimizes 12 to 16 normalized operating inputs, including reactor temperatures and residence times, the reagent concentration, the cooling profile, and the crystallizer residence times, with 60 simulations, which solves Problem (Sim) approximately. The objective $-(\dot{m}/10^{-4} + 2d_{50})$, with the production rate $\dot{m}$ in kg/s and the median crystal size d_{50} in μm, rewards both quantities, and a solubility path constraint prevents crystallization before the crystallizer; all 36 configurations satisfy it. These inner optimizations take 91 to 1,019 s per configuration, so evaluating every configuration takes 5.3 h. All methods read the stored results of these inner optimizations, so they evaluate the same $\Phi(\boldsymbol{\xi})$, and each evaluation is charged its recorded inner-optimization time. The reference configuration is a PFR followed by a batch reactor and the batch crystallizer, $\boldsymbol{\xi} = (3, 1, 1)$, with $f_{\text{ref}} = -487.842$; the second-best configuration replaces the PFR by a CSTR and lies 0.8% above it. The budgets are $B = 10$, 15, and 20, with the shared initialization, seeds, unaccepted-response values, and target rule of the process benchmarks, and `D-SDA` uses the same unit L_∞ neighborhood. Supplementary Section S5 lists the 36 objective values and the data provenance.

3.3 Catalytic distillation column with live NLP evaluations

The `Catalytic Column` benchmark designs a catalytic distillation column that produces ethyl *tert*-butyl ether (ETBE) from ethanol and a mixture of butenes, for which Liñán et al. (2020b) formulated a GDP model and solved it with `D-SDA`. The column has a condenser, 20 internal stages, and a reboiler, and the reflux returns to the top internal stage. The GDP decisions place the ethanol feed at stage f_1, catalyst on three stages $c_1 < c_2 < c_3$, the butenes feed at stage f_2, and the boil-up return at stage b, with $2 \le f_1 \le c_1 < c_2 < c_3 \le f_2 \le b \le 21$. Internal stages below the boil-up return do not exist, so b also sets the number of trays. Each existing stage is an equilibrium stage with tray hydraulics, and each catalytic stage adds the reaction kinetics and a 120 °C limit on the catalyst temperature. Every decision chooses a stage from an ordered list, so this model meets the ordering condition of the external-variable reformulation (Ovalle et al., 2025). We use a corrected version of the published model in which thermodynamic, property, hydraulic, and cost errors are fixed and the pure-component data are fitted to open reference data (Supplementary Section S6). The objective is the negative annual profit, in \$/yr, of a column whose bottoms product contains at least 95 mol% ETBE, at 2010 prices; it is positive because the butenes left in the distillate earn no credit. With the configuration fixed, the model is an NLP with about 2,600 variables and 2,700 equations, which `CONOPT` (Drud, 1994) solves from the initialization of the published model.

The search methods use the six gaps $\mathbf{g} = (f_1{-}2,\ c_1{-}f_1,\ c_2{-}c_1{-}1,\ c_3{-}c_2{-}1,\ f_2{-}c_3,\ b{-}f_2) \in \{0, \dots, 6\}^6$ as external variables, so that every point of the box satisfies the order of the stages. The GDP admits 100,947 configurations, and the bound of six on each gap, which keeps every design of a pilot within 5% of the pilot's best, leaves 54,159 of them in the box. Of the 117,649 points of this box, the 54,159 with $\sum_i g_i \le 17$ are admissible; the others would place the boil-up return below the last internal stage and are charged to the budget without an NLP solve, like the logical rejections of `CSTR` and `Dynamic`. Unlike in the other process cases, no stored table of responses exists: each evaluation solves the NLP of its configuration when the configuration is first requested. Because these responses are deterministic, they are stored and reused when another run requests the same configuration, and each evaluation is charged the recorded solve time of its configuration. A configuration is accepted when `CONOPT` returns an optimal or locally optimal solution with normal completion within 300 s. The 3,499 configurations solved in the campaign took a median of 4.8 s and a mean of 7.9 s, so solving the 54,159 admissible configurations of the box would take about 119 h, and all 100,947 configurations of the GDP about 220 h.

The budgets are $B = 25$, 50, and 100 evaluations, and each seed has seven shared initial configurations drawn uniformly from the admissible configurations, one more than the number of external variables, which replace the three Latin hypercube points of the other benchmarks. `D-SDA` starts from the best of them and uses the separable neighborhood, which changes one external variable by ± 1 and corresponds to the 2-norm neighborhood of the original `D-SDA` study of this column (Liñán et al., 2020b). The unit L_∞ neighborhood of the other benchmarks would contain 728 configurations in six dimensions, more than seven times the

largest budget, whereas the separable neighborhood contains at most 12. Because D-SDA moves in the gap coordinates, one move shifts every stage after the changed gap, so its neighbors differ from the neighbors in stage positions. RBFOpt, BAM, NOMAD, and MISO keep the settings of the process benchmarks, FMQA-SA has a time limit of 7,200 s instead of 1,800 s because the solve times of its evaluations count toward it, and uniform sampling draws without replacement from its own random stream. No global optimum is known, so the reference is the best accepted configuration found by any run or by a pilot of 202 configurations, $(f_1, c_1, c_2, c_3, f_2, b) = (8, 8, 12, 17, 17, 21)$ with $f_{\text{ref}} = 24{,}895.55$ \$/yr, and the target rule is that of the process benchmarks. Its first gap lies on the bound of the box, so we also solved the 747 configurations outside the box within two stages of it: none is better, and the best lies 0.14% above it (Supplementary Section S6). The 210 runs ran as three concurrent single-core jobs on the server described in Section 3.6 with GAMS and CONOPT.

We also solved three MINLP forms of the column with DICOPT (Viswanathan and Grossmann, 1990), SBB (GAMS Development Corporation, 2026), BARON (Sahinidis, 1996), and SCIP (Bolusani et al., 2024) in GAMS, each on one core of the same server with a limit of 3,600 s, from the initialization and initial configuration of the published model. These forms contain all 100,947 configurations of the GDP. The first is the MINLP form published with the D-SDA study, which multiplies the switched equations by binary variables (Liñán et al., 2020b). The other two are the Big-M and Hull reformulations that the GAMS extended mathematical programming framework (Ferris et al., 2009) generates from a disjunctive form of the corrected model with one disjunction for each stage decision. Both reformulations are valid only after two modeling steps. Each nonlinear equation of a disjunct defines a residual variable on every stage, so that every disjunct constraint is linear and the Hull reformulation never evaluates terms such as $1/\rho_L$ at zero. Each disjunct constraint is also divided by an interval bound on its residual, so that $M = 1$ relaxes every constraint whose disjunct is inactive (Supplementary Section S6). These interval bounds are finite only if the liquid is denser than the vapor, so the two reformulations also bound the mass densities of the liquid (at least 200 kg/m^3) and the vapor (5 to 150 kg/m^3), which the GDP evaluations and the published form do not. For any vapor composition, the vapor bounds cannot bind within the model's bounds on pressure, temperature, and compressibility factor, and every validated solution has a liquid density of at least 529 kg/m^3. With the binary variables fixed, the Big-M and Hull reformulations reproduce the GDP objective at the six feasible configurations we tested, the published form reproduces it at four of them and reaches other local optima at the remaining two, and all forms agree that three other configurations are locally infeasible.

3.4 Controlled benchmark: Grid-GDP

To study search behavior without NLP subproblems, we also consider 21 two-dimensional test functions adapted from the Quantum Hamiltonian Descent benchmark collection (Wu Group, n.d.). Each is sampled at 31 equally spaced values per coordinate, giving 961 configurations. For each benchmark function, let $[\ell_x, u_x]$ and $[\ell_y, u_y]$ denote the coordinate bounds, and let $I = J = \{0, \ldots, 30\}$ be the corresponding index sets. The Grid optimizers receive $(i, j) \in \{0, \ldots, 30\}^2$ directly. Relative to the one-based convention in Section 2.1, $(\xi_1, \xi_2) = (i + 1, j + 1)$ is only a relabeling of alternatives. FMQA-SA encodes each Grid index in five unsigned bits and rejects decoded values of 31. The physical coordinates are defined as

$$a_i = \ell_x + \frac{i(u_x - \ell_x)}{30}, \qquad b_j = \ell_y + \frac{j(u_y - \ell_y)}{30}. \tag{6}$$

Let f denote the original analytical objective function and let L and U be its minimum and maximum over the 31×31 lattice. Both constants are obtained by exhaustive preprocessing of all 961 lattice points prior to any search. A normalized target tolerance of 10^{-5} corresponds to $10^{-5}(U - L)$ in raw objective units, so equal normalized tolerances do not need to define equally large target sets. Supplementary Table S1 gives target-set sizes for all functions. Full-precision normalization constants are available in the repository

(Supplementary Section S8). Each benchmark function is represented by the following GDP:

$$
\begin{aligned}
\min_{x,y,Y^x,Y^y} \quad & F(x,y) = \frac{f(x,y) - L}{U - L} \\
\text{s.t.} \quad & \bigvee_{i \in I} \begin{bmatrix} Y_i^x \\ x = a_i \end{bmatrix}, \qquad \bigvee_{j \in J} \begin{bmatrix} Y_j^y \\ y = b_j \end{bmatrix}, \\
& \text{ExactlyOne}(Y_0^x, \ldots, Y_{30}^x), \\
& \text{ExactlyOne}(Y_0^y, \ldots, Y_{30}^y), \\
& \ell_x \le x \le u_x, \qquad \ell_y \le y \le u_y, \\
& Y_i^x, Y_j^y \in \{\text{False}, \text{True}\}.
\end{aligned}
\tag{7}
$$

The two exclusive disjunctions contain 62 coordinate-selection Boolean indicators. The objective retains the explicit mathematical expression. The conventional GDP methods therefore receive the expression and its disjunctive structure, whereas the DFO methods observe only the objective values at the coordinates they select.

Fixing both coordinate disjunctions fixes x and y, leaving no remaining original continuous degrees of freedom. The measured fixed-point `IPOPT` (Wächter and Biegler, 2006) evaluations therefore consist of model preparation, solver invocation, and verification. The unfixed objective functions remain nonlinear and generally nonconvex, so Big-M and Hull produce MINLP representations. The complete mathematical definitions and coordinate domains of the 21 benchmark functions are provided in Supplementary Sections S1-S2 (Table S1) (SECQUOIA, 2026a; Wu Group, n.d.).

3.5 Search protocol and compared algorithms

3.5.1 Shared initialization and evaluation budgets

In the process, `Lomustine`, and Grid-GDP benchmarks, all configuration-search methods and sampling controls use precomputed configuration responses during the search; the `Catalytic Column` solves each configuration when it is first requested (Section 3.3). For process cases, accepted responses are objective values from `IPOPT` under the initialization and acceptance rules in Section 3.6. Because `IPOPT` returns local solutions, these values can exceed the subproblem value $\Phi(\boldsymbol{\xi})$.

Each Grid-GDP benchmark uses ten seeds, each defining a different initial design, and evaluation budgets $B \in \{10, 30, 50, 70, 100\}$. The primary Grid comparison comprises six search implementations and two sampling controls, giving $21 \times 10 \times 5 \times 8 = 8{,}400$ runs. The process benchmarks comprise 14 model-budget conditions, each evaluated with ten paired seeds, giving 140 runs per method and 980 runs across the six search implementations and uniform sampling (Table 1). The `Lomustine` benchmark adds three budgets with ten seeds each, 210 runs in total, and the `Catalytic Column` another 210 runs with seven initial configurations per seed (Section 3.3). In all other benchmarks, for each problem-budget-seed combination, three distinct initial configurations are generated using seeded Latin hypercube sampling, mapped to integer coordinates, and shared in the same order across all methods and sampling controls. This common design controls the first three evaluations; subsequent sampling, response handling, and stopping rules follow the individual implementations. All process cases use seeds 0-9, with no replacement of failed starts.

Methods may terminate before exhausting B; only Grid `D-SDA` receives the reference target as a stopping criterion. Detailed settings and worker time limits are provided in the Supplementary Section S3. Every objective evaluation consumes budget, including initialization, repeated configurations, and unaccepted evaluations; internal cache lookups that do not request an evaluation are free. Target attainment uses the recorded acceptance status and original objective values, while each method receives the following values for unaccepted evaluations. In all process cases, `D-SDA`, `RBFOpt`, `NOMAD`, and `MISO` assign a penalty of 10^6 to unaccepted evaluations. `BAM` represents these evaluations by `NaN`, indicating that no accepted objective value is available. For surrogate training, `FMQA-SA` scales the accepted objective values to $[0, 1]$ and assigns the unaccepted evaluations a value of 2.

3.5.2 BAM

BAM uses local algebraic surrogate models optimized by BARON to generate candidate configurations (Ma et al., 2023; The Optimization Firm, 2026). All external variables are integers. The three shared configurations and their responses are supplied without an additional starting point. The remaining evaluation budget is $B-3$, with other settings at their defaults. On Grid-GDP, the next ten planned evaluations use a Sobol sequence, which distributes points across the domain, subject to the remaining budget. In the process experiments, BAM stops at its own evaluation limit after the B-th evaluation.

3.5.3 Budget-limited D-SDA

We use a budget-limited implementation of the previously developed D-SDA (Liñán et al., 2020a), applying its neighborhood search and discrete line search over the GDP external integer lattice as in LD-SDA (Ovalle et al., 2025). It evaluates the unit L_∞ neighborhood in fixed enumeration order, selects the best improving neighbor, and continues along the resulting direction through a discrete line search; on the six-dimensional Catalytic Column, it uses the separable neighborhood instead (Section 3.3), and Supplementary Section S14 repeats every other D-SDA run with the separable neighborhood. Neighborhood and line-search moves require an absolute objective improvement greater than 10^{-5}. As in the original D-SDA and LD-SDA implementations, a configuration is never evaluated twice: when a neighbor or line-search point was already evaluated in the same run, including the shared initial configurations, D-SDA reuses its stored value, and no evaluation is charged. If the evaluation budget B is exhausted, the best solution found is retained.

For the process benchmarks, D-SDA starts with the best accepted configuration among the three initial points evaluated and stops when this descent ends. For Grid-GDP benchmarks, initial configurations are ranked by objective value and successive descents are attempted while evaluation budget remains. After each completed descent, the search stops if the best observed objective satisfies the common target criterion in Equation (8); otherwise, it tries the next ranked initial configuration. For all process benchmarks, D-SDA searches within the declared integer bounds without precomputed feasibility filtering, using the evaluation accounting and unaccepted-response values stated above. To check that the single process descent does not handicap D-SDA, we repeated the 140 process runs with the Grid restart rule, and with further descents from uniformly drawn unvisited configurations until the budget is exhausted (Supplementary Section S13).

3.5.4 FMQA-SA

FMQA-SA uses the FM training and QUBO construction code of SECQUOIA (n.d.). At each iteration, a new second-order rank-eight FM is fitted to all evaluated configurations. Finite precomputed responses are min-max scaled to $[0, 1]$ (or zero when all finite values are identical), while non-finite evaluations receive a training target of 2.0.

Integer coordinates are encoded in unsigned binary, and the fitted FM is converted to a QUBO. Binary candidates are generated on CPU using D-Wave's classical simulated annealing sampler (D-Wave Systems, n.d.). The lowest-energy decoded candidate within the coordinate bounds that has not been previously evaluated is selected; if none is available, a candidate is drawn uniformly from the remaining unevaluated configurations. Candidates are not screened for the feasibility of logical or nonlinear constraints before evaluation. The new observation is appended, and FM fitting and candidate generation repeat until the evaluation budget B or the 1,800 s method time limit is reached.

3.5.5 MISO

MISO uses the CPTV strategy of Müller (2016) as implemented in soogo, with a cubic radial basis function model with a linear polynomial tail, integer coordinates, and a median low-pass filter on large objective values. The three shared configurations and their responses initialize the model. Because soogo does not accept pure-integer domains in its coordinate-perturbation step, our adapter applies the same dynamically dimensioned search perturbation on the integer lattice and disables the continuous local search, which has no continuous coordinates to act on. A linear tail needs $d+1$ affinely independent points, so when the shared initial configurations do not span the space, as with the three initial configurations of the three-dimensional Small Batch and Lomustine cases, uniformly drawn admissible configurations, charged to the budget, are

added until they do. A run stops at the budget, or after ten consecutive iterations without a new evaluation when no unevaluated candidate remains.

3.5.6 NOMAD

NOMAD (Audet et al., 2022), called through PyNomadBBO, runs mesh adaptive direct search with all external variables declared integer. It starts from the best of the three shared initial configurations, whose stored value it reuses without charge, with other settings at their defaults. With integer variables, NOMAD stops when its mesh reaches the integer resolution, so, like D-SDA, it can end before the budget. NOMAD sometimes terminated with a segmentation fault after its last new evaluation. Such a run is repeated with larger evaluation caps, and a rerun is accepted only if it evaluates exactly the same configurations in the same order; 123 of the 1,190 runs completed this way. Twenty-one Grid runs at $B = 100$ crashed on every attempt after identical evaluations; they are scored on the evaluations completed before the crash, and 16 of them had already reached the target (Supplementary Section S12). The Lomustine harness repeats a crashed run once; 12 runs of its option-order and capital-cost repetitions, none in the source order of Table 3, crashed on both attempts after identical evaluations and are scored in the same way (Supplementary Section S5).

3.5.7 RBFOpt

RBFOpt uses MSRSM, automatic RBF selection, and the default genetic candidate-search option of version 4.3.0 (Costa and Nannicini, 2018; Nannicini, 2021). All external variables are integers, and the three shared initial configurations are supplied with the additional package initialization disabled. Search stops at the evaluation budget or the native iteration and stall limits. Other surrogate-minimization operations use standalone Bonmin (Bonami et al., 2008) and IPOPT (Wächter and Biegler, 2006) executables. These calls contribute to search time, but not the objective-evaluation budget.

3.5.8 Sampling controls and full enumeration

Uniform sampling evaluates the same three initial configurations and then draws uniformly without replacement from the remaining declared integer domain. It uses a separate deterministic random stream, no objective-dependent selection, and no logical or NLP-feasibility screening before evaluation. The Grid BAM-prefix control repeats BAM's ten predetermined sampling points in the same order after the three shared starts, then samples uniformly without replacement, stopping at the budget. This control isolates the contribution of the fixed initial design of BAM. Full enumeration evaluates every configuration of the DFO domain once. It reaches the target by construction, because f_{ref} is the best accepted value among these configurations, and its cost is the sum of their measured point costs.

3.6 Execution environment and conventional GDP comparisons

All experiments were conducted on a server with an AMD EPYC 7643 processor (48 cores, 2.3 GHz) and 1 TB RAM. All optimization methods and numerical libraries were configured for single-threaded execution. The software comprised Python 3.11.16, Pyomo 6.10.0 (Bynum et al., 2021), RBFOpt 4.3.0, BAM 5.0.0, soogo 2.1.0, NOMAD 4.6.0 called through PyNomadBBO 4.6.0, PyTorch 2.8.0 (Paszke et al., 2019), Bonmin 1.8.9, IPOPT 3.14.19 with MUMPS 5.8.1 (Amestoy et al., 2001), and CONOPT 4.39.1. The process and Grid comparisons called IPOPT and SCIP 9.2.4 (Bolusani et al., 2024) through the General Algebraic Modeling System (GAMS) 52.2.0 (GAMS Development Corporation, 2025). The Catalytic Column used GAMS 54.3.1 with SCIP 10.0.3 and BARON 26.5.27 (Section 3.3), and its check of the configurations outside the search box used GAMS 54.2.2 with the same CONOPT.

Fixed-disjunction NLP evaluations. We fix the original Boolean GDP decisions and evaluate the resulting subproblems using IPOPT with the MUMPS linear solver. For the Grid-GDP benchmarks, all 20,181 configurations were evaluated individually, while the process benchmarks used fixed-configuration evaluations of the corresponding process models. IPOPT used a tolerance of 10^{-8} and a solver time limit of 60 s per configuration. Process acceptance requires a finite objective, a maximum absolute violation of any constraint or variable bound not greater than 10^{-5}, and the accepted solver and termination statuses specified in the

Supplementary Section S3.1. Time-limited candidates remain unaccepted even when feasible. `CSTR` sizes 5-30 use the mass-balance/zero-recycle initialization. `Column` and `Small Batch` use their saved initialization files, and `Dynamic` uses fresh clones of the unsolved discretized template.

Conventional GDP comparisons. Conventional comparisons include the `GDPopt` methods GLOA, LOA, and RIC, as well as the Big-M and Hull reformulations. `GDPopt` uses `SCIP` for MILP master problems and `IPOPT` for NLP subproblems (Chen et al., 2022; Pyomo Development Team, n.d.). Because fixing both Grid disjunctions also fixes x and y, the Grid subproblems are solved as linear problems by `SCIP`, and `IPOPT` is not called in those runs. The Big-M and Hull reformulations are solved directly using `SCIP`. All conventional GDP methods use the original GDP domains, a time limit of 1,800 s per instance, and a relative optimality gap of 10^{-5}; the MINLP solvers of the `Catalytic Column` use a limit of 3,600 s and a relative gap of zero (Section 3.3).

3.7 Target attainment and cumulative success profiles

Let f_{best} denote the lowest feasible objective value observed so far within the evaluation budget, and let f_{ref} denote the reference objective value. For process benchmarks, f_{best} is computed from evaluations that meet the solver acceptance criteria in Section 3.6. A run reaches the reference target when

$$|f_{\text{best}} - f_{\text{ref}}| \leq 10^{-5} \max\{1, |f_{\text{ref}}|\}. \tag{8}$$

For the process experiments, f_{ref} is the lowest accepted objective value among the precomputed fixed-configuration NLP evaluations, so reaching the target means matching the best configuration that full enumeration of the stored responses would find. Runs that terminate with an execution failure are counted as unsuccessful, even if the target was observed earlier; the exception is a `NOMAD` run that crashes on every attempt after evaluating the same configurations, which is scored on the evaluations completed before the crash (Section 3.5.6). For process runs, the reported solution is the best accepted configuration recorded within budget, including the final evaluation. Reported objectives and bounds of the process benchmarks use six significant figures; all target and bound checks use unrounded values. For the full-process-GDP comparison, a feasible objective at or below $f_{\text{ref}} + 10^{-5} \max\{1, |f_{\text{ref}}|\}$ counts as achievement, so solutions better than the reference are also included.

For each evaluation budget B, let $\mathcal{R}_B$ denote the set of problem-seed instances compared across methods, and let s identify a solution method. For each instance $r \in \mathcal{R}_B$, H_{sBr} equals one if the run reaches the target and zero otherwise. Let E_{sBr} be the index of the first objective evaluation at which the target is reached, with $E_{sBr} = +\infty$ if the target is not reached. At a threshold of N objective evaluations, where $0 \leq N \leq B$, the absolute evaluation-count profile is

$$C_{sB}^{\text{eval}}(N) = \frac{1}{|\mathcal{R}_B|} \sum_{r \in \mathcal{R}_B} \mathbf{1}\{H_{sBr} = 1,\ E_{sBr} \leq N\}. \tag{9}$$

Here, $\mathbf{1}\{\cdot\}$ equals one when the stated conditions hold and zero otherwise, and $|\mathcal{R}_B|$ is the number of problem-seed instances. Thus, $C_{sB}^{\text{eval}}(N)$ gives the fraction of runs that reach the target within N evaluations. For the Grid-GDP benchmarks, $|\mathcal{R}_B| = 210$ at each budget; for each process benchmark and budget, $|\mathcal{R}_B| = 10$. All runs, including target misses, remain in the denominator.

Experiments at different evaluation budgets are analyzed separately, so a larger-budget trajectory need not extend a smaller-budget one. These profiles count raw evaluations, without the best-solver normalization of performance profiles (Dolan and Moré, 2002) or the dimension normalization of data profiles (Moré and Wild, 2009).

NLP subproblems solved. An evaluation counts as an NLP subproblem solve when its configuration passes logical and FBBT preprocessing and `IPOPT` is called, whether or not the result is accepted. For `GDPopt`, we count the `IPOPT` calls confirmed in its execution logs. Internal NLP counts are unavailable for the direct Big-M and Hull solves.

3.8 Runtime accounting and uncertainty

For a recorded lookup-based run with n objective evaluations and measured lookup-call times, the estimated full-run completion time is

$$\widehat{T}_{\text{complete}} = T_{\text{recorded}} - \sum_{q=1}^{n} t_q^{\text{lookup}} + \sum_{q=1}^{n} c(\boldsymbol{\xi}^{(q)}) + t_{\text{prep}}. \tag{10}$$

Here, T_{recorded} is the measured runtime of the search based on lookup, t_q^{lookup} is the measured lookup-processing interval of its qth objective call, and $\boldsymbol{\xi}^{(q)}$ is the external integer configuration evaluated in that call. The point cost $c(\boldsymbol{\xi}^{(q)})$ is measured separately for the same benchmark model, problem size and configuration, including configuration-specific preparation, preprocessing, invocation of the solver when required, and validation. The term t_{prep} accounts for the preparation of a single model that is not already included in the point costs. Search time includes surrogate fitting, candidate generation, auxiliary optimization, and recording performed within the method clock. Throughout this manuscript, full-run completion denotes completion of this recorded search stage with reconstructed evaluation costs, excluding final result extraction and export. Every recorded objective call contributes its point cost, including initialization, repeated visits, and preprocessing rejections; the same stored measurement is used each time a configuration is revisited. Completion times therefore combine recorded search paths with one stored cost measurement per configuration.

Grid-GDP timers. Grid-GDP enumeration and objective normalization are excluded from the timing analysis, and $t_{\text{prep}} = 0$ because configuration-specific preparation is included in point costs. Recorded lookup-call times are subtracted from the method runtime for all compared Grid methods. The 21 `NOMAD` runs that crashed on every attempt have no recorded times and are left out of the Grid timing summaries.

Process timers. Structurally rejected configurations incur measured preparation and rejection costs without an NLP solution (Supplementary Section S3.1). For `Dynamic`, the original model-template construction and discretization cost is included once per run through t_{prep}; additional campaign-level preparation is not charged repeatedly. All in-budget evaluations have measured point costs, and estimated full-run completion times are available for all 980 primary process runs.

Conventional GDP completion times are measured separately and are not augmented with the DFO point costs. For Grid-GDP, conventional GDP timings include per-run preparation, reformulation, solution, and validation, but exclude shared-template preparation and lattice enumeration. Full-process-GDP timings include model-template preparation.

Statistical comparisons. With ten seeds per process condition, we compare two methods with the two-sided exact McNemar test on their seed-paired target outcomes, and we call differences that this test does not resolve at the 0.05 level comparable (Supplementary Table S17). The tests of `RBFOpt` against `D-SDA` are descriptive and not adjusted for multiple comparisons, and the 70 exploratory tests that involve `NOMAD` or `MISO` use a Holm correction (Supplementary Section S12). Grid comparisons use a percentile bootstrap that resamples the 21 functions (Supplementary Section S10).

4 Computational Results and Discussion

4.1 Process GDP case studies

4.1.1 Target attainment

Table 2 reports target attainment for the 14 process conditions. Across the 60 paired `CSTR` runs, `RBFOpt` reaches the target in 45, `MISO` in 43, `NOMAD` in 21, and `D-SDA` in 16. `RBFOpt` succeeds in 30 runs in which `D-SDA` misses, and `D-SDA` in one in which `RBFOpt` misses (exact paired test $p < 10^{-7}$; Supplementary Table S17). With ten seeds per size, the difference is significant at the 0.05 level only for `CSTR5` and `CSTR30`, but it has the

same sign at every size. On Dynamic at $B = 12$, RBFOpt reaches all ten targets and D-SDA four ($p = 0.031$); D-SDA reaches all ten from $B = 20$ onward.

MISO shows the same advantage over D-SDA with a different profile across sizes. It succeeds in 28 CSTR runs in which D-SDA misses, against one in the other direction ($p < 10^{-6}$), and it reaches all 30 targets on CSTR20 through CSTR30, but only 13 of 30 on CSTR5 through CSTR15, where RBFOpt reaches 21. Pooled over the CSTR sizes, MISO and RBFOpt disagree in 16 runs, seven in favor of MISO ($p = 0.80$). NOMAD reaches more CSTR targets than D-SDA (7 against 2 discordant runs, $p = 0.18$) but fewer than RBFOpt (5 against 29, $p < 10^{-4}$). Like D-SDA, it stops before the budget in many runs (77 of 140) because its mesh reaches the integer resolution. Exploratory per-condition tests of NOMAD and MISO against D-SDA, RBFOpt, and each other find no contrast that remains significant after a Holm correction over the 70 tests (Supplementary Section S12).

On the two smallest design models, D-SDA and RBFOpt are comparable. D-SDA reaches five Column targets and RBFOpt four ($p = 1$), and on Small Batch they reach eight and five ($p = 0.38$). MISO reaches all ten Column targets and NOMAD seven; on Small Batch they reach seven and six. BAM reaches all ten CSTR5 targets, compared with eight for RBFOpt ($p = 0.50$). The only Column hits of BAM and uniform sampling occur in seed 6, whose shared initial design contains the target.

The single-descent D-SDA stops before exhausting its budget in 87 of the 140 process runs. Restarting it from the remaining shared initial configurations, or from uniformly drawn unvisited configurations until the budget is spent, adds one target, raising target attainment from 79 to 80 of 140 runs; both restart variants improve the final best objective in 14 runs (Supplementary Section S13).

FMQA-SA reaches 42 of 140 process targets, compared with 37 for uniform sampling. In the small process domains, the 100 annealed samples often contain no new configuration within the bounds, so 52.9% of the FMQA-SA proposals come from its uniform fallback (Supplementary Section S3.2).

Supplementary Table S12 reports the search times, which exclude configuration-evaluation costs. Median search times stay below 0.06 s for D-SDA and uniform sampling and below 1 s for BAM. For RBFOpt, they grow with the budget, from 0.06 s on CSTR5 to 26.3 s on Dynamic at $B = 50$. MISO needs 0.66 to 158 s because it refits its model and scores a large candidate pool at every iteration, whereas NOMAD stays below 2 s. FMQA-SA needs 3.7 to 29.8 s because it refits its factorization machine after every evaluation.

Supplementary Figure S1 shows the evaluation at which each successful run first reaches the target.

4.1.2 NLP subproblems solved

Figure 2 compares the number of NLP subproblems that each run solves with the number that full enumeration of the configuration domain requires. On CSTR5 through CSTR20 and on Dynamic, RIC and LOA solve exactly as many subproblems as full enumeration, and their execution logs show that every subproblem fixes a different configuration (Supplementary Section S11). RIC also enumerates all 325 admissible CSTR25 configurations and solves 455 of the 465 CSTR30 subproblems before the time limit (Supplementary Table S24). The DFO methods solve at most 38 subproblems in any run, and MISO and RBFOpt reach the CSTR30 target in ten and nine of ten runs with medians of 29.5 and 31.5, 6 to 7% of the enumeration count.

Enumeration does not guarantee that a logic-based method reaches the target. RIC and LOA miss the CSTR10 target after solving all 55 of its subproblems, and RIC misses the CSTR25 target after solving all 325. These misses come from local NLP solutions obtained with the GDPopt initialization, which differs from the initialization used to build the lookup tables (Supplementary Section S3.1). GLOA solves two to four subproblems per CSTR model and misses every CSTR target.

4.1.3 Completion time

In wall time, the advantage of fewer subproblem solves depends on how much each solve costs. Configuration evaluations have median costs of 0.4 to 2.5 s in these models (Supplementary Table S19). Full enumeration therefore takes from 10.18 s for Small Batch to 1,719.00 s for CSTR30 (Table 1). Table 2 gives the median completion times of the DFO methods. On Small Batch, the DFO medians of 3.49 to 9.09 s are close to the enumeration time, and on Dynamic at $B = 50$, RBFOpt and MISO need 65.14 and 195.36 s, more than enumerating all 72 configurations. On CSTR30, MISO and RBFOpt reach the target in ten and nine of ten runs with medians of 107.33 and 101.96 s, about one sixteenth of the enumeration time, while NOMAD and D-SDA reach four and three targets with medians of 71.23 and 47.17 s.

Table 2: Process GDP models: target hits and median full-run completion times of the six DFO methods and uniform sampling, with ten runs per method and condition.

Model	N	B	Measure	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
CSTR	5	8	Target hits (of 10)	2	4	8	4	**10**	3	4
			Median completion time (s)	10.67	10.71	10.72	11.33	10.67	13.92	**9.44**
	10	15	Target hits (of 10)	3	3	**6**	2	1	2	1
			Median completion time (s)	**17.61**	21.25	21.79	21.87	18.54	27.07	17.75
	15	22	Target hits (of 10)	2	2	**7**	**7**	1	1	2
			Median completion time (s)	**25.68**	33.92	37.59	39.79	27.67	42.82	27.84
	20	29	Target hits (of 10)	3	2	7	**10**	2	1	2
			Median completion time (s)	**32.31**	38.17	45.03	56.03	39.37	59.10	40.12
	25	36	Target hits (of 10)	3	6	8	**10**	2	1	1
			Median completion time (s)	**45.49**	56.56	62.29	78.56	50.47	74.70	51.17
	30	43	Target hits (of 10)	3	4	9	**10**	2	1	1
			Median completion time (s)	**47.17**	71.23	101.96	107.33	115.41	101.19	65.20
Column	–	13	Target hits (of 10)	5	7	4	**10**	1	4	1
			Median completion time (s)	35.44	31.93	54.60	31.98	27.50	41.82	**21.69**
Small Batch	–	10	Target hits (of 10)	**8**	6	5	7	2	6	4
			Median completion time (s)	4.23	4.97	5.12	5.62	4.23	9.09	**3.49**
Dynamic	–	12	Target hits (of 10)	4	8	**10**	**10**	0	2	0
			Median completion time (s)	12.22	12.36	12.29	11.80	9.95	14.47	**7.80**
		15	Target hits (of 10)	6	**10**	**10**	**10**	3	3	3
			Median completion time (s)	15.32	15.48	16.10	15.44	11.53	18.39	**8.71**
		20	Target hits (of 10)	**10**	**10**	**10**	**10**	3	3	3
			Median completion time (s)	20.02	20.85	20.54	42.42	14.07	25.23	**12.27**
		30	Target hits (of 10)	**10**	**10**	**10**	**10**	4	4	4
			Median completion time (s)	20.02	22.00	34.19	106.24	21.91	36.24	**19.38**
		37	Target hits (of 10)	**10**	**10**	**10**	**10**	7	5	5
			Median completion time (s)	**20.02**	22.00	42.42	111.61	24.99	46.05	24.38
		50	Target hits (of 10)	**10**	**10**	**10**	**10**	**10**	6	6
			Median completion time (s)	**20.02**	21.99	65.14	195.36	33.98	61.31	33.12

Target hits: runs whose best accepted objective is within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1, the lowest accepted objective of the DFO domain (Equation (8)). Completion time: search time plus the measured cost of every evaluated configuration and, for Dynamic, one model preparation (Equation (10)); medians are over all ten runs, including target misses. Table 1 gives the enumeration times, and Supplementary Table S12 the search times alone. N: number of potential reactors of CSTR (–: not applicable); B: evaluation budget; Uniform: uniform sampling without replacement. Bold: most target hits and shortest median completion time in each row, including ties.

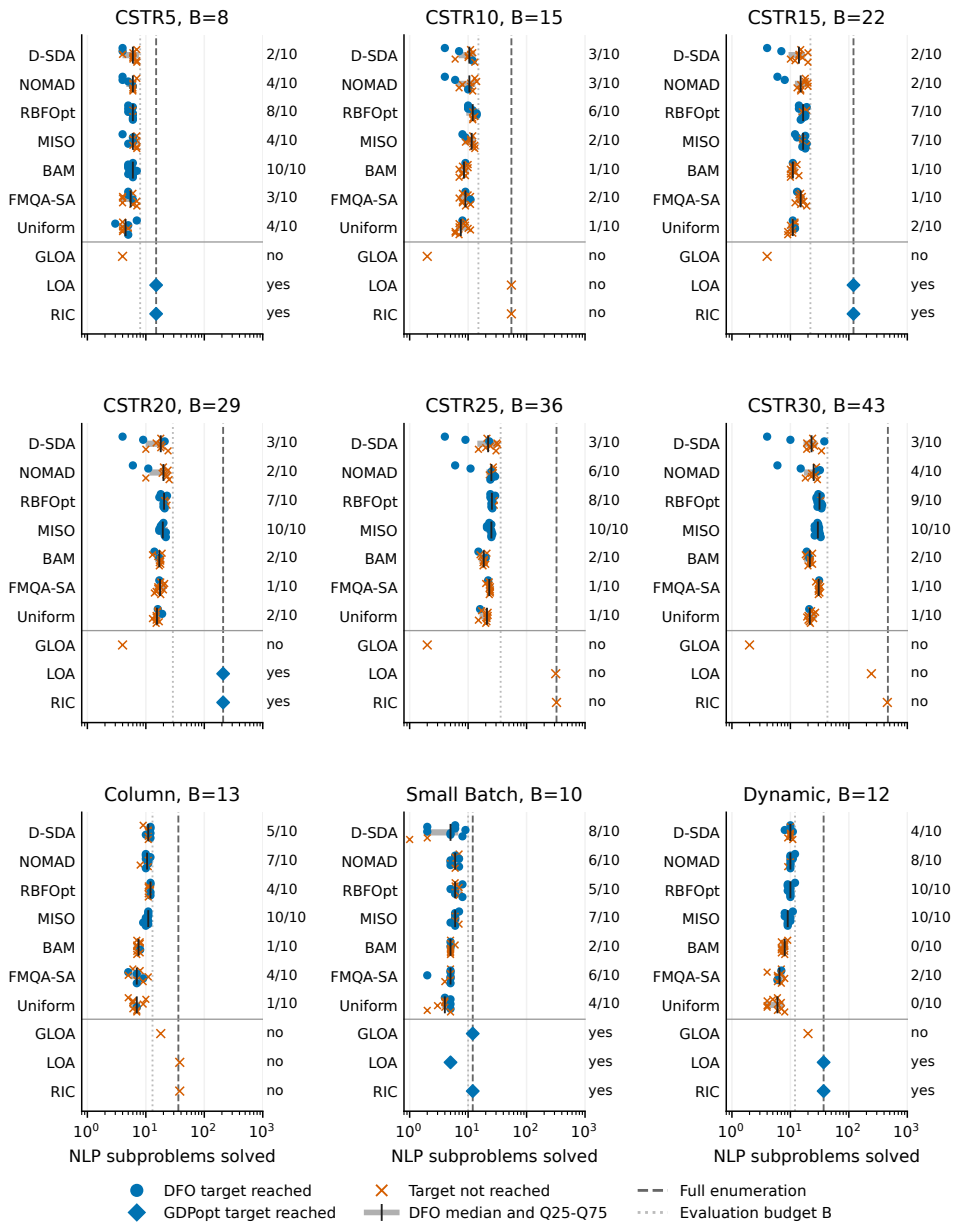


Figure 2: NLP subproblems solved per run on the nine process models, by the six DFO methods and uniform sampling (ten seeds each, at the evaluation budget B of each panel title) and by the single run of each `GDPopt` method: GLOA (global logic-based outer approximation), LOA (logic-based outer approximation), and RIC (relaxation with integer cuts). Dots mark DFO runs that reach the reference target of Equation (8), an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1; diamonds mark `GDPopt` runs whose feasible objective is at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$ (Section 3.7); crosses mark the other runs; gray bars span the 25th to 75th percentiles of the ten DFO seeds, with a tick at the median. The right-hand labels give the target hits out of ten of each DFO method and the outcome, yes or no, of each `GDPopt` run. The dashed line marks the NLP subproblems needed to evaluate every configuration of the DFO domain, and the dotted line the evaluation budget B. Configurations rejected by logical or feasibility-based bound tightening (FBBT) preprocessing are not counted as NLP subproblems. `GDPopt` searches the full `Column` box of 225 configurations, so its count can exceed the enumeration line of the 64-configuration DFO domain. The direct Big-M and Hull solves are not shown because their internal NLP counts are unavailable. Uniform: uniform sampling without replacement.

Search overhead and stopping rules also shape completion times. For `Dynamic` at $B = 50$, search overhead contributes 42.7% of the mean completion time of `RBFOpt`, compared with less than 0.1% for `D-SDA`, and `MISO` spends a median of 158 s of its 195.36 s in search. Across these ten runs, `D-SDA` and `NOMAD` evaluate 167 and 170 configurations because they stop at a local optimum or at the integer mesh resolution, whereas `RBFOpt`, `MISO`, `BAM`, and `FMQA-SA` each evaluate 500. At $B = 30$, `RBFOpt` reaches all ten `Dynamic` targets in fewer evaluations than `D-SDA` (Supplementary Figure S1), but it continues to the budget, and its median completion time is 34.19 s, compared with 20.02 s for `D-SDA`, 22.00 s for `NOMAD`, and 106.24 s for `MISO`. On `Column`, where `MISO` reaches all ten targets, the median completion times are 31.93 s for `NOMAD`, 31.98 s for `MISO`, 35.44 s for `D-SDA`, and 54.60 s for `RBFOpt`.

Figure 3 places the conventional GDP methods on the same time axis, and Supplementary Table S23 lists their objectives; Supplementary Figure S2 shows the other eight conditions. Hull reaches the target on seven of the nine models, RIC, LOA, and Big-M on five each, and GLOA only on `Small Batch`. Big-M and Hull solve `Small Batch` in about 0.5 s, faster than any DFO run, and they are the only conventional methods that reach the `CSTR30` target, both stopping at the 1,800 s time limit. No conventional method reaches the `Column` target.

All six `CSTR` GLOA runs report `optimal` termination, yet none comes within the relative target tolerance of 10^{-5} (Equation (8)); their objectives exceed the references by 2.2% on `CSTR5` and by 32% on `CSTR30`. Their reported lower bounds exceed independently checked feasible objectives. An earlier configuration that used `SCIP` as a global solver for both master problems and subproblems returned the same six objectives within a relative difference of 3×10^{-5}, so these misses do not come from the local NLP solver. Supplementary Section S9 reports these and the other bound inconsistencies, which affect the interpretation of reported optimality but not the target counts.

4.2 Simulation-based lomustine case

Table 3 reports the `Lomustine` results. `RBFOpt` and `MISO` reach the reference configuration in eight and seven of ten runs with ten evaluations and in all ten runs with 20, and their successful runs find it at a median of the fifth to sixth and the sixth to seventh evaluation. `NOMAD` reaches it in six runs at every budget because, when the budget allows more, it stops after a median of 12.5 evaluations. `D-SDA` reaches it once with ten evaluations: its unit L_∞ neighborhood in three dimensions contains up to 26 configurations, so ten evaluations do not complete the first neighborhood search. With the separable neighborhood, which has at most six neighbors, `D-SDA` reaches the reference in two, five, and five runs at the three budgets (Supplementary Section S14). `BAM` misses every target with ten evaluations, and all ten of its runs end at the same PFR–PFR configuration, 11% above the reference. Of the 96 misses over all methods and budgets, 41 end at this PFR–PFR configuration and 27 at the CSTR–batch configuration, 0.8% above the reference.

In this case, an evaluation costs minutes of simulation, so the completion time is governed by the number of evaluations. Median search times stay below 13 s, whereas the median completion times with ten evaluations range from 1.0 to 1.5 h, compared with 5.3 h for evaluating all 36 configurations. Search adds less than 0.7 s per evaluation in every run, so the methods differ in completion time through the evaluations they choose; `MISO`, for example, completes its ten-evaluation runs in a median of 1.13 h, about as fast as any method.

Because the order of the categorical alternatives is a convention, we repeated the 210 runs with five other orders, including the reversed order and an order by degree of back-mixing (Supplementary Section S5). The order changes the results of several methods. With 20 evaluations, `BAM` reaches the reference in 0 to 10 of the ten runs depending on the order, `D-SDA` in 3 to 8, and `MISO` in 6 to 10, whereas `RBFOpt` reaches it in 9 or 10 runs in every order. When the objective is the capital cost of each flowsheet at its optimized operation, the all-batch flowsheet is the reference and `BAM` reaches it in every run.

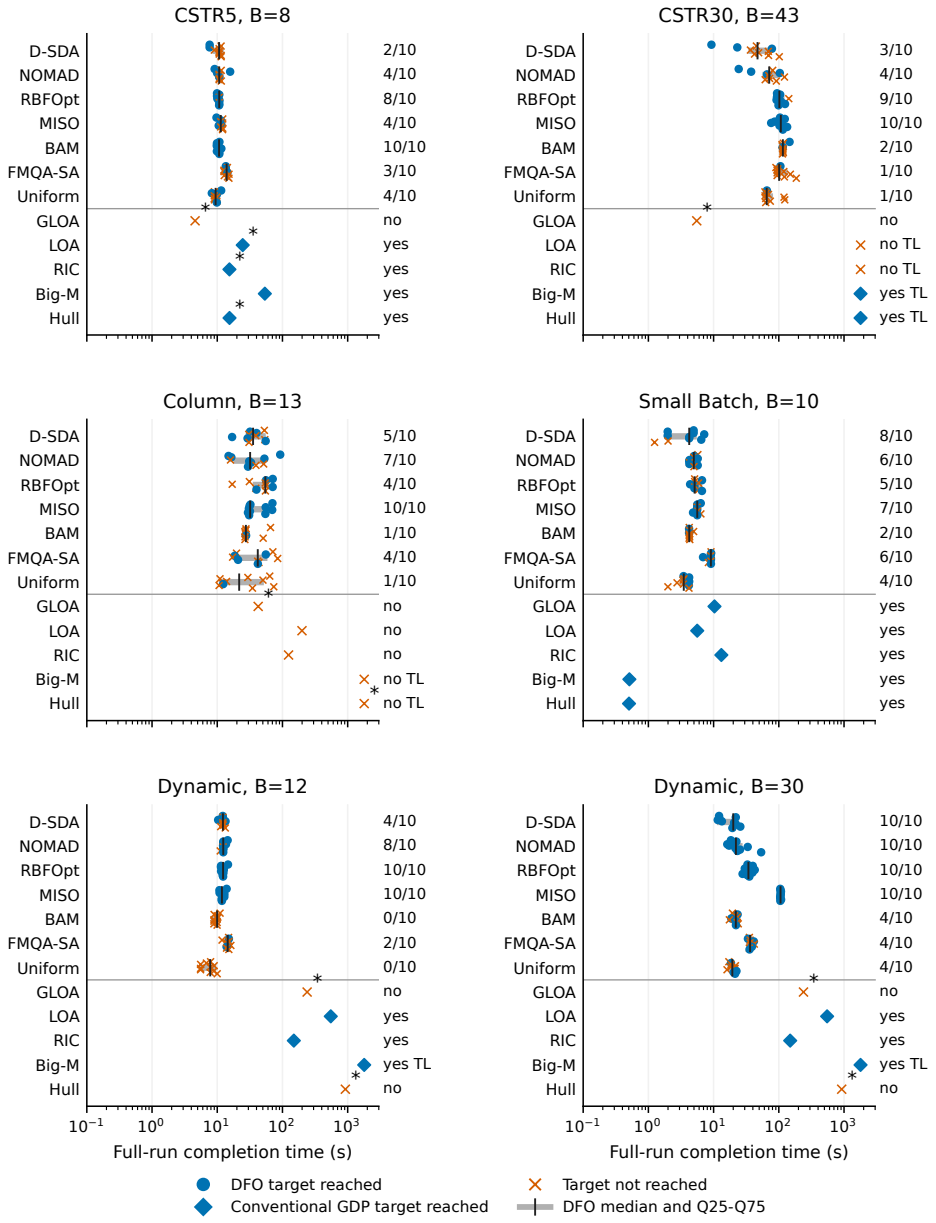


Figure 3: Full-run completion times on six process conditions, by the six DFO methods and uniform sampling (ten seeds each, at the evaluation budget B of each panel title) and by the single run of each conventional GDP method: GLOA, LOA, and RIC of `GDPopt` (defined in Figure 2) and the Big-M and Hull reformulations solved by `SCIP`. DFO times add the search time to the measured cost of each evaluated configuration (Equation (10)); conventional GDP times are measured and include model preparation. Dots mark DFO runs that reach the reference target of Equation (8), an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref} of Table 1; diamonds mark conventional GDP runs whose feasible objective is at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$ (Section 3.7); crosses mark the other runs; gray bars span the 25th to 75th percentiles of the ten DFO seeds, with a tick at the median. The right-hand labels give the target hits out of ten of each DFO method and the outcome, yes or no, of each conventional run, with TL for runs that report termination at the 1,800 s time limit. Asterisks mark conventional runs whose reported lower bound exceeds a checked feasible objective (Supplementary Section S9). The conventional GDP methods have no evaluation budget, so both `Dynamic` panels show the same conventional runs. Supplementary Figure S2 shows the other eight conditions. Uniform: uniform sampling without replacement.

Table 3: `Lomustine`: target hits and median full-run completion times of the six DFO methods and uniform sampling, with ten runs per method and budget.

B	Measure	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
10	Target hits (of 10)	1	6	**8**	7	0	2	2
	Median completion time (h)	1.15	**1.05**	1.24	1.13	1.36	1.47	1.48
15	Target hits (of 10)	4	6	9	**10**	4	3	4
	Median completion time (h)	1.68	**1.22**	1.70	1.82	1.90	2.39	2.11
20	Target hits (of 10)	4	6	**10**	**10**	6	7	5
	Median completion time (h)	1.97	**1.22**	2.44	2.66	2.48	3.09	2.91

Objective: $-(\dot{m}/10^{-4} + 2d_{50})$, with the production rate $\dot{m}$ in kg/s and the median crystal size d_{50} in μm (Section 3.2). Target hits: runs whose best objective is within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of $f_{\text{ref}} = -487.842$, the objective of the reference configuration, a plug-flow reactor (PFR) followed by a batch reactor and the batch crystallizer, which is the best of the 36 simulated configurations (Equation (8)). Completion time: search time plus the recorded wall time of the `PharmaPy` simulation-based optimization of every evaluated configuration; medians are over all ten runs, including target misses. Evaluating all 36 configurations takes 5.3 h. B: evaluation budget; Uniform: uniform sampling without replacement. Bold: most target hits and shortest median completion time for each budget, including ties.

4.3 Catalytic distillation column

Table 4 reports the `Catalytic Column` results. No method reaches the reference with 25 evaluations; with 100 evaluations, `D-SDA` and `MISO` reach it in three of ten runs and `RBFOpt` in one. Targets are rare because the 54,159 admissible configurations of the search box are more than 500 times the largest budget and the seven initial configurations are spread over the whole space. The eight best designs accepted in the campaign share the ethanol feed and the first catalytic stage at stage 8, the butenes feed at the last catalytic stage, and the boil-up return at stage 20 or 21, so a run must first reach this region and then refine three stage positions within it.

Most runs that miss the reference with 100 evaluations end close to it. Nine of them, three of `D-SDA`, five of `NOMAD`, and one of `RBFOpt`, end at $(8, 8, 13, 17, 17, 21)$, whose objective is 1.29 \$/yr (0.005%) above the reference. This design differs from the reference only in the second catalytic stage, but moving that stage from 13 to 12 changes two gaps, (g_3, g_4) from $(4, 3)$ to $(3, 4)$. The separable neighborhood of `D-SDA` therefore does not contain the reference, and the descent stops one stage away from it; the unit L_∞ neighborhood contains this diagonal move, but with 728 neighbors it does not fit the budgets. The choice of external variables and neighborhood thus decides which designs are neighbors, as the order of the alternatives did for `Lomustine`: stage positions as coordinates would make this move a single step, but most of their box violates the order of the stages. `MISO` misses by 9.37 to 653 \$/yr, and its median gap of 34.5 \$/yr is close to that of `NOMAD`. `BAM` evaluates, after the shared initial configurations, an initial design of its own that is the same in every seed, and its last point, $(6, 7, 10, 13, 14, 18)$, lies 4.7% above the reference; in six of the ten runs with 100 evaluations, `BAM` finds nothing better.

An evaluation solves an NLP in a median of 4.8 s, whereas the median search overhead per evaluation ranges from 2 ms for uniform sampling to 0.31 s for `MISO` and 0.65 s for `FMQA-SA`, so the completion times are governed by the number of evaluations. With 100 evaluations, the median completion times range from 8.25 min for uniform sampling to 11.4 min for `NOMAD`, and `D-SDA` and `MISO` need 8.71 and 10.5 min. Solving every admissible configuration of the box would take about 119 h, so these runs evaluate 0.2% of them, and the median runs of `D-SDA`, `NOMAD`, and `MISO` end within 0.14% of the reference.

No MINLP solver reaches the reference or comes within 15% of it (Supplementary Table S10). On the published MINLP form, `SBB` stops after 10 s at $(4, 4, 5, 6, 6, 9)$, next to the initial configuration and 15.4% above the reference. `DICOPT` ends without a feasible design on all three forms after 211 to 902 s, and `SBB` finds none on the Big-M and Hull reformulations. `BARON` and `SCIP` declare the published form infeasible after 255 and 9 s, and `BARON` declares the Big-M and Hull reformulations infeasible after 214 and 362 s. `SCIP` finds no feasible design on the Big-M and Hull reformulations and stops at the 3,600 s time limit with a lower bound of -5.7×10^7 \$/yr. These infeasibility claims are wrong: with the binary variables fixed at

the reference design, BARON still declares both reformulations infeasible, although a CONOPT solution satisfies every constraint and bound of the Big-M reformulation within 10^{-6}. When started from a feasible design, both solvers instead certify that design as the global optimum of the published form, and they do so both for the design of the pilot, at 25,271.33 \$/yr, and for the reference, at 24,895.55 \$/yr, so the first certificate is contradicted by the second design. In every MINLP form, the equations of stages that do not exist and of stages without catalyst stay in the model, multiplied by zero or relaxed, so the solvers must bound terms such as Arrhenius rates over the whole temperature range and inverse powers of vanishing liquid flows; in the Big-M reformulation, the constraint coefficients span 35 orders of magnitude. The GDP evaluation used by the search methods solves only the equations of the selected configuration, and every one of the 210 runs returned accepted designs.

Table 4: Catalytic Column: target hits, gaps to the reference, and median full-run completion times of the six DFO methods and uniform sampling, with ten runs per method and budget.

B	Measure	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
25	Target hits (of 10)	0	0	0	0	0	0	0
	Median gap (\$/yr)	1,720	**843**	1,970	1,140	1,170	1,900	1,400
	Median completion time (min)	2.82	2.76	2.38	2.51	2.52	2.94	**2.32**
50	Target hits (of 10)	0	0	**1**	**1**	0	0	0
	Median gap (\$/yr)	1,020	**351**	1,110	946	1,170	1,250	1,190
	Median completion time (min)	5.75	5.35	4.43	5.05	5.13	5.04	**4.00**
100	Target hits (of 10)	**3**	0	1	**3**	0	0	0
	Median gap (\$/yr)	**1.29**	30.4	417	34.5	1,170	1,020	1,120
	Median completion time (min)	8.71	11.4	9.07	10.5	9.52	10.1	**8.25**

Target hits: runs whose best objective, the negative annual profit, is within $10^{-5}\max\{1, |f_{\mathrm{ref}}|\}$ of $f_{\mathrm{ref}} = 24{,}895.55$ \$/yr, the objective of the reference design, which places the ethanol feed and the first catalytic stage at stage 8, the second and third catalytic stages at stages 12 and 17, the butenes feed at stage 17, and the boil-up return at stage 21 (Equation (8)). Gap: best objective of a run minus f_{ref}, zero for a run that reaches the reference. Completion time: search time plus the recorded CONOPT solve time of every evaluated configuration. Gaps and completion times are medians over all ten runs, including target misses. Solving all 54,159 admissible configurations of the search box would take about 119 h. B: evaluation budget; Uniform: uniform sampling without replacement. Bold: most target hits when any run reaches the reference, smallest median gap, and shortest median completion time for each budget, including ties.

4.4 Controlled search behavior on Grid-GDP

Grid-GDP isolates search behavior on explicit two-dimensional functions whose configurations need no NLP solve. Table 5 reports the separate experiments at the five budgets. MISO reaches the largest target fraction at $B = 30$, 70, and 100, while RBFOpt leads at $B = 50$ and BAM at $B = 10$. At $B = 100$, MISO and RBFOpt reach 86.2% and 81.4% of the targets, NOMAD 60.0%, and D-SDA 48.1%, with BAM and FMQA-SA in between and uniform sampling at 21.4%. The NOMAD fraction at $B = 100$ includes 16 targets reached before a reproducible solver crash; counting those 21 crashed runs as misses would give 52.4%. The paired bootstrap intervals in Supplementary Section S10 exclude zero for RBFOpt against D-SDA at both $B = 10$ and $B = 100$.

At $B = 10$, BAM and its prefix control each reach 52.9% because nine of the 21 functions have their target at the domain center, which BAM evaluates as the fifth point of its Sobol design. By $B = 100$, the model-based search of BAM adds 23 hits over the prefix control (Figure 4). D-SDA levels off at 48.1% from $B = 70$ onward, once the descents from its three initial configurations end, and its median number of evaluations stays at 37 for every budget from $B = 50$ onward. Excluding Xin-She Yang 3, whose target set covers 870 of the 961 lattice points, and Bohachevsky 2, which nearly duplicates Sum of Squares on the lattice, preserves this ordering.

Because the explicit expression is available, the direct Big-M and Hull reformulations solved with SCIP reach all 21 Grid targets in a median of 0.66 and 0.53 s per function, less than the median completion time of every DFO method (Supplementary Table S13). The GDPopt methods reach 14 or 15 of the 21 targets, and in diagnostic runs their preprocessing rejected the feasible reference configurations of the three

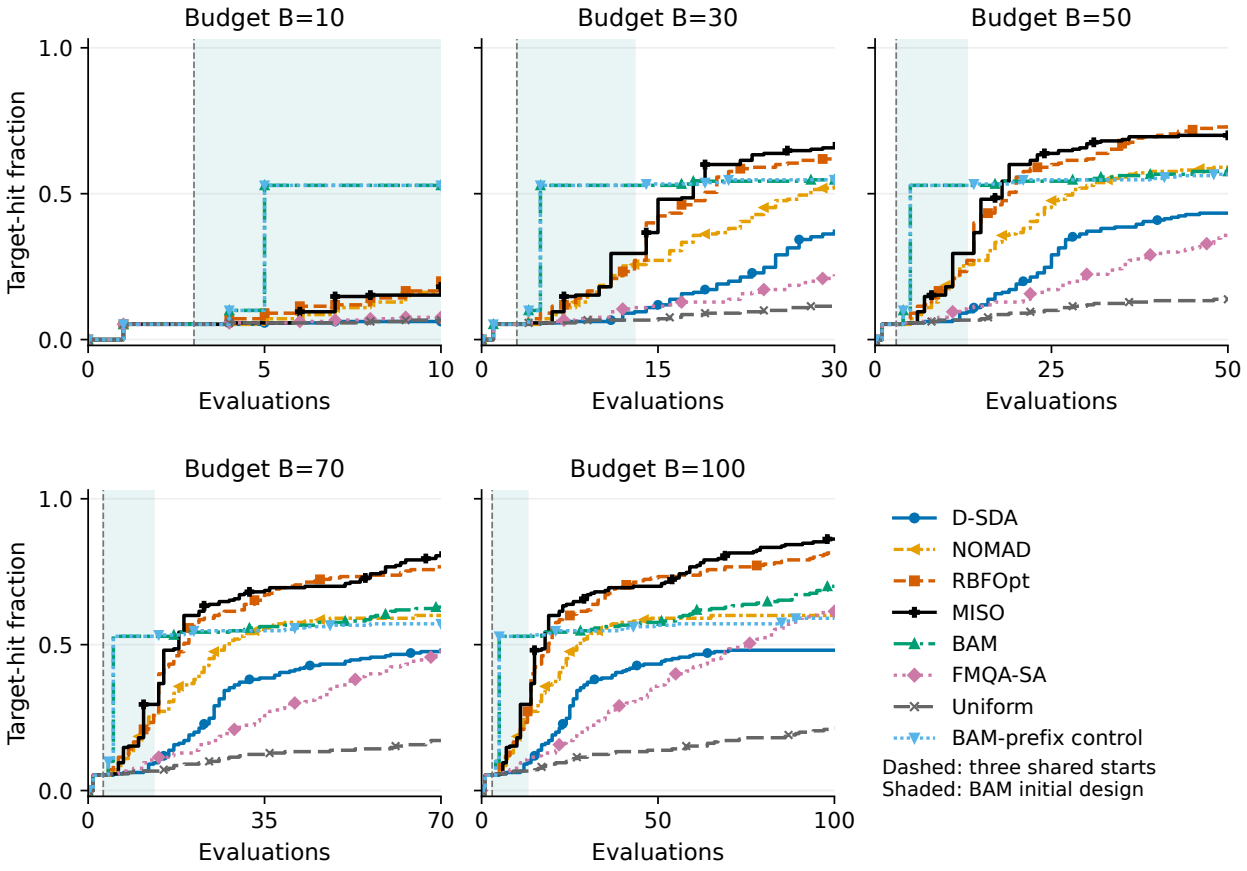


Figure 4: Grid-GDP: fraction of the 210 runs (21 test functions and ten seeds) that have reached the reference target of Equation (8), a configuration whose normalized objective is within 10^{-5} of zero, within a given number of evaluations (Equation (9)), at the five evaluation budgets B. Each budget is a separate experiment; initial and repeated evaluations count toward the budget, and runs that miss the target remain in the denominator. The dashed line marks the end of the three shared initial evaluations, and the shaded band the ten predetermined points of `BAM`, which the `BAM`-prefix control repeats before it samples uniformly (Section 3.5.8). Uniform: uniform sampling without replacement. At $B = 100$, `MISO` reaches the largest fraction, followed by `RBFOpt`.

Table 5: Grid-GDP: target hits of the six DFO methods, uniform sampling, and the BAM-prefix control at five evaluation budgets, as percentages of 210 runs.

	Target hits (% of 210 runs)							
B	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform	BAM-prefix control
10	6.2	18.6	20.0	18.1	**52.9**	7.6	6.7	**52.9**
30	37.1	52.9	61.9	**67.1**	54.8	22.4	11.9	54.8
50	43.3	59.0	**73.3**	70.0	57.6	36.2	13.8	56.7
70	48.1	60.0	76.7	**81.4**	62.9	48.6	17.1	57.1
100	48.1	60.0	81.4	**86.2**	70.0	61.4	21.4	59.0

Target hits: runs that evaluate a configuration whose normalized objective is within 10^{-5} of zero, the normalized minimum over the lattice (Equation (8)); the 210 runs are 21 test functions with ten seeds each. Each budget is a separate experiment, and the budget B counts the three shared initial evaluations and repeated configurations. BAM-prefix control: the ten predetermined points of BAM after the three shared initial configurations, then uniform sampling (Section 3.5.8). Uniform: uniform sampling without replacement. At $B = 10$, all 111 BAM hits precede its surrogate-guided selection: 11 occur at the shared initial configurations and 100 at two of its predetermined points. The NOMAD percentage at $B = 100$ includes 16 targets reached before a solver crash that recurred on every attempt. Bold: largest percentage in each row, including ties.

functions that all of them miss. Grid-GDP therefore tests search behavior; for these explicit functions, a direct reformulation is the method of choice.

4.5 Discussion

No single method reaches the most targets in every case. The ranking depends on the size of the configuration space, on where the target lies relative to the initial configurations and to the configurations without accepted objectives, and on how each method uses its evaluation budget.

On `CSTR`, the recycle/feed ordering makes roughly half of each configuration box logically inadmissible, and the target lies at the corner (N, N), on the edge of the admissible region. On `CSTR10` through `CSTR30`, 31 of the 36 `D-SDA` misses end at the configuration $(5, 1)$, whose volume of 3.13020 is the same for every N. Because the reference volume decreases with N, this local minimum lies 8.3% above the reference for `CSTR10` and 13.0% above it for `CSTR30`, which explains why the `D-SDA` gaps grow with size. Restarted descents mostly end at $(5, 1)$ or at the target, so the misses reflect the large basin of this local minimum, and additional budget does not remove them. `NOMAD`, which polls configurations around its incumbent, ends 32 of its 33 misses on these sizes at $(5, 1)$ as well. `RBFOpt` and `MISO` can propose configurations far from their incumbents and reach the corner in nine and ten of the ten `CSTR30` runs. A global surrogate does not guarantee success on small budgets, however: `MISO` reaches 13 of the 30 targets on `CSTR5` through `CSTR15`, with 8 to 22 evaluations, compared with 21 for `RBFOpt`. On `Column`, four of the five successful `D-SDA` runs reach the target in one or two improving moves, and the fifth starts at it. With only 29 and 12 accepted configurations and budgets of 13 and 10 evaluations, `Column` and `Small Batch` give a surrogate model few observations to learn from, and local descent remains competitive with `RBFOpt`. `MISO` nevertheless reaches all ten `Column` targets; in the nine runs whose initial design does not contain the target, it finds it between the 7th and 11th of its 13 evaluations. On Grid-GDP, the early lead of `BAM` reflects the symmetry of the test functions with respect to its initial design, and `FMQA-SA` improves on uniform sampling there, but in the small process domains, where more than half of its proposals come from its uniform fallback, it reaches only five more targets than uniform sampling (42 against 37 of 140). From $B = 30$ onward, the two radial-basis-function methods, `MISO` and `RBFOpt`, reach the most Grid targets.

The methods also use additional evaluations differently. `D-SDA` terminates when its descents reach configurations without an improving neighbor, and `NOMAD` when its mesh reaches the integer resolution, so additional budget helps them only until then. On Grid, `NOMAD` stops after a median of 31.5 evaluations from $B = 50$ onward, and its target fraction stays at 60.0% from $B = 70$ to $B = 100$. The surrogate-based methods and the sampling controls have no comparable termination test and use the full budget in every Grid run, so their Grid target-hit rates keep increasing between $B = 70$ and $B = 100$, by 4.8 percentage points for `RBFOpt` and `MISO`, 7.1 for `BAM`, and 12.9 for `FMQA-SA`, whereas the `D-SDA` rate stays at 48.1%. Once a method has found the target, however, the remaining evaluations add cost without improving the result: `RBFOpt` and `MISO` reach all ten `Dynamic` targets at $B = 12$, yet their median completion times grow from 12.29 and 11.80 s at $B = 12$ to 65.14 and 195.36 s at $B = 50$, whereas `NOMAD`, which stops early, needs 21.99 s at $B = 50$. Additional search time also does not guarantee more targets: on Grid at $B = 100$, `FMQA-SA` spends a median of 62.04 s on search and reaches fewer targets than `BAM`, which spends 0.43 s. For `MISO`, it does: with a median of 32.70 s, about four times the search time of `RBFOpt`, it reaches 181 targets against 171 (Supplementary Table S13). How much search time is worthwhile depends on the cost of an evaluation. `FMQA-SA` spends about 0.5 s on search per evaluation, comparable to the evaluations in these models, while `RBFOpt` spends 17 to 98 ms in median but more as observations accumulate. `MISO` spends 52 ms to 1.86 s per evaluation, and on `Dynamic` its search per evaluation exceeds the median evaluation cost of 0.84 s; `NOMAD` spends 3 to 97 ms (Supplementary Table S19). When each evaluation requires a longer optimization or a process simulation, this overhead becomes small, and the number of subproblem solves becomes the main cost.

Because the initial configurations change with the seed and the surrogate-based methods are randomized, repeated runs can return different solutions, and the spread of these solutions differs between methods. `RBFOpt` returns at most three distinct final objectives across the ten seeds of each `CSTR`, `Small Batch`, and `Dynamic` condition; on `CSTR10` through `CSTR30`, its runs either reach the target or stop 8.1 to 12.5% above it. `MISO` and `NOMAD` are similarly consistent, with at most five and three distinct final objectives per condition, and the `MISO` misses on `CSTR10` and `CSTR15` lie 6.0 to 10.6% above the reference. `BAM` reaches the exact target in only 18 of the 60 `CSTR` runs, but all of its final objectives lie within 0.7% of the reference. `FMQA-SA` and uniform sampling spread more widely, with up to ten distinct final objectives in one condition and gaps of up to 17%. With a relative tolerance of 1%, `BAM` would succeed in all 60 `CSTR` runs, compared with 45

for RBFOpt, 43 for MISO, 21 for NOMAD, and 19 for D-SDA (Supplementary Table S18), so the strict target separates exact attainment from near-optimal solutions that may suffice in a design study.

The comparison with enumeration and conventional GDP methods indicates when each approach is preferable. For small models such as Small Batch and the Grid functions, enumeration or a direct reformulation finds the optimum in seconds or less. As the configuration space grows, logic-based methods that solve one subproblem per admissible configuration and direct reformulations that reach the time limit become expensive, while surrogate search keeps its number of subproblem solves near the budget, as on CSTR30. Dynamic lies between these cases: RIC reaches its target in 149.39 s after solving all 37 admissible subproblems, and RBFOpt and MISO reach it in every run at $B = 12$ in medians of 12.29 and 11.80 s.

The Lomustine case shows that the configuration search carries over when a simulator defines each subproblem and conventional GDP methods do not apply. Its ranking agrees with the CSTR and Grid results: the two radial-basis-function methods reach the reference most often, NOMAD stops early, and the size of the D-SDA neighborhood limits local descent on small budgets. Because each evaluation costs minutes, the search overhead that slows MISO on Dynamic is negligible here. Unordered alternatives can still be encoded as external variables, but their order is then part of the problem the search sees: with 20 evaluations, only RBFOpt reached the reference in at least nine of ten runs in every order.

The neighborhood of D-SDA trades the number of evaluations per step against the moves it can make. Repeating every stored-response D-SDA run with the separable neighborhood, which changes one external variable at a time, shows no overall winner (Supplementary Section S14). The separable neighborhood reaches more targets when the budget barely covers the L_∞ neighborhood, as on Grid at $B = 10$ (22 against 13 of 210 runs) and on Lomustine (99 against 90 of 180 runs over the six orders). It reaches fewer when a descent must move diagonally: on CSTR, 7 against 16 of 60 runs, because 18 of its misses stop on the edge of the admissible region that leads to the target. The L_∞ neighborhood therefore suits two external variables with budgets that cover its eight neighbors, and the separable neighborhood suits more external variables or smaller budgets, which is why the Catalytic Column uses it.

The Catalytic Column tests the search where enumeration is impractical and every evaluation is a live NLP solve. With 100 evaluations, 0.2% of the configurations in the search box, the median runs of D-SDA, MISO, and NOMAD end within 0.14% of the best known design, and local descent is competitive with the surrogate methods here because the good designs form one region: six of the ten D-SDA descents end at the reference or one stage from it. Within that region, the external variables and the neighborhood decide which designs are neighbors: in the gap coordinates, the runner-up and the reference are two coordinates apart although they differ in one stage, so the separable neighborhood stops local descent 1.29 \$/yr short of the reference, as the order of categorical alternatives changed the results of Lomustine. The MINLP forms of the same model keep the equations of every possible stage and catalyst position, and the solvers fail in different ways: SBB stops at its limit on function evaluations outside their domains, DICOPT when its NLP subproblems fail or are all infeasible, SCIP at the time limit on the Big-M and Hull forms, and BARON and SCIP with infeasibility claims or certificates that the feasible designs contradict. A plausible common cause is that terms harmless in a solved configuration, such as reaction rates on stages without catalyst and hydraulics with vanishing flows, make the relaxations poorly scaled, with constraint coefficients that span 35 orders of magnitude in the Big-M form; the size of the full MINLP, which contains all 100,947 configurations, may contribute as well. Reported bounds were also unreliable in the process models, where GLOA terminated as optimal on every CSTR network while missing the reference by 2.2 to 32%. The GDP model avoids these terms because each evaluation contains only the equations of the selected configuration, so the external-variable representation lets a black-box method optimize a model that its direct MINLP reformulations cannot solve reliably.

Except for the Catalytic Column, these conclusions rest on configuration spaces of 25 to 961 configurations, on NLP responses obtained with one initialization per model, and on completion times that combine recorded search paths with one cost measurement per configuration. The Catalytic Column reference is the best design found and has no reliable optimality certificate, because the global solvers that certify it also certify a worse design, and its responses also come from one initialization. Because the CSTR budgets grow with N, the CSTR series does not isolate the effect of size. With ten seeds per condition, the per-condition comparisons that involve NOMAD or MISO are not significant after a Holm correction over the 70 exploratory tests, and the per-condition tests of RBFOpt against D-SDA are not adjusted for multiple comparisons, so rankings within a single condition are descriptive. The NOMAD and MISO search times were measured in a

later campaign on a shared server, so small timing differences between them and the other methods should not be overinterpreted.

5 Conclusions

This study used GDP to model process design alternatives and derivative-free methods to search their configurations, connected by the external-variable reformulation, and compared this pairing with local descent, logic-based and MINLP solvers, and full enumeration. The experiments lead to five conclusions.

First, the pairing solves GDP models to which conventional methods do not apply or on which they failed in these experiments. On the `Lomustine` flowsheet, whose unit models are simulations, `RBFOpt` and `MISO` found the best of 36 configurations in every run with 20 evaluations. On the `Catalytic Column`, `D-SDA` and `MISO` reached the best known design in three of ten runs with 100 of the 54,159 configurations in its search box, whereas no MINLP solver came within 15% of it on the published, Big-M, or Hull formulations. These formulations keep the equations of absent stages and catalyst, while each GDP evaluation contains only the equations of the selected configuration.

Second, surrogate search reaches near-optimal configurations with a small fraction of the NLP subproblems that logic-based methods solve. On `CSTR5` through `CSTR20` and on `Dynamic`, RIC and LOA solved one subproblem for each admissible configuration, and RIC solved 455 of the 465 `CSTR30` subproblems before its time limit, whereas `MISO` and `RBFOpt` reached the `CSTR30` target in ten and nine of ten runs with medians of 29.5 and 31.5 subproblems, 6 to 7% of full enumeration.

Third, the benefit over local descent depends on the landscape. On `CSTR`, most `D-SDA` descents end at one local minimum whose gap to the reference grows with the network size, and restarts add a single target; the direct search of `NOMAD` ends at the same configuration in 32 of its 33 misses on the larger networks. `RBFOpt` and `MISO`, which interpolate all observations with radial basis functions, reached the target in 45 and 43 of 60 runs. On the two smallest design models, `D-SDA` and `RBFOpt` perform comparably, and `MISO` reached all ten `Column` targets. On the `Catalytic Column`, where the good designs form one region, `D-SDA` and `MISO` reached the best known design equally often, and the choice of external variables and neighborhood, which decides which designs are neighbors, kept other runs 1.29 \$/yr short of it.

Fourth, the stochastic methods differ in how their solutions spread across runs. `RBFOpt` and `MISO` either reached the target or stopped at a local solution 6.0 to 12.5% above it on the larger `CSTR` instances, whereas `BAM` rarely reached the exact target but always ended within 0.7% of it. Reporting only exact target attainment can therefore hide reliable near-optimal performance.

Fifth, more search per evaluation can buy targets, but its value depends on the cost of an evaluation. `MISO` reached the most targets overall, 181 of 210 on Grid at $B = 100$ and 120 of 140 in the process conditions, against 171 and 114 for `RBFOpt`, but on `Dynamic` at $B = 50$ its model refits made it three times slower than `RBFOpt`. `D-SDA` and `NOMAD`, in contrast, stop early, `D-SDA` when no neighbor improves and `NOMAD` when its mesh reaches the integer resolution, and leave budget unused.

For future GDP instances, these results suggest enumerating the configurations or solving a direct Big-M or Hull reformulation when the model is small, as for Grid-GDP and `Small Batch`. When the configuration space is too large to enumerate and contains large basins of local minima, as for the larger `CSTR` networks, surrogate-based configuration search with radial basis functions reaches good solutions with a small fraction of the subproblem solves; `MISO` is preferable when evaluations are expensive relative to its search overhead, and `RBFOpt` when they are cheap. For local descent, the unit L_∞ neighborhood suits two external variables, and the separable neighborhood suits more variables or budgets smaller than the $3^d - 1$ neighbors of the L_∞ neighborhood. When the subproblems are simulations, or when the MINLP reformulations keep equations that are ill-defined for absent units, the GDP model is best solved by configuration search, which needs only the value of each configuration. Local descent then suffices when the good designs form one region, as in the `Catalytic Column`, and surrogate search is the safer choice when the landscape has several basins. The wall-time advantage of searching fewer configurations grows with the cost of each evaluation, so testing these expectations on live simulation-based subproblems, in which every evaluation runs the inner optimization, is the next step.

6 Acknowledgement

The authors acknowledge and express sincere gratitude to the Davidson School of Chemical Engineering at Purdue University for providing a supportive research environment and funding.

Data and code availability

The code and data are available in the Black-Box-for-GDP repository (SECQUOIA, 2026a). Supplementary Section S8 identifies the archived inputs, including the CSTR updates, Column logical-domain runs, conventional GDP comparisons, `D-SDA` reruns, the `Lomustine` runs, and the `Catalytic Column` runs and MINLP comparisons.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used ChatGPT and Codex (OpenAI), including the GPT-6 Astra model, to assist with drafting, organizing, and revising the manuscript and developing analysis and plotting code. The authors also used Claude Code (Anthropic), with the Claude Opus 5.5 model, to review the manuscript, revise the Discussion and Conclusions, and develop analysis, plotting, and experiment code. The resulting text and code were checked and revised by the authors, who remain responsible for the article's content.

References

Abramson, M.A., Audet, C., Chrissis, J.W., Walston, J.G., 2009. Mesh adaptive direct search algorithms for mixed variable optimization. Optim. Lett. 3, 35–47. doi:10.1007/s11590-008-0089-2.

Amestoy, P.R., Duff, I.S., Koster, J., L'Excellent, J.Y., 2001. A fully asynchronous multifrontal solver using distributed dynamic scheduling. SIAM Journal on Matrix Analysis and Applications 23, 15–41. doi:10.1137/S0895479899358194.

Audet, C., Le Digabel, S., Rochon Montplaisir, V., Tribes, C., 2022. Algorithm 1027: NOMAD version 4: Nonlinear optimization with the MADS algorithm. ACM Trans. Math. Softw. 48, 35:1–35:22. doi:10.1145/3544489.

Baptista, R., Poloczek, M., 2018. Bayesian optimization of combinatorial structures, in: Proceedings of the 35th International Conference on Machine Learning, pp. 462–471.

Barhate, Y., Laky, D.J., Casas-Orozco, D., Reklaitis, G.V., Nagy, Z.K., 2025. Hybrid rule-based and optimization-driven framework for the synthesis of end-to-end optimal pharmaceutical processes. AIChE Journal , e18888doi:10.1002/aic.18888.

Bergamini, M.L., Aguirre, P., Grossmann, I., 2005. Logic-based outer approximation for globally optimal synthesis of process networks. Comput. Chem. Eng. 29, 1914–1933. doi:10.1016/j.compchemeng.2005.04.003.

Bolusani, S., Besançon, M., Bestuzheva, K., Chmiela, A., Dionísio, J., Donkiewicz, T., van Doornmalen, J., Eifler, L., Ghannam, M., Gleixner, A., Graczyk, C., Halbig, K., Hedtke, I., Hoen, A., Hojny, C., van der Hulst, R., Kamp, D., Koch, T., Kofler, K., Lentz, J., Manns, J., Mexi, G., Mühmer, E., Pfetsch, M.E., Schlösser, F., Serrano, F., Shinano, Y., Turner, M., Vigerske, S., Weninger, D., Xu, L., 2024. The SCIP Optimization Suite 9.0. ZIB Report 24-02-29. Zuse Institute Berlin. URL: https://nbn-resolving.org/urn:nbn:de:0297-zib-95528.

Bonami, P., Biegler, L.T., Conn, A.R., Cornuéjols, G., Grossmann, I.E., Laird, C.D., Lee, J., Lodi, A., Margot, F., Sawaya, N., Wächter, A., 2008. An algorithmic framework for convex mixed integer nonlinear programs. Discrete Optimization 5, 186–204. doi:10.1016/j.disopt.2006.10.011.

Bynum, M.L., Hackebeil, G.A., Hart, W.E., Laird, C.D., Nicholson, B.L., Siirola, J.D., Watson, J.P., Woodruff, D.L., 2021. Pyomo: Optimization Modeling in Python. volume 67 of *Springer Optimization and Its Applications*. 3 ed., Springer. doi:10.1007/978-3-030-68928-5.

Casas-Orozco, D., Laky, D., Wang, V., Abdi, M., Feng, X., Wood, E., Laird, C., Reklaitis, G.V., Nagy, Z.K., 2021. PharmaPy: An object-oriented tool for the development of hybrid pharmaceutical flowsheets. Computers & Chemical Engineering 153, 107408. doi:10.1016/j.compchemeng.2021.107408.

Casas-Orozco, D., Laky, D.J., Wang, V., Abdi, M., Feng, X., Wood, E., Reklaitis, G.V., Nagy, Z.K., 2023. Techno-economic analysis of dynamic, end-to-end optimal pharmaceutical campaign manufacturing using PharmaPy. AIChE Journal 69, e18142. doi:10.1002/aic.18142.

Chen, Q., Johnson, E.S., Bernal, D.E., Valentin, R., Kale, S., Bates, J., Siirola, J.D., Grossmann, I.E., 2022. Pyomo.GDP: an ecosystem for logic based modeling and optimization development. Optim. Eng. 23, 607–642. doi:10.1007/s11081-021-09601-7.

Costa, A., Nannicini, G., 2018. RBFOpt: an open-source library for black-box optimization with costly function evaluations. Math. Program. Comput. 10, 597–629. doi:10.1007/s12532-018-0144-7.

Cozad, A., Sahinidis, N.V., Miller, D.C., 2014. Learning surrogate models for simulation-based optimization. AIChE J. 60, 2211–2227. doi:10.1002/aic.14418.

D-Wave Systems, n.d. dwave-samplers: Classical algorithms for solving binary quadratic models. URL: https://github.com/dwavesystems/dwave-samplers/releases/tag/1.6.0. Version 1.6.0; classical CPU simulated annealing; accessed October 15, 2026.

Dolan, E.D., Moré, J.J., 2002. Benchmarking optimization software with performance profiles. Math. Program. 91, 201–213. doi:10.1007/s101070100263.

Drud, A.S., 1994. CONOPT—a large-scale GRG code. ORSA Journal on Computing 6, 207–216. doi:10.1287/ijoc.6.2.207.

Ferris, M.C., Dirkse, S.P., Jagla, J.H., Meeraus, A., 2009. An extended mathematical programming framework. Computers & Chemical Engineering 33, 1973–1982. doi:10.1016/j.compchemeng.2009.06.013.

GAMS Development Corporation, 2025. General Algebraic Modeling System (GAMS). URL: https://www.gams.com/. release 52.2.0, Fairfax, VA, USA.

GAMS Development Corporation, 2026. SBB solver manual. URL: https://www.gams.com/latest/docs/S_SBB.html. gAMS release 54.

Ghouse, J.H., Chen, Q., Zamarripa, M.A., Lee, A., Burgard, A.P., Grossmann, I.E., Miller, D.C., 2018. A comparative study between gdp and nlp formulations for conceptual design of distillation columns, in: Computer Aided Chemical Engineering, pp. 865–870. doi:10.1016/B978-0-444-64241-7.50139-7.

Grossmann, I.E., Lee, S., 2003. Generalized convex disjunctive programming: Nonlinear convex hull relaxation. Comput. Optim. Appl. 26, 83–100. doi:10.1023/A:1025154322278.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization. International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/1412.6980.

Kirkpatrick, S., Gelatt, Jr., C.D., Vecchi, M.P., 1983. Optimization by simulated annealing. Science 220, 671–680. doi:10.1126/science.220.4598.671.

Kitai, K., Guo, J., Ju, S., Tanaka, S., Tsuda, K., Shiomi, J., Tamura, R., 2020. Designing metamaterials with quantum annealing and factorization machines. Phys. Rev. Res. 2, 013319. doi:10.1103/PhysRevResearch.2.013319.

Kocis, G.R., Grossmann, I.E., 1988. Global optimization of nonconvex mixed-integer nonlinear programming (minlp) problems in process synthesis. Industrial & Engineering Chemistry Research 27, 1407–1421. doi:10.1021/ie00080a013.

Laky, D., Casas-Orozco, D., 2022. Superstructure synthesis and simulation-optimization of pharmaceutical manufacturing processes using PharmaPy. AIChE Annual Meeting, Phoenix, AZ, paper 183b.

Larson, J., Menickelly, M., Wild, S.M., 2019. Derivative-free optimization methods. Acta Numerica 28, 287–404. doi:10.1017/S0962492919000060.

Liñán, D.A., Bernal, D.E., Ricardez-Sandoval, L.A., Gómez, J.M., 2020a. Optimal design of superstructures for placing units and streams with multiple and ordered available locations. part i: A new mathematical framework. Computers & Chemical Engineering 137, 106794. doi:10.1016/j.compchemeng.2020.106794.

Liñán, D.A., Bernal, D.E., Ricardez-Sandoval, L.A., Gómez, J.M., 2020b. Optimal design of superstructures for placing units and streams with multiple and ordered available locations. part ii: Rigorous design of catalytic distillation columns. Computers & Chemical Engineering 139, 106845. doi:10.1016/j.compchemeng.2020.106845.

Ma, K., Rios, L.M., Bhosekar, A., Sahinidis, N.V., Rajagopalan, S., 2023. Branch-and-Model: a derivative-free global optimization algorithm. Comput. Optim. Appl. 85, 337–367. doi:10.1007/s10589-023-00466-3.

Moré, J.J., Wild, S.M., 2009. Benchmarking derivative-free optimization algorithms. SIAM J. Optim. 20, 172–191. doi:10.1137/080724083.

Müller, J., 2016. MISO: mixed-integer surrogate optimization framework. Optim. Eng. 17, 177–203. doi:10.1007/s11081-015-9281-2.

Nannicini, G., 2021. On the implementation of a global optimization method for mixed-variable problems. Open J. Math. Optim. 2, 1–25. doi:10.5802/ojmo.3.

Nicholson, B., Siirola, J.D., Watson, J.P., Zavala, V.M., Biegler, L.T., 2018. pyomo.dae: a modeling and automatic discretization framework for optimization with differential and algebraic equations. Mathematical Programming Computation 10, 187–223. doi:10.1007/s12532-017-0127-0.

Oh, C., Tomczak, J.M., Gavves, E., Welling, M., 2019. Combinatorial Bayesian optimization using the graph Cartesian product, in: Advances in Neural Information Processing Systems.

Ovalle, D., Liñán, D.A., Lee, A., Gómez, J.M., Ricardez-Sandoval, L., Grossmann, I.E., Bernal Neira, D.E., 2025. Logic-Based Discrete-Steepest Descent: A solution method for process synthesis Generalized Disjunctive Programs. Comput. Chem. Eng. 195, 108993. doi:10.1016/j.compchemeng.2024.108993.

Park, Y., Bernal Neira, D.E., 2026. A computational study of Ising-based solvers for discrete landscape exploration in process optimization. AIChE Journal , e70656doi:10.1002/aic.70656.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., Chintala, S., 2019. PyTorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems, pp. 8024–8035.

Peng, Z., Lee, A., Bernal Neira, D.E., 2024. Addressing discrete dynamic optimization via a logic-based discrete-steepest descent algorithm, in: 2024 IEEE 63rd Conference on Decision and Control (CDC), IEEE. pp. 1664–1669. doi:10.1109/CDC56724.2024.10886477.

Ploskas, N., Sahinidis, N.V., 2022. Review and comparison of algorithms and software for mixed-integer derivative-free optimization. J. Glob. Optim. 82, 433–462. doi:10.1007/s10898-021-01085-0.

Pyomo Development Team, n.d. GDPopt logic-based solver. URL: https://pyomo.readthedocs.io/en/6.10.0/explanation/solvers/gdpopt.html. Version 6.10.0; accessed October 20, 2026.

Raman, R., Grossmann, I.E., 1994. Modelling and computational techniques for logic based integer programming. Comput. Chem. Eng. 18, 563–578. doi:10.1016/0098-1354(93)E0010-7.

Regis, R.G., Shoemaker, C.A., 2007. A stochastic radial basis function method for the global optimization of expensive functions. INFORMS J. Comput. 19, 497–509. doi:10.1287/ijoc.1060.0182.

Rendle, S., 2010. Factorization machines, in: 2010 IEEE International Conference on Data Mining, pp. 995–1000. doi:10.1109/ICDM.2010.127.

Sahinidis, N.V., 1996. BARON: A general purpose global optimization software package. J. Glob. Optim. 8, 201–205. doi:10.1007/BF00138693.

SECQUOIA, 2026a. Black-Box-for-GDP: benchmark code, experimental data, and analysis. GitHub. URL: https://github.com/SECQUOIA/Black-Box-for-GDP/tree/de6ab9395e5121f27db3b95844b1cd381264367d. commit de6ab939; accessed 2 November 2026.

SECQUOIA, 2026b. pd_ising: Lomustine superstructure PharmaPy simulation workflow and data. URL: https://github.com/SECQUOIA/pd_ising/tree/633cf923f07abba88970fa1fa816c4d06769751e/ds-mfg. pull request 28; commit 633cf923f07abba88970fa1fa816c4d06769751e; accessed October 31, 2026.

SECQUOIA, n.d. bbo_via_fmqa: Black-box optimization via factorization machines and quantum annealing. URL: https://github.com/SECQUOIA/bbo_via_fmqa/tree/eb3a20d4b5a3773d17ff31d72a35d51f58e84e56. QHD_Data branch; commit eb3a20d4b5a3773d17ff31d72a35d51f58e84e56; upstream PyTorch FM and BQM code; accessed October 19, 2026.

The Optimization Firm, 2026. BAM python interface documentation: bam-solver 5.0.0. URL: https://pypi.org/project/bam-solver/5.0.0/. Version 5.0.0, released September 11, 2026; executed engine 2026.9.10; accessed October 20, 2026.

Türkay, M., Grossmann, I.E., 1996. Logic-based MINLP algorithms for the optimal synthesis of process networks. Comput. Chem. Eng. 20, 959–978. doi:10.1016/0098-1354(95)00219-7.

Viswanathan, J., Grossmann, I.E., 1990. A combined penalty function and outer-approximation method for MINLP optimization. Computers & Chemical Engineering 14, 769–782. doi:10.1016/0098-1354(90)87085-4.

Wächter, A., Biegler, L.T., 2006. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. Mathematical Programming 106, 25–57. doi:10.1007/s10107-004-0559-y.

Wu Group, n.d. Quantum Hamiltonian descent: 2D test functions. https://jiaqileng.github.io/quantum-hamiltonian-descent/nonconvex2d.html. Source collection for the 21 adapted finite grids used here; accessed September 29, 2026.

Supplementary Material: Benchmark Definitions, Execution Protocol, and Uncertainty

The Grid-GDP functions are adapted from the Quantum Hamiltonian Descent collection; each instance is defined by the expression and domain implemented there, which can differ from other functions known by the same name (SECQUOIA, 2026a; Wu Group, n.d.).

S1 Coordinate domains and normalization

The coordinate mapping and objective normalization follow Section 3.4 of the main paper. Both coordinate endpoints are included, and L and U are the minimum and maximum over the 961 lattice points. All trigonometric arguments are in radians. Full-precision normalization constants and target coordinates are archived in `grid_inventory.json` (Section S8).

Table S1: Grid-GDP: coordinate domains and target-set sizes of the 21 test functions of Section S2.

Function	x domain	y domain	n_{target}
Ackley	$[-5, 5]$	$[-5, 5]$	1
Alpine 1	$[0, 10]$	$[0, 10]$	1
Alpine 2	$[0, 1000]$	$[0, 1000]$	1
Bohachevsky 2	$[-100, 100]$	$[-100, 100]$	1
Three-hump camel	$[-5, 5]$	$[-5, 5]$	1
Csendes (squared variant)	$[-1, 1]$	$[-1, 1]$	36
Deflected corrugated spring	$[0, 10]$	$[0, 10]$	1
Drop-Wave	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Easom	$[-10, 10]$	$[-10, 10]$	1
Griewank	$[-600, 600]$	$[-600, 600]$	1
Hölder table	$[-10, 10]$	$[-10, 10]$	4
Hosaki	$[0, 5]$	$[0, 6]$	1
Lévy	$[-10, 10]$	$[-10, 10]$	2
Lévy 13	$[-10, 10]$	$[-10, 10]$	4
Michalewicz	$[0, \pi]$	$[0, \pi]$	1
Rastrigin	$[-5.12, 5.12]$	$[-5.12, 5.12]$	1
Rosenbrock	$[-2, 2]$	$[-1, 3]$	2
Shubert (variant)	$[0, 10]$	$[0, 10]$	1
Styblinski–Tang	$[-5, 5]$	$[-5, 5]$	1
Sum of squares	$[-10, 10]$	$[-10, 10]$	1
Xin-She Yang 3 (variant)	$[-20, 20]$	$[-20, 20]$	870

x domain, y domain: coordinate ranges, each sampled at 31 equally spaced points that include both endpoints, which gives a 31×31 lattice of 961 configurations and 62 Boolean indicators. n_{target}: number of lattice points whose normalized objective $F = (f - L)/(U - L)$ is at most 10^{-5}, where L and U are the minimum and maximum of f over the lattice; a run reaches the reference target of Equation (8) of the main paper when it evaluates one of these points.

S2 Grid-GDP objective functions

The implemented functions below use the domains in Table S1 (SECQUOIA, 2026a).

1. Ackley

$$f(x, y) = -20 \exp\left(-0.2\sqrt{\frac{x^2 + y^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x) + \cos(2\pi y)}{2}\right) + 20 + e. \tag{S1}$$

2. Alpine 1

$$f(x,y) = |x\sin x + 0.1x| + |y\sin y + 0.1y|. \tag{S2}$$

3. Alpine 2

$$f(x,y) = -\sqrt{xy}\,\sin x \sin y. \tag{S3}$$

4. Bohachevsky 2

$$f(x,y) = x^2 + 2y^2 - 0.3\cos(3\pi x)\cos(4\pi y) + 0.3. \tag{S4}$$

5. Three-hump camel

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2. \tag{S5}$$

6. Csendes (squared variant)

$$f(x,y) = c(x) + c(y), \qquad c(t) = \begin{cases} t^6\big[2 + \sin(1/t)\big]^2, & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{S6}$$

The GDP uses branch-fixed auxiliaries $r_t = 1/t$ for $t \neq 0$ and $r_t = 0$ otherwise, avoiding division by zero without an epsilon shift.

7. Deflected corrugated spring

With $r = \sqrt{(x-5)^2 + (y-5)^2}$,

$$f(x,y) = 0.1r^2 - \cos(5r). \tag{S7}$$

8. Drop-Wave

$$f(x,y) = -\frac{1 + \cos\Big(12\sqrt{x^2+y^2}\Big)}{0.5(x^2+y^2) + 2}. \tag{S8}$$

9. Easom

$$f(x,y) = -\cos x \cos y\, \exp\big(-[(x-\pi)^2 + (y-\pi)^2]\big)\,. \tag{S9}$$

10. Griewank

$$f(x,y) = \frac{x^2+y^2}{4000} - \cos x \cos\left(\frac{y}{\sqrt{2}}\right) + 1. \tag{S10}$$

11. Hölder table

$$f(x,y) = -\left|\sin x \cos y\, \exp\left(\left|1 - \frac{\sqrt{x^2+y^2}}{\pi}\right|\right)\right|. \tag{S11}$$

12. Hosaki

$$f(x,y) = \left(1 - 8x + 7x^2 - \frac{7}{3}x^3 + \frac{1}{4}x^4\right) y^2 \exp(-y). \tag{S12}$$

13. Lévy

Define $w_x = 1 + (x-1)/4$ and $w_y = 1 + (y-1)/4$. Then

$$\begin{aligned} f(x,y) = \; & \sin^2(\pi w_x) \\ & + (w_x - 1)^2\big[1 + 10\sin^2(\pi w_x + 1)\big] \\ & + (w_y - 1)^2\big[1 + \sin^2(2\pi w_y)\big]. \end{aligned} \tag{S13}$$

14. Lévy 13

$$\begin{aligned} f(x,y) = \; & \sin^2(3\pi x) + (x-1)^2\big[1 + \sin^2(3\pi y)\big] \\ & + (y-1)^2\big[1 + \sin^2(2\pi y)\big]. \end{aligned} \tag{S14}$$

15. Michalewicz

$$f(x,y) = -\sin x \, \sin^{20}\!\left(\frac{x^2}{\pi}\right) - \sin y \, \sin^{20}\!\left(\frac{2y^2}{\pi}\right). \tag{S15}$$

16. Rastrigin

$$f(x,y) = x^2 - 10\cos(2\pi x) + y^2 - 10\cos(2\pi y) + 20. \tag{S16}$$

17. Rosenbrock

$$f(x,y) = (1-x)^2 + 100(y - x^2)^2. \tag{S17}$$

18. Shubert (variant)

$$\begin{aligned} f(x,y) = \; & \big[\cos(2x+1) + 2\cos(3x+2) + 3\cos(4x+3)\big] \\ & \times \big[\cos(2y+1) + \cos(y+2)\big]. \end{aligned} \tag{S18}$$

19. Styblinski–Tang

$$f(x,y) = \frac{1}{78}\,\frac{1}{2}\left(x^4 - 16x^2 + 5x + y^4 - 16y^2 + 5y\right). \tag{S19}$$

20. Sum of squares

$$f(x,y) = x^2 + 2y^2. \tag{S20}$$

21. Xin-She Yang 3 (variant)

$$f(x,y) = \exp\!\left(-\left[\left(\frac{x}{15}\right)^6 + y^6\right]\right) + 2\exp\!\big(-(x^2+y^2)\big)\cos^2 x \cos^2 y. \tag{S21}$$

S3 Executable protocol and implementation settings

Python and `NumPy` use the job seed, and `RBFOpt` also receives `rand_seed`. `BAM` exposes no separate native random-seed option: its process wrapper seeds Python/`NumPy`, while the Grid seed identifies the common initial design. `FMQA-SA` uses three `NumPy SeedSequence(seed)` streams for FM initialization, annealing, and uniform unseen-point fallback; each training or annealing invocation draws a seed from its stream. `PyTorch` uses deterministic CPU operations and one thread.

Grid `D-SDA` enumerates L_∞ neighbors lexicographically from $(-1, 0, 1)$, excluding the zero direction and out-of-bounds points; ties select the first point.

The RBFOpt options are rbf=auto, global_search_method=genetic, do_init_strategy=False, ga_base_population_size=400, and ga_num_generations=20.

BAM receives x0=None and max_evals=B-3; its local models use ALAMO/BARON, whose internal versions are not reported by the package.

The FMQA-SA time limit is checked between operations and through training/sampling callbacks. Each FM uses identity activation, zero bias, and normal initial weights with mean zero and standard deviation 0.01. Training uses Adam (Kingma and Ba, 2015), 1,000 full-batch epochs, mean squared error, and learning rate 0.01. Coordinate ξ_j uses $\lfloor \log_2 u_j \rfloor + 1$ unsigned bits, where $u_j > 0$ is its upper bound; QUBO coefficients are divided by the largest absolute entry of the quadratic matrix. D-Wave's SimulatedAnnealingSampler in dwave-samplers 1.6.0 uses 100 reads, 1,000 sweeps, and a geometric inverse-temperature schedule (D-Wave Systems, n.d.). All process workers and the Grid FMQA-SA worker have a 2,100 s watchdog. The Grid RBFOpt/BAM trace watchdogs are disabled (max_run_seconds=0); archived Grid D-SDA uses budget, target, or local-search termination without an additional method-clock limit.

S3.1 Versions and independently timed process evaluations

Software versions are given in Section 3.6 of the main paper.

The conventional GDP process models retain their archived initializations, including constructor defaults for CSTR30; the updated DFO evaluation table is identified separately in Section S8.

The acceptance rule in gdp_models/evaluation.py requires all of the following: a finite objective; termination in {optimal, locallyOptimal, globallyOptimal}; solver status other than error or aborted; interface status SOLVESTAT=1 and MODELSTAT$\in \{1, 2\}$; and a finite maximum primal violation no greater than 10^{-5}. Time-limit, iteration-limit, evaluation-limit, and error outcomes are not accepted, even if a finite feasible candidate is available.

The primal check uses unscaled violations of original constraints and initialized-variable bounds, including constraint rows deactivated during preprocessing. Nonfinite objectives, constraint bodies, or initialized-variable values preclude acceptance; unused uninitialized variables are skipped. Preprocessing rejection or worker failure supplies no accepted objective.

The CSTR initializer cstr_box_ipopt.py:initialize_feasible runs after preprocessing and checks the mass-balance/zero-recycle start at 10^{-8} without changing constraints or bounds. All process evaluations use cold starts; NLP solutions are not transferred between configurations. Objectives, acceptance status, and NLP point costs are linked to the same recorded attempt by model, CSTR size, and coordinate, without successful-retry selection or cross-size time substitution.

S3.2 Observed proposal and initialization behavior

Table S2 pools FMQA-SA proposals across budgets and seeds, excluding the shared initialization and including terminal callbacks. Fallback supplies 52.89% of process proposals; the uniform comparison evaluates the complete procedure, without isolating the contributions of learning, annealing, and fallback.

Table S2: Grid-GDP and process GDP models: proposals and uniform fallback proposals of `FMQA-SA`, pooled over all budgets and seeds.

Benchmark	Runs	Proposals	Fallbacks	Rate (%)
Grid-GDP	1,050	51,450	9,522	18.51
`CSTR`	60	1,350	369	27.33
`Column`	10	100	86	86.00
`Small Batch`	10	70	58	82.86
`Dynamic`	60	1,460	1,063	72.81
Process total	140	2,980	1,576	52.89

Runs: number of `FMQA-SA` runs (Grid-GDP: 21 functions, five budgets, and ten seeds; process models: ten seeds for each `CSTR` size or budget). Proposals: configurations that `FMQA-SA` selects after the three shared initial configurations, so a run with evaluation budget B makes $B - 3$ proposals. Fallbacks: proposals drawn uniformly from the unevaluated configurations because no annealed candidate is a new configuration within the coordinate bounds. Rate: fallbacks as a percentage of proposals.

S4 Process model scale, units, and discrete decisions

Table S3 counts the original GDP before reformulation or presolve, including all active disjunct branches and fixed continuous variables. `Dynamic` is discretized with 30 finite elements per stage and three Radau points. Binary counts refer to the indicator variables already constructed by the model, before Boolean-to-binary conversion. All nine models have zero additional nonbinary integer variables.

Table S3: Process GDP models: numbers of variables, constraints, and disjunctions of the original GDP models.

Model	N	n_c	n_b	m_a	m_l	n_d
`CSTR`	5	56	20	100	12	10
	10	106	40	195	22	20
	15	156	60	290	32	30
	20	206	80	385	42	40
	25	256	100	480	52	50
	30	306	120	575	62	60
`Column`	–	446	30	637	16	15
`Small Batch`	–	19	18	34	3	9
`Dynamic`	–	1,647	27	3,276	31	9

Counts: original GDP before reformulation or presolve, including all active disjunct branches and fixed continuous variables; `Dynamic` is discretized with 30 finite elements per stage and three Radau collocation points. N: number of potential reactors of `CSTR` (–: not applicable). n_c: continuous variables; n_b: binary indicator variables constructed by the model; m_a: algebraic constraints; m_l: logical constraints; n_d: disjunctions.

The `Column` objective is $10^3(Q_c + Q_b) + 10^3 n_{\text{active}}$, the weighted cost proxy of the source model, which we did not convert to an annualized cost. The `Dynamic` objective is $-\int_0^9 x(t)^2\,dt$.

S5 Lomustine simulation-based benchmark

The `Lomustine` configurations and responses come from the corrected-model tables of SECQUOIA (2026b), generated with `PharmaPy` at commit 3194efd. For each superstructure option, the source ran `PyNomadBBO`

from a stored initial point on the normalized operating inputs with 60 simulations and stored the best objective, the constraint values, the equipment sizes, and the wall time of the inner optimization. The inner search accepts a point when the solubility path constraint holds. The heptane mass-fraction specification of the source is not enforced, because the fixed three-hour vaporizer operation cannot reach it within the pressure bounds. Table S7 lists the 36 two-reactor configurations; the four single-reactor semibatch options of the source do not fit the three external variables and are excluded. The configurations with a batch crystallizer have the lowest objective for every reactor pair, and the choice of the first reactor changes the objective by up to a factor of 2.9. The vendored table, case module, run records, and analysis are listed in Section S8.

Option order and objective. The external variables index categorical alternatives, so we repeated the 210 runs with the five other orders of Table S4, using the same seeds, initial designs in index space, and budgets. Table S5 reports the hits in the source order and their range over the six orders. `NOMAD` ended with a segmentation fault after identical evaluations on both attempts in 12 runs over all campaigns; as for the Grid runs of Section S12, these runs are scored on the evaluations completed before the crash. Table S6 repeats the source-order runs with the bare-module capital cost of each flowsheet at its optimized operation as the objective, whose reference is the all-batch flowsheet at \$82,766. This cost is evaluated at the operating point that maximizes production and crystal size, and it is not itself optimized over the operating inputs.

Table S4: `Lomustine`: the six orders of the reactor and crystallizer alternatives used as external-variable indices.

Order	Reactors	Crystallizers
Source	batch, CSTR, PFR	batch, 1, 2, 3
Reversed	PFR, CSTR, batch	3, 2, 1, batch
Back-mixing	batch, PFR, CSTR	batch, 3, 2, 1
Random 1	PFR, batch, CSTR	1, 2, batch, 3
Random 2	batch, PFR, CSTR	batch, 3, 1, 2
Random 3	CSTR, PFR, batch	2, 1, batch, 3

Source: order of the source enumeration, used in the main comparison. Back-mixing: alternatives ordered by degree of back-mixing. Random 1 to 3: three further permutations. CSTR: continuous stirred-tank reactor; PFR: plug-flow reactor. Crystallizers: batch, or the number of mixed-suspension, mixed-product-removal (MSMPR) stages.

Table S5: `Lomustine`: target hits of the six DFO methods and uniform sampling in the source order and over the six option orders, with ten runs per method, budget, and order.

		Target hits (of 10)						
B		`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform
10	Source order	1	6	8	7	0	2	2
	Fewest over orders	1	5	2	4	0	2	0
	Most over orders	6	6	8	7	2	7	4
15	Source order	4	6	9	10	4	3	4
	Fewest over orders	3	5	7	5	0	3	2
	Most over orders	8	9	10	10	10	10	4
20	Source order	4	6	10	10	6	7	5
	Fewest over orders	3	5	9	6	0	7	4
	Most over orders	8	9	10	10	10	10	7

Target hits: runs that reach the reference target of Equation (8) of the main paper, here the reference configuration. Source order: target hits in the order of the source enumeration, used in the main comparison. Fewest and most over orders: smallest and largest number of target hits over the six orders of Table S4. B: evaluation budget; Uniform: uniform sampling without replacement.

Table S6: `Lomustine` with the capital-cost objective: target hits of the six DFO methods and uniform sampling in the source order, with ten runs per method and budget.

	Target hits (of 10)						
B	`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform
10	3	9	7	8	**10**	2	4
15	6	9	8	**10**	**10**	3	7
20	9	9	**10**	**10**	**10**	5	8

Target hits: runs that reach the reference of the capital-cost objective, the all-batch configuration at \$82,766. Capital-cost objective: bare-module capital cost of each flowsheet at its optimized operation. B: evaluation budget; Uniform: uniform sampling without replacement. Bold: most target hits for each budget, including ties.

Table S7: `Lomustine`: objective of every configuration after the simulation-based optimization of its operating inputs, and the wall time of that optimization.

		Objective				Wall time (s)			
R01	R02	batch	1 MSMPR	2 MSMPR	3 MSMPR	batch	1 MSMPR	2 MSMPR	3 MSMPR
	batch	-170.68	-131.70	-146.86	-148.92	91	310	538	835
batch	CSTR	-230.52	-175.25	-175.27	-166.13	146	348	650	970
	PFR	-268.46	-209.36	-213.29	-202.26	159	348	651	975
	batch	-483.85	-360.16	-365.91	-345.28	152	331	656	936
CSTR	CSTR	-331.90	-244.39	-245.34	-239.50	159	367	660	1,019
	PFR	-429.81	-315.22	-321.46	-304.62	206	408	717	1,017
	batch	**-487.84**	-362.45	-361.55	-348.06	208	347	655	886
PFR	CSTR	-351.71	-249.82	-259.47	-244.65	183	388	659	983
	PFR	-433.41	-318.34	-324.18	-306.21	211	393	629	907

Objective: $-(\dot{m}/10^{-4} + 2d_{50})$, with the production rate $\dot{m}$ in kg/s and the median crystal size d_{50} in μm. Wall time: wall time of the simulation-based optimization of the operating inputs. R01, R02: first and second reactor; CSTR: continuous stirred-tank reactor; PFR: plug-flow reactor. batch, 1 to 3 MSMPR: batch crystallizer or one to three mixed-suspension, mixed-product-removal (MSMPR) stages. Bold: reference configuration, which has the lowest objective.

S6 Catalytic distillation column

Corrected model. We checked every equation block, parameter table, and bound of the published model against standard forms and literature values, and the `Catalytic Column` benchmark uses the model with the errors fixed. The corrections concern the vapor–liquid equilibrium and the equilibrium constant of the reaction, the mixture liquid density (Rackett equation with the Spencer–Danner mixing rule) and surface tension (Macleod–Sugden), the tray hydraulics, and the objective, which counted the annualization factor twice and credited the impurities in the bottoms at the ETBE price. The pure-component data of isobutene, 1-butene, and ethanol are fitted to open reference equations of state, and the ETBE data, for which no reference equation exists, agree with open measurements of its vapor pressure and density. The bottoms must contain at least 95 mol% ETBE, the fuel-grade specification, the catalytic stages stay at or below 120 °C, the maximum operating temperature of the Amberlyst 15 catalyst, and the prices are 2010 market prices. The repository lists each finding with its effect on the published optimum; with the 95 mol% specification, the published optimal configuration is infeasible, because its column is too short.

Evaluation campaign. The 210 runs made 12,015 evaluations: 6,921 returned accepted objectives, 2,789 were logical rejections without an NLP solve, and 2,305 returned unaccepted NLP responses. Eight `D-SDA` runs stopped before their budgets because no neighbor improved, and all other runs used their full budgets. The runs requested 3,499 distinct configurations, of which 2,652 were accepted, and solving them took 7.7 h.

The completion times of the runs sum to 20.2 h, but the campaign took 2.8 h on three cores because a configuration requested by several runs was solved once. Each run evaluated distinct configurations, so the reuse comes from different runs requesting the same configuration.

Search box. The GDP admits 100,947 configurations, and the gap box $g_i \leq 6$ of the search methods keeps 54,159 of them, including every design within 5% of the best of a pilot of 202 configurations. The reference has $g_1 = f_1 - 2 = 6$, on the bound of the box, so a better design could lie just outside it. We therefore solved, with the initialization and acceptance rule of the campaign, all 747 configurations outside the box within two stages of the reference in every stage position. Of these, 255 are accepted and 492 are locally infeasible, and none is better than the reference: the best, $(9, 9, 13, 17, 17, 21)$ at 24,930.96 \$/yr, lies 0.14% above it. This check ran on a workstation with the `CONOPT` version of the campaign (Section 3.6 of the main paper), and every solve finished within 97 s, below the campaign's 300 s limit.

MINLP forms. The published MINLP form makes the stage decisions binary variables, multiplies switched equations by these variables or by the stage-existence indicator, and keeps the property and kinetic equations on every stage. The disjunctive form behind the Big-M and Hull reformulations has five disjunctions for each internal stage, one per decision, whose terms collect the equations that the GDP model switches on or off (Table S8). The property and feed-enthalpy equations hold on every stage, as in the published form, but the kinetics stay in the catalyst disjunction; with the kinetics on every stage, the bounds of the reaction rate restricted the temperature of a stage without catalyst, and with the binary variables of $(3, 4, 5, 11, 11, 14)$ fixed, the Big-M reformulation stopped at 32,075 \$/yr, a point from which the GDP form continues to 30,822 \$/yr. Each nonlinear disjunct equation $h(\mathbf{x}) = 0$ or $h(\mathbf{x}) \leq 0$ is replaced by the definition $s\,r = h(\mathbf{x})$, which holds on every stage, and by the linear disjunct constraint $r = 0$ or $r \leq 0$; a linear disjunct equation is divided by s. The scale s is 1.01 times the largest absolute value of h over the variable bounds, computed by interval arithmetic after feasibility-based bound tightening with the constraints that hold for every configuration. Every scaled disjunct constraint therefore lies in $[-1, 1]$ at every feasible point, so $M = 1$ is a valid Big-M value for all of them, and the bound tightening adds 101 implied variable bounds that are written into the model. The scales range from 0.10 to 1.3×10^{18}; the largest belongs to the rate equation, whose products of activities, adsorption constant, and rate span this range over the variable bounds. The interval bounds of the flooding and weeping correlations are finite only if the liquid is denser than the vapor, which the source bounds do not imply, so the disjunctive form bounds the liquid and vapor mass densities by $\rho_L \geq 200$ and $5 \leq \rho_V \leq 150$ kg/m^3; every validated solution has $\rho_L \geq 529$ and $\rho_V \leq 34$ kg/m^3. These bounds belong only to the Big-M and Hull reformulations; the GDP model, its fixed-configuration evaluations, and the published MINLP form do not impose them. Within the model's bounds (pressure 9.5 to 10 bar, temperature 200 to 445 K, compressibility factor 0.5 to 1.3) and for any vapor composition, whose molar mass lies between 46.07 and 102.18 g/mol, the vapor density lies between 9 and 123 kg/m^3, so the vapor bounds cannot bind and only the liquid bound restricts the reformulations. The `GAMS` convex hull reformulation accepts only inequalities, so each disjunct equality is written as two inequalities, and Big-M of the pair equals Big-M of the equality. Table S9 compares the forms with the binary variables fixed.

MINLP solvers. Table S10 lists the results of the MINLP solvers. `SBB` stops on the Big-M and Hull reformulations when the NLP relaxations of its nodes exceed the limit on function evaluations outside their domains. Table S11 lists the global optimality certificates that `SCIP` and `BARON` return when they start from a feasible design of the published form; the `GAMS Examiner` confirms that both starting designs satisfy every constraint and bound within 10^{-6}.

Table S8: `Catalytic Column`: the five disjunctions of each internal stage k in the disjunctive form behind the Big-M and Hull reformulations.

Indicator	Indicator true	Indicator false
stage k exists	vapor–liquid equilibrium, tray hydraulics, pressure drop, flooding and weeping limits, tray height	vapor composition and flow pass through, no pressure drop, no tray height
catalyst on stage k	kinetics, catalyst hydraulics and flooding, reaction terms, 120 °C limit	no reaction, no catalyst pressure drop
ethanol feed at k	feed flow	no feed flow
butenes feed at k	feed flow	no feed flow
boil-up return at k	boil-up flow	no boil-up flow

k: internal stage of the column. Indicator: Boolean indicator of one disjunction. Indicator true, Indicator false: equations enforced when the indicator is true or false. The property and feed-enthalpy equations hold on every stage and belong to no disjunction.

Table S9: `Catalytic Column`: objectives of four model forms with the binary variables fixed at nine configurations, solved by `CONOPT` from the initialization of the published model.

	Objective (\$/yr)			
$(f_1, c_1, c_2, c_3, f_2, b)$	GDP	Published MINLP	Big-M	Hull
$(8, 8, 12, 17, 17, 21)$	24,895.55	24,895.55	24,895.55	24,895.55
$(7, 8, 11, 16, 16, 20)$	25,271.33	25,271.33	25,271.33	25,271.33
$(3, 4, 5, 11, 11, 14)$	30,822.02	30,822.02	30,822.02	30,822.02
$(4, 4, 9, 10, 14, 17)$	28,373.98	28,373.98	28,373.98	28,373.98
$(3, 6, 12, 14, 19, 21)$	31,116.60	32,320.55	31,116.60	31,116.60
$(2, 3, 9, 10, 20, 21)$	35,745.43	35,194.65	35,745.43	35,745.43
$(4, 4, 6, 8, 8, 10)$		infeasible		
$(3, 4, 5, 6, 8, 9)$		infeasible		
$(13, 13, 15, 16, 17, 19)$		infeasible		

$(f_1, c_1, c_2, c_3, f_2, b)$: configuration given by the stages of the ethanol feed f_1, the three catalytic stages $c_1 < c_2 < c_3$, the butenes feed f_2, and the boil-up return b; the first row is the reference configuration. GDP: the GDP model evaluated by the search methods; Published MINLP: the MINLP form published with the original `D-SDA` study of this column; Big-M and Hull: the reformulations of the disjunctive form of Table S8. infeasible: every form reports the configuration as locally infeasible.

Table S10: `Catalytic Column`: outcomes and solve times of four MINLP solvers on three model forms, with one run per solver and form started from the initial configuration $(3, 4, 5, 6, 8, 9)$ of the published model.

Form	Solver	Outcome	Time (s)
Published	`DICOPT`	no feasible design; every NLP subproblem infeasible, iteration limit	686
	`SBB`	28,717.22 \$/yr at $(4, 4, 5, 6, 6, 9)$	10
	`BARON`	declared infeasible	255
	`SCIP`	declared infeasible	9
Big-M	`DICOPT`	no feasible design; NLP subproblem failed	902
	`SBB`	no feasible design; domain violation limit	38
	`BARON`	declared infeasible	214
	`SCIP`	no feasible design; lower bound -5.7×10^7 \$/yr	3,600+
Hull	`DICOPT`	no feasible design; NLP subproblem failed	211
	`SBB`	no feasible design; domain violation limit	130
	`BARON`	declared infeasible	362
	`SCIP`	no feasible design; lower bound -5.7×10^7 \$/yr	3,600+

Outcome: objective and configuration $(f_1, c_1, c_2, c_3, f_2, b)$ of the design found, with configurations as in Table S9, or the reason no feasible design was returned; the reference is 24,895.55 \$/yr. Time: wall-clock time of the `GAMS` job; each run uses one core and a 3,600 s time limit. Published: the published MINLP form; Big-M and Hull: the reformulations of the disjunctive form. 3,600+: stopped at the time limit.

Table S11: `Catalytic Column`: global optimality certificates that `SCIP` and `BARON` return for the published MINLP form, with one run per solver from each of two feasible starting designs.

	`SCIP`			`BARON`		
Starting design	Status	Objective ($/yr)	Time (s)	Status	Objective ($/yr)	Time (s)
$(7, 8, 11, 16, 16, 20)$	optimal	25,271.33	8	optimal	25,271.33	172
$(8, 8, 12, 17, 17, 21)$	optimal	24,895.55	10	optimal	24,895.55	270

Starting design: configuration $(f_1, c_1, c_2, c_3, f_2, b)$ as in Table S9; the first is a design from the pilot of 202 configurations and the second the reference configuration. Status, Objective: status and objective reported by the solver. Time: solve time.

S7 Completion costs

Table S12 reports the search times in all 14 process conditions; the completion times are in Table 2 and the enumeration times in Table 1 of the main paper. Table S13 reports the Grid-GDP costs at $B = 100$. `MISO` reaches the most Grid targets, but its median search time of 32.70 s is about four times that of `RBFOpt`, whereas `D-SDA` and `NOMAD` stop after medians of 37 and 31.5 evaluations and have the lowest median completion times, 18.38 and 18.50 s (`NOMAD` over its 189 timed runs).

Per-function Grid results at $B = 100$ and complete seed-level outcomes are available in the analysis files linked in Section S8.

Table S12: Process GDP models: search times of the six DFO methods and uniform sampling, with ten runs per method and condition.

Model	N	B	Search time	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
CSTR	5	8	Median (s)	0.0105	0.0367	0.0596	0.659	0.0360	3.66	0.00460
			25th percentile (s)	0.0102	0.0323	0.0590	0.658	0.0151	3.64	0.00387
			75th percentile (s)	0.0111	1.13	0.0597	0.669	0.0816	3.67	0.00478
	10	15	Median (s)	0.0137	1.83	0.302	0.677	0.137	8.00	0.00807
			25th percentile (s)	0.0130	0.469	0.288	0.675	0.126	7.98	0.00772
			75th percentile (s)	0.0147	1.88	0.315	0.683	0.158	8.08	0.00812
	15	22	Median (s)	0.0169	0.825	1.00	3.29	0.150	12.4	0.0113
			25th percentile (s)	0.0156	0.0949	0.492	1.21	0.140	12.3	0.0108
			75th percentile (s)	0.0172	1.82	3.26	7.50	0.162	12.4	0.0116
	20	29	Median (s)	0.0207	1.63	2.82	10.8	0.167	16.9	0.0139
			25th percentile (s)	0.0189	0.521	0.632	9.02	0.165	16.8	0.0121
			75th percentile (s)	0.0218	1.92	4.01	15.8	0.313	16.9	0.0148
	25	36	Median (s)	0.0258	1.08	2.47	17.6	0.401	21.3	0.0161
			25th percentile (s)	0.0238	0.170	1.86	16.4	0.392	21.3	0.0147
			75th percentile (s)	0.0299	2.32	4.43	18.8	0.449	21.4	0.0186
	30	43	Median (s)	0.0585	1.90	4.99	18.1	0.425	25.8	0.0241
			25th percentile (s)	0.0556	1.76	2.95	14.0	0.404	25.7	0.0224
			75th percentile (s)	0.0610	7.65	10.3	19.3	0.451	26.0	0.0257
Column	–	13	Median (s)	0.0120	0.0423	0.226	0.672	0.0177	6.85	0.00716
			25th percentile (s)	0.0116	0.0358	0.205	0.669	0.0168	6.81	0.00711
			75th percentile (s)	0.0124	0.0461	0.239	0.681	0.155	6.89	0.00748
Small Batch	–	10	Median (s)	0.0102	0.0475	0.170	0.668	0.0169	4.87	0.00536
			25th percentile (s)	0.00971	0.0333	0.159	0.666	0.0161	4.86	0.00506
			75th percentile (s)	0.0103	0.0540	0.180	0.670	0.0191	4.89	0.00545
Dynamic	–	12	Median (s)	0.0119	0.0419	0.186	0.671	0.0173	6.16	0.00589
			25th percentile (s)	0.0113	0.0377	0.182	0.670	0.0172	6.14	0.00540
			75th percentile (s)	0.0125	0.0630	0.189	0.676	0.0174	6.20	0.00658
		15	Median (s)	0.0123	0.0896	0.296	0.679	0.139	8.04	0.00740
			25th percentile (s)	0.0115	0.0627	0.291	0.676	0.128	8.01	0.00622
			75th percentile (s)	0.0131	1.09	0.321	0.683	0.156	8.09	0.00760
		20	Median (s)	0.0133	1.49	0.367	25.0	0.143	11.2	0.00987
			25th percentile (s)	0.0119	0.447	0.355	24.6	0.130	11.1	0.00927
			75th percentile (s)	0.0154	6.90	0.381	30.1	0.156	11.2	0.00991
		30	Median (s)	0.0149	1.77	7.61	79.7	0.355	17.4	0.0141
			25th percentile (s)	0.0124	1.44	4.88	79.4	0.204	17.2	0.0135
			75th percentile (s)	0.0157	9.24	13.7	80.5	0.364	17.5	0.0145
		37	Median (s)	0.0134	1.77	10.1	81.0	0.376	21.7	0.0173
			25th percentile (s)	0.0118	1.43	5.25	79.7	0.364	21.6	0.0171
			75th percentile (s)	0.0160	9.23	14.2	82.9	0.383	21.8	0.0174
		50	Median (s)	0.0135	1.77	26.3	158	0.732	29.8	0.0233
			25th percentile (s)	0.0129	1.42	17.0	157	0.685	29.7	0.0229
			75th percentile (s)	0.0153	9.21	37.2	160	0.751	30.1	0.0235

Search time: the time a method spends selecting configurations, which excludes the cost of evaluating them; the full-run completion times and target hits are in Table 2 of the main paper. Percentiles are over the ten runs. N: number of potential reactors of CSTR (–: not applicable); B: evaluation budget; Uniform: uniform sampling without replacement.

Table S13: Grid-GDP: target hits, full-run completion times, summed evaluation costs, and search times of the six DFO methods and the two sampling controls with a budget of 100 evaluations, with 210 runs per method.

		Full-run completion time (s)			Summed evaluation cost (s)			Search time (s)		
Method	Target hits (of 210)	Median	25th pct.	75th pct.	Median	25th pct.	75th pct.	Median	25th pct.	75th pct.
`D-SDA`	101	18.38	12.42	37.06	18.38	12.41	37.05	0.0024	0.0023	0.0026
`NOMAD`[a]	126	18.50	13.46	35.00	15.20	11.09	30.82	2.04	1.79	2.28
`RBFOpt`	171	70.81	54.83	93.09	60.84	44.31	82.15	7.58	4.12	14.82
`MISO`	181	94.82	79.22	116.05	63.21	44.03	82.37	32.70	25.19	39.38
`BAM`	147	64.42	48.74	81.67	63.97	48.34	81.22	0.43	0.37	0.51
`FMQA-SA`	129	124.13	105.57	139.26	62.34	43.98	77.52	62.04	61.58	62.53
Uniform	45	60.51	43.33	78.60	60.51	43.33	78.59	0.0026	0.0026	0.0027
`BAM`-prefix control	124	61.81	40.96	78.33	61.80	40.95	78.33	0.0026	0.0026	0.0027

Target hits: runs that reach the reference target of Equation (8) of the main paper, the normalized reference objective zero within 10^{-5}; the 210 runs are 21 test functions with ten seeds each. Full-run completion time: search time plus summed evaluation cost (Equation (10) of the main paper). Summed evaluation cost: sum of the separately measured costs of every evaluation in the run, including repeated configurations. Search time: method time excluding the evaluation cost. Median, 25th pct., 75th pct.: median and linearly interpolated 25th and 75th percentiles over all runs, including target misses; percentiles of the two components need not sum to percentiles of the completion time. Uniform: uniform sampling without replacement after the three shared initial configurations; `BAM`-prefix control: the ten predetermined sampling points of `BAM` after the three shared initial configurations, followed by uniform sampling without replacement.

[a] `NOMAD` times cover the 189 runs with recorded times; the 21 runs that crashed on every attempt have none, but their target hits are counted (Section S12).

S8 Data sources and reproducibility

The publication snapshot contains the source code, experimental data, analysis results, and reproduction instructions for every experiment in this article. The `CSTR` updates are the penalty of 10^6 for unaccepted responses, analyzed in `results/cstr_penalty_1e6_analysis_20261022`, and the `CSTR30` evaluation table rebuilt with the mass-balance/zero-recycle initialization, in `results/cstr30_publication_20261025`; the `Column` runs on its 64-configuration logical domain with $B = 13$ are in `results/column_logical_b13_analysis_20261025`. The updated `GDPopt` analysis and execution records contain the 63 Grid and 27 process runs configured with `SCIP` for MIP and `IPOPT` for NLP subproblems. The three presolve-disabled LOA diagnostic runs are excluded. The same records contain the paired comparison with the earlier configuration that used `SCIP` for both MIP and NLP subproblems, in which GLOA returned the same `CSTR` objectives within a relative difference of 3×10^{-5}. Seed spread, tolerance sensitivity, search overhead, and `GDPopt` enumeration counts are generated by `review_comparison/revision_diagnostics.py` from the retained run records and execution logs. The `D-SDA` rerun, including the process restart variants, is `experiments/dsda_no_revisit/run.py`, which replays the sealed lookup package of the adopted `D-SDA` rerun. The `NOMAD` and `MISO` runs were executed at source revision c4d7db6; the returned records contain the original archive, its SHA-256 checksum, and hash-checked analysis tables of every run. All `D-SDA` runs, including the restart variants, were rerun at source revision c186408, where no configuration is evaluated twice; with re-evaluation switched back on, the same code reproduces all 1,190 published `D-SDA` runs exactly, and the returned records hold every run. The `Lomustine` case is `experiments/lomustine_simulation`, which vendors the corrected-model table of SECQUOIA (2026b) and runs the seven methods through the harness in `experiments/catalytic_distillation_live`; the run records and analysis are in `results/lomustine_simulation_20261031`. The `Catalytic Column` model and its review are in `gdp_models/catalytic_distillation`, the live runs are `experiments/catalytic_distillation_live`, and their returned records, with the original archive, its SHA-256 checksum, and an audit of every run, are in `results/catalytic_distillation_live_20261101`. The MINLP forms, their generators, the fixed-configuration validation, and the solver runs are `experiments/catalytic_distillation_minlp`; the runs reported here are in `results/catalytic_distillation_minlp_dantzig_20261101`, and an earlier run of the same experiments on a workstation with `GAMS` 54.2.2, in `results/catalytic_di`

stillation_minlp_20261101, returned the same outcome for every solver, form, and configuration. The D-SDA neighborhood comparison of Section S14 is experiments/dsda_separable, with results in results/dsda_separable_20261101, and the check of the Catalytic Column configurations outside the search box is in results/catalytic_distillation_outside_box_20261102. The seven-method tables and figures are generated from these tables and the published records by review_comparison/seven_methods.py and the review_comparison/joint_process_*.py scripts, without new solver calls.

S9 Feasibility and bound consistency of conventional GDP results

Of the 45 conventional GDP runs in Table S23, 21 have flagged bounds (Table S14) relative to the checked solutions in Table S15. These results combine the 27 new SCIP–IPOPT GDPopt runs with the 18 retained direct SCIP runs. Unrounded values, residuals, and source records are linked in Section S8.

For each minimization model, let f_w be the recorded objective of its best independently checked saved feasible solution, and let L denote a reported lower bound. We apply the same rule to all five methods:

$$\tau_w = 10^{-5}\max\{1, |f_w|\}, \qquad \Delta = L - f_w, \qquad \rho = \Delta/\tau_w, \qquad \dagger \iff \rho > 1. \tag{S22}$$

Missing lower bounds remain missing, not zero.

Saved vectors are loaded into the original GDP to reevaluate objectives and check active constraints, variable bounds, integrality, disjunctions, and logical validity without optimization. Violations are unscaled original-equation residuals, not reformulation residuals. All 43 saved feasible vectors satisfy the 10^{-5} threshold; the largest violation is 9.90×10^{-7}. The two Big-M runs without accepted feasible solutions remain in the 45-run denominator. This numerical check establishes neither exact feasibility nor global optimality.

Eight of the 21 discrepancies are only 1.09–3.54 times the tolerance and are sensitive to numerical feasibility conventions. These conflicting bounds and optimal reports provide no independent global certificate or intrinsic method ranking; their cause in GDPopt or its subsolvers remains unresolved.

Table S14: Process GDP models: the 21 conventional GDP runs whose reported lower bound exceeds the recorded objective of the comparison solution of the same model by more than the audit tolerance.

Model	N	Method	ID	Reported L	Unit	$v_{\max}$	C	ρ
CSTR	5	GLOA	S02	3.13020	L	9.85×10^{-12}	C01	$2.23 \times 10^{3\dagger}$
		LOA	S03	3.06201	L	2.60×10^{-11}	C01	$1.29^{\dagger}$
		RIC	S01	3.06201	L	2.60×10^{-11}	C01	$1.29^{\dagger}$
		Hull	S05	3.06201	L	9.34×10^{-7}	C01	$1.09^{\dagger}$
	10	GLOA	S07	3.44796	L	3.11×10^{-15}	C02	$1.93 \times 10^{4\dagger}$
		LOA	S08	2.96605	L	1.62×10^{-10}	C02	$2.65 \times 10^{3\dagger}$
		RIC	S06	2.96605	L	1.62×10^{-10}	C02	$2.65 \times 10^{3\dagger}$
		Hull	S10	2.88951	L	9.37×10^{-7}	C02	$3.54^{\dagger}$
	15	GLOA	S12	3.56196	L	1.01×10^{-15}	C03	$2.59 \times 10^{4\dagger}$
		LOA	S13	2.82939	L	1.33×10^{-15}	C03	$1.30^{\dagger}$
		RIC	S11	2.82939	L	1.33×10^{-15}	C03	$1.30^{\dagger}$
	20	GLOA	S17	3.61612	L	4.60×10^{-11}	C04	$2.92 \times 10^{4\dagger}$
		LOA	S18	2.79941	L	1.58×10^{-15}	C04	$1.67^{\dagger}$
		RIC	S16	2.79941	L	1.58×10^{-15}	C04	$1.67^{\dagger}$
	25	GLOA	S22	3.64786	L	2.22×10^{-14}	C05	$3.12 \times 10^{4\dagger}$
		RIC	S21	2.79125	L	1.38×10^{-10}	C05	$3.53 \times 10^{2\dagger}$
	30	GLOA	S27	3.66872	L	7.11×10^{-15}	C06	$3.25 \times 10^{4\dagger}$
Column	–	GLOA	S32	20,355.1	none	1.15×10^{-13}	C07	$4.65 \times 10^{3\dagger}$
		Hull	S35	20,044.2	none	1.00×10^{-8}	C07	$3.06 \times 10^{3\dagger}$
Dynamic	–	GLOA	S42	-197.140	none	9.85×10^{-12}	C09	$7.01 \times 10^{3\dagger}$
		Hull	S45	-196.931	none	1.00×10^{-7}	C09	$7.11 \times 10^{3\dagger}$

ID: run identifier; C: identifier of the comparison solution of the same model in Table S15, with recorded objective f_w. Reported L: lower bound reported by the run. Unit: unit of the objective, as in Table 1 of the main paper; none, no physical unit. $v_{\max}$: largest unscaled violation of the original GDP constraints and variable bounds at the run's saved solution. $\rho = (L - f_w)/\tau_w$: excess of the reported lower bound over f_w in multiples of the audit tolerance $\tau_w = 10^{-5}\max\{1, |f_w|\}$. N: number of potential reactors of `CSTR` (–: not applicable). GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; all three are `GDPopt` methods. Hull: the reformulation solved directly by `SCIP`. †: $\rho > 1$. Displayed values are rounded; flags use unrounded values.

Table S15: Process GDP models: comparison solutions of Table S14, one for each model.

Model	N	Method	ID	Recorded f_w	Unit	$v_{\max}$
`CSTR`	5	Big-M	C01	3.06198	L	9.40×10^{-7}
	10	Big-M	C02	2.88941	L	9.76×10^{-7}
	15	Hull	C03	2.82936	L	9.47×10^{-7}
	20	Hull	C04	2.79936	L	9.90×10^{-7}
	25	Hull	C05	2.78143	L	9.90×10^{-7}
	30	Big-M	C06	2.76957	L	9.68×10^{-9}
`Column`	–	RIC	C07	19,449.9	none	1.01×10^{-13}
`Small Batch`	–	Big-M	C08	167,428	$	9.00×10^{-7}
`Dynamic`	–	Big-M	C09	-212.009	none	1.00×10^{-7}

Comparison solution: saved solution with the lowest recorded objective f_w among the conventional GDP runs on the model whose saved solutions pass the independent feasibility check in the original GDP at 10^{-5}. ID: identifier used in column C of Table S14. Method: conventional GDP method that returned the solution. Unit: unit of the objective, as in Table 1 of the main paper; none, no physical unit. $v_{\max}$: largest unscaled violation of the original GDP constraints and variable bounds at the solution. N: number of potential reactors of `CSTR` (–: not applicable). RIC: relaxation with integer cuts of `GDPopt`; Big-M and Hull: reformulations solved directly by `SCIP`. Displayed values are rounded.

Additional tightened-feasibility check for `CSTR15`. An `IPOPT` solve kept the discrete configuration and starting values of an archived LOA solution and targeted an original-GDP residual of 10^{-8} (SECQUOIA, 2026a). That solution differs from the comparison solution C03 of Table S15. Only exactly satisfied constant rows were omitted from the computational clone; the independent check retained all original constraints. The returned objective was 2.82939, with maximum unscaled violation 1.11×10^{-16}. The new GLOA lower bound exceeds this objective by 0.732565, or approximately 2.59×10^4 times the audit tolerance. The discrepancy persists under the tighter check, but its cause remains unresolved.

Additional formulation check for `Dynamic` Hull. The saved Big-M point has objective -212.009, approximately 15.0776 below the Hull lower bound -196.931. It satisfies both the original GDP and a reconstructed Hull formulation at 10^{-5}, with maximum violation approximately 10^{-7}. The cause of the discrepancy remains unresolved.

S10 Additional candidate-search and sampling comparisons

Figure S1 shows the evaluation at which each process run that reaches the target first reaches it.

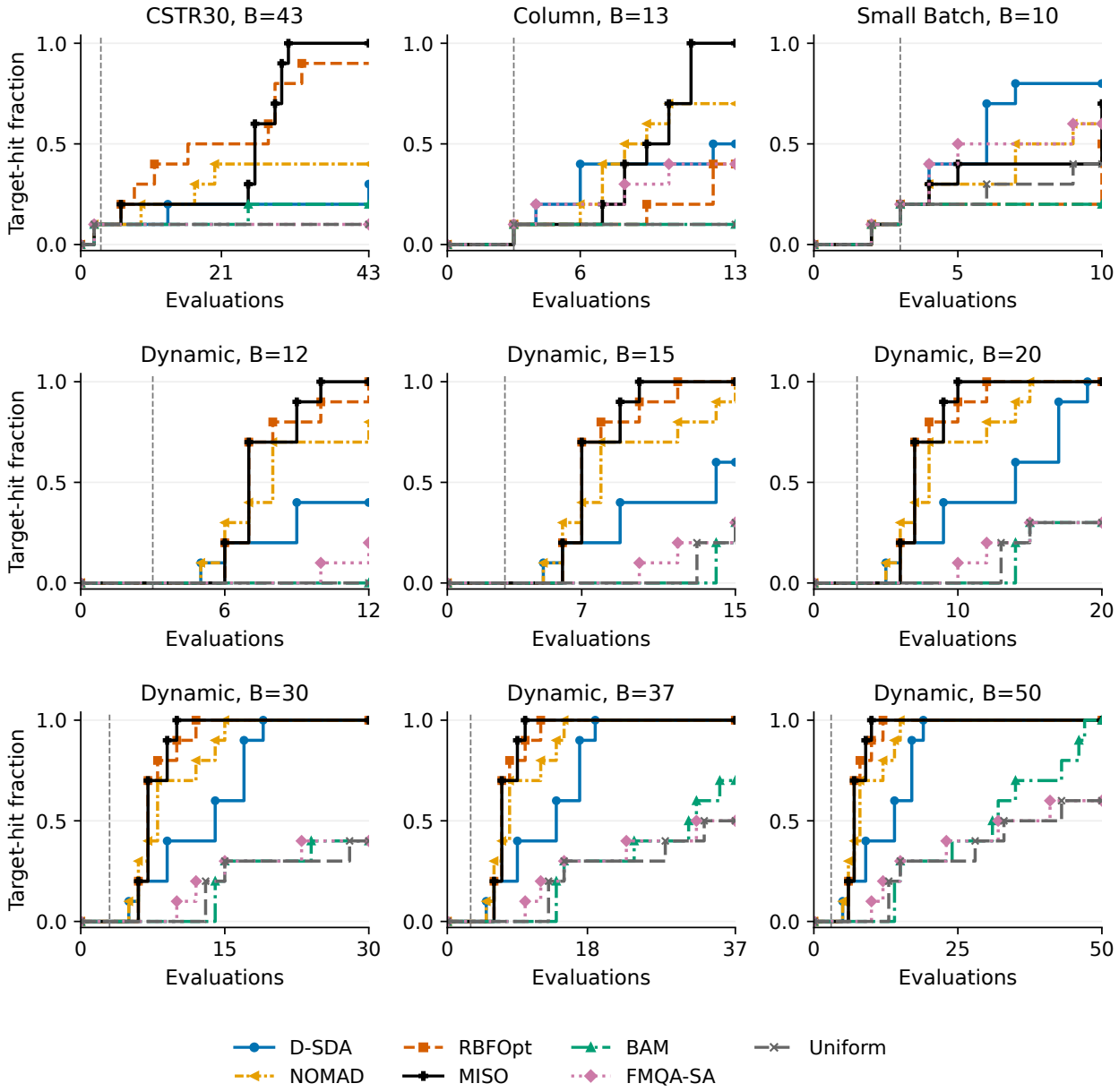


Figure S1: Fraction of the ten seeds whose run has reached the reference target of Equation (8) of the main paper within a given number of evaluations, for nine process conditions and the six DFO methods and uniform sampling. Top row: `CSTR30` at $B = 43$, `Column` at $B = 13$, and `Small Batch` at $B = 10$; middle and bottom rows: `Dynamic` at $B = 12$, 15, 20, 30, 37, and 50, where B is the evaluation budget. Every evaluation counts, including the three shared initial configurations, unaccepted evaluations, and repeated configurations, and the dashed vertical line at three evaluations marks the end of the shared initial evaluations. Each curve steps up at the evaluation in which a run first observes an accepted objective that meets the target; runs that miss the target remain in the denominator. Methods are distinguished by color, line style, and marker. Uniform: uniform sampling without replacement. Target hits in all 14 process conditions are listed in Table 2 of the main paper.

S10.1 Paired differences and sampling controls

The paired comparisons quantify differences in target-hit rates under shared initial designs, matching case, budget, and seed. Positive differences favor the first method and are expressed in percentage points. For Grid, the descriptive percentile bootstrap resamples the 21 intact function vectors, each containing ten paired seed differences, because the ten seeds of a function share its landscape and are not independent draws. For Process, paired seeds are resampled within each of the 14 conditions; the six related `Dynamic` budgets are not pooled for inference. The seed is 20261020 with 2,000 bootstrap replications and deterministic case/seed ordering. Intervals are exploratory and unadjusted for multiple comparisons; ten-seed boundary cases may yield degenerate intervals, and zero inclusion does not establish equivalence.

An interval entirely above zero favors the first method under this resampling; an interval spanning zero leaves the direction uncertain. There is no preset cutoff for practical importance. For example, `BAM` matches its prefix control at $B = 10$ but exceeds it by 10.95 percentage points at $B = 100$, so initial sampling alone does not account for its final target-hit rate.

Table S16: Grid-GDP and process GDP models: seed-paired differences in the target-hit percentage of selected pairs of methods, with 95% percentile bootstrap intervals.

				95% interval (pp)		Seed outcomes			
Benchmark	B	Contrast	Difference (pp)	Lower	Upper	Both	First only	Second only	Neither
Grid-GDP	10	`D-SDA` − `RBFOpt`	-13.81	-24.29	-6.19				
	100	`D-SDA` − `RBFOpt`	-33.33	-48.10	-19.52				
	10	`RBFOpt` − Uniform	13.33	5.71	24.29				
	100	`RBFOpt` − Uniform	60.00	42.86	75.71				
	10	`BAM` − `BAM`-prefix control	0.00	0.00	0.00				
	100	`BAM` − `BAM`-prefix control	10.95	0.48	23.81				
	10	`FMQA-SA` − Uniform	0.95	-0.95	2.86				
	100	`FMQA-SA` − Uniform	40.00	28.57	52.38				
`Small Batch`	10	`FMQA-SA` − Uniform	20	-30	60	2	4	2	2
`Dynamic`	12	`RBFOpt` − `D-SDA`	60	30	90	4	6	0	0

Difference: percentage of runs of the first method that reach the reference target of Equation (8) of the main paper minus that of the second, in percentage points (pp), so positive values favor the first method. 95% interval: lower and upper limits of the percentile bootstrap interval from 2,000 replications; the intervals are not adjusted for multiple comparisons. Paired runs: 210 per Grid-GDP budget (21 functions and ten seeds), with the 21 functions resampled; ten seeds per process condition, resampled within the stated condition only. Seed outcomes: numbers of seeds in which both methods, only the first, only the second, or neither reaches the target, listed for the process conditions. B: evaluation budget; Uniform: uniform sampling without replacement; `BAM`-prefix control: the ten predetermined sampling points of `BAM` after the three shared initial configurations, followed by uniform sampling without replacement.

S10.2 Exact paired tests for process conditions

Table S17 lists the seed-paired target outcomes of `RBFOpt` and `D-SDA` in every process condition and of `BAM` and `RBFOpt` on `CSTR5`. The p-values come from the two-sided exact McNemar test, which uses only the pairs in which exactly one method reaches the target. They are descriptive and not adjusted for multiple comparisons. In the 60 paired `CSTR` runs, `RBFOpt` reaches the target in 30 of the 31 discordant pairs.

Table S17: Process GDP models: seed-paired target outcomes of RBFOpt and D-SDA and of BAM and RBFOpt, with ten seed pairs per condition.

Model	N	B	Both	First only	Second only	Neither	p
First RBFOpt, second D-SDA							
CSTR	5	8	2	6	0	2	0.0312
	10	15	2	4	1	3	0.375
	15	22	2	5	0	3	0.0625
	20	29	3	4	0	3	0.125
	25	36	3	5	0	2	0.0625
	30	43	3	6	0	1	0.0312
	all	–	15	30	1	14	2.98×10^{-8}
Column	–	13	3	1	2	4	1
Small Batch	–	10	4	1	4	1	0.375
Dynamic	–	12	4	6	0	0	0.0312
		15	6	4	0	0	0.125
		20	10	0	0	0	1
		30	10	0	0	0	1
		37	10	0	0	0	1
		50	10	0	0	0	1
First BAM, second RBFOpt							
CSTR	5	8	8	2	0	0	0.500

Both, First only, Second only, Neither: numbers of seeds in which both methods, only the first, only the second, or neither reaches the reference target of Equation (8) of the main paper. p: two-sided exact McNemar p-value, which uses only the seeds in which exactly one method reaches the target; the p-values are descriptive and not adjusted for multiple comparisons. N: number of potential reactors of CSTR (–: not applicable); all: the six sizes pooled (60 seed pairs); B: evaluation budget (–: several budgets, in the pooled row).

S10.3 Grid suite exclusions

Excluding Xin-She Yang 3 and Bohachevsky 2 preserves the main descriptive pattern. For the sampled Bohachevsky 2 lattice, $\cos(3\pi a_i) = 1$ at every coordinate, leaving an objective nearly redundant with sampled Sum of Squares after normalization. Without both functions, $B = 100$ rates are 84.7% for MISO, 79.5% for RBFOpt, 66.8% for BAM, 57.9% for FMQA-SA, 55.8% for NOMAD, 54.7% for the BAM-prefix control, 42.6% for D-SDA, and 17.9% for uniform. At $B = 10$, BAM and its prefix control each retain 47.9%, compared with 15.8% for RBFOpt, 14.7% for MISO, and 12.6% for NOMAD.

S11 Seed spread, tolerance sensitivity, and search overhead

Table S18 compares exact target attainment with relaxed relative tolerances for the 980 primary process runs. The relative gap is $(f_{\text{best}} - f_{\text{ref}})/\max\{1, |f_{\text{ref}}|\}$, the normalization of the target rule in Equation (8) of the main paper. The exact target-hit rows apply the target tolerance of 10^{-5}. Distinct finals is the largest number of different final objectives among the ten seeds of any condition in the family, and the smallest and largest miss-gap rows cover every run that misses the exact target. Table S19 compares the median configuration-evaluation cost with the median search time per evaluation. Both are medians over runs of per-run averages, namely the summed point cost or the stored search time divided by the number of evaluations, over all seven methods for the evaluation cost. The 21 NOMAD Grid runs without a recorded search time are left out of its search median.

Every NLP subproblem solved by GDPopt in the process comparisons fixes a different configuration, as recorded in its execution logs before each subproblem. RIC and LOA solve one subproblem for each of the $N(N+1)/2$ logically admissible configurations of CSTR5 to CSTR20 and for each of the 37 admissible Dynamic configurations, and RIC also covers all 325 admissible configurations of CSTR25.

Table S18: Process GDP models: exact and relaxed target attainment and the spread of final objectives of the six DFO methods and uniform sampling, with ten runs per method and condition.

Model		D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
CSTR	Exact target hits (of 60)	16	21	45	43	18	9	11
	Runs with gap $\leq 0.1\%$	16	21	45	43	18	9	11
	Runs with gap $\leq 1\%$	19	21	45	43	60	23	25
	Runs with gap $\leq 5\%$	27	27	47	49	60	41	48
	Distinct finals	5	3	3	5	3	10	10
	Smallest miss gap (%)	0.224	2.23	2.23	2.23	0.224	0.169	0.224
	Largest miss gap (%)	13.0	13.0	12.5	10.6	0.692	17.2	14.3
Column	Exact target hits (of 10)	5	7	4	10	1	4	1
	Runs with gap $\leq 0.1\%$	5	7	4	10	1	4	1
	Runs with gap $\leq 1\%$	8	9	5	10	1	4	2
	Runs with gap $\leq 5\%$	10	10	9	10	10	10	9
	Distinct finals	4	3	6	1	2	4	8
	Smallest miss gap (%)	0.536	0.536	0.536	–	1.05	1.05	0.536
	Largest miss gap (%)	1.14	1.14	5.92	–	1.05	2.76	5.22
Small Batch	Exact target hits (of 10)	8	6	5	7	2	6	4
	Runs with gap $\leq 0.1\%$	8	6	5	7	2	6	4
	Runs with gap $\leq 1\%$	8	6	5	7	2	6	4
	Runs with gap $\leq 5\%$	8	6	5	7	2	6	4
	Distinct finals	2	2	3	2	3	5	4
	Smallest miss gap (%)	22.2	6.64	6.64	6.64	6.64	6.64	6.64
	Largest miss gap (%)	22.2	6.64	22.2	6.64	8.23	22.2	22.2
Dynamic	Exact target hits (of 60)	50	58	60	60	27	23	21
	Runs with gap $\leq 0.1\%$	50	58	60	60	27	23	21
	Runs with gap $\leq 1\%$	50	58	60	60	27	23	21
	Runs with gap $\leq 5\%$	54	60	60	60	46	42	38
	Distinct finals	5	3	1	1	4	7	6
	Smallest miss gap (%)	2.07	2.07	–	–	2.07	2.07	2.07
	Largest miss gap (%)	36.3	2.66	–	–	7.01	36.3	30.7

Exact target hits: runs that reach the reference target of Equation (8) of the main paper (relative tolerance 10^{-5}), out of the runs of each model (CSTR: six sizes and ten seeds; Column and Small Batch: ten seeds; Dynamic: six budgets and ten seeds). Runs with gap $\leq 0.1\%$, $\leq 1\%$, $\leq 5\%$: runs whose relative gap $(f_{\text{best}} - f_{\text{ref}})/\max\{1, |f_{\text{ref}}|\}$ is at most the stated value, where f_{best} is the best accepted objective of the run and f_{ref} the reference objective. Distinct finals: largest number of different final objectives among the ten seeds of any condition of the model. Smallest and largest miss gap: smallest and largest relative gap over the runs that miss the exact target. Uniform: uniform sampling without replacement. –: no run misses the exact target.

Table S19: Grid-GDP and process GDP models: median evaluation cost and median search time per evaluation of the six DFO methods and uniform sampling.

		Median search time per evaluation (ms)						
Benchmark	Median evaluation cost (s)	D-SDA	NOMAD	RBFOpt	MISO	BAM	FMQA-SA	Uniform
Grid-GDP	0.390	0.0777	50.5	31.1	231	3.77	604	0.0336
CSTR	1.46	1.24	72.2	24.2	287	9.01	572	0.517
Column	2.50	0.941	3.25	17.4	51.7	1.36	527	0.551
Small Batch	0.422	1.02	4.75	17.0	66.8	1.69	487	0.536
Dynamic	0.836	0.893	96.9	98.4	1,860	9.34	567	0.471

Evaluation cost: summed recorded cost of the evaluated configurations divided by the number of evaluations of the run; its median pools the runs of all seven methods. Search time per evaluation: search time, the method time excluding the evaluation cost, divided by the number of evaluations of the run. Medians are over runs; the 21 NOMAD Grid-GDP runs without a recorded search time are left out of its median. Uniform: uniform sampling without replacement.

S12 NOMAD and MISO runs

The NOMAD and MISO runs use the stored responses, shared initial configurations, seeds, budgets, penalties, and reference targets of the other five methods. Neither method receives the target as a stopping criterion. They were executed after the other methods, on a shared server with six workers, each restricted to one logical CPU on a different physical core and to one OpenMP and BLAS thread; the cores were not reserved exclusively. The server used NumPy 2.4.6 and SciPy 1.17.1; the other versions are given in Section 3.6 of the main paper. As for the other methods, completion times add the measured search time of each run to the recorded cost of every evaluated configuration, so they do not depend on the speed of the server's NLP solves.

Termination. All 1,190 MISO runs use their full budget. NOMAD stops before the budget in 77 of the 140 process runs and in 697 of the 1,050 Grid runs, including all 210 runs at $B = 100$, where it stops after a median of 31.5 evaluations. Its Grid target fraction is therefore 60.0% at both $B = 70$ and $B = 100$.

NOMAD crashes and reruns. Of the 1,190 NOMAD runs, 1,046 completed on the first attempt. The other 144 ended with a segmentation fault after their last new evaluation. Of these, 110 completed on the first rerun and 13 on the second, each after evaluating the same configurations in the same order as the crashed attempt, and 21 crashed on all three attempts after identical evaluations. All 21 are Grid runs at $B = 100$, and 16 of them had reached the target before the crash. Scoring them on the evaluations completed before the crash gives NOMAD 526 of 1,050 Grid targets and 92 of 140 process targets; counting all 21 as misses gives 510 Grid targets, or 52.4% instead of 60.0% at $B = 100$. These 21 runs have no recorded search time, so the NOMAD timing summaries at Grid $B = 100$ use the other 189 runs. For a completed rerun, the search time is that of the completed attempt; the time spent in crashed attempts is not recorded. No MISO run failed.

Exploratory paired tests. Table S20 compares NOMAD and MISO with D-SDA, RBFOpt, and each other in every process condition, a family of 70 exact McNemar tests defined after the runs were completed. With a Holm correction over the 70 tests, the smallest adjusted p-value is 0.547, for MISO against NOMAD on CSTR20, so no contrast is significant at the 0.05 level. Ten seeds give each test little power, and a nonsignificant test does not establish equivalence. The six Dynamic budgets reuse the same seeds and are not independent replications. The last row pools the six CSTR sizes for the descriptive comparisons in the main text; it is not part of the corrected family.

Table S20: Process GDP models: seed-paired target outcomes of NOMAD and MISO against D-SDA, RBFOpt, and each other, with ten seed pairs per condition.

Model	N	B		NOMAD vs D-SDA	NOMAD vs RBFOpt	MISO vs D-SDA	MISO vs RBFOpt	MISO vs NOMAD
CSTR	5	8	Discordant seeds	2:0	1:5	2:0	0:4	1:1
			p	0.500	0.219	0.500	0.125	1
	10	15	Discordant seeds	0:0	1:4	0:1	0:4	0:1
			p	1	0.375	1	0.125	1
	15	22	Discordant seeds	0:0	0:5	5:0	1:1	5:0
			p	1	0.0625	0.0625	1	0.0625
	20	29	Discordant seeds	0:1	0:5	7:0	3:0	8:0
			p	1	0.0625	0.0156	0.250	0.00781
	25	36	Discordant seeds	4:1	2:4	7:0	2:0	4:0
			p	0.375	0.688	0.0156	0.500	0.125
	30	43	Discordant seeds	1:0	1:6	7:0	1:0	6:0
			p	1	0.125	0.0156	1	0.0312
Column	–	13	Discordant seeds	3:1	4:1	5:0	6:0	3:0
			p	0.625	0.375	0.0625	0.0312	0.250
Small Batch	–	10	Discordant seeds	1:3	2:1	1:2	2:0	3:2
			p	0.625	1	1	0.500	1
Dynamic	–	12	Discordant seeds	4:0	0:2	6:0	0:0	2:0
			p	0.125	0.500	0.0312	1	0.500
		15	Discordant seeds	4:0	0:0	4:0	0:0	0:0
			p	0.125	1	0.125	1	1
		20	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		30	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		37	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
		50	Discordant seeds	0:0	0:0	0:0	0:0	0:0
			p	1	1	1	1	1
CSTR	all	–	Discordant seeds	7:2	5:29	28:1	7:9	24:2
			p	0.180	3.86×10^{-5}	1.12×10^{-7}	0.804	1.05×10^{-5}

Discordant seeds: a:b, the numbers of seeds in which only the first or only the second method of the column reaches the reference target of Equation (8) of the main paper. p: unadjusted two-sided exact McNemar p-value; the smallest Holm-adjusted p-value of the 70 per-condition tests is 0.547. N: number of potential reactors of CSTR; all: the six sizes pooled (60 seed pairs), which is not part of the 70 per-condition tests; B: evaluation budget (–: several budgets, in the pooled row).

S13 Process D-SDA restarts

The process D-SDA of the main comparison descends once from the best of the three shared initial configurations (S). Variant R applies the Grid rule and descends from all three shared initial configurations, best first, while budget remains. Variant RR continues after R with descents from uniformly drawn unvisited configurations until the budget is exhausted, and it uses the full budget in all 140 runs. No variant receives the reference target. All variants read the stored responses and, like the main comparison, never evaluate a configuration twice. Table S21 shows that R and RR evaluate 1,055 and 1,249 more configurations than S and add one target, on seed 4 of CSTR30, so target attainment rises from 79 to 80 of the 140 runs. Both restart variants improve the final objective in the same 14 CSTR runs. In each of them, S ends at $(5, 1)$, 8.3 to 13.0% above the reference, and the restarts end at the target in the one new hit and 0.4 to 8.6% above it in the other 13. On CSTR10 to CSTR30, 31 of the 36 single-descent misses end at $(5, 1)$.

Table S21: Process GDP models: target hits, full-budget runs, median evaluations, and median full-run completion times of three D-SDA restart variants, with ten runs per variant and condition.

			Target hits (of 10)			Full budget (of 10)		Median evaluations			Median completion time (s)		
Model	N	B	S	R	RR	S	R	S	R	RR	S	R	RR
CSTR	5	8	2	2	2	8	10	8	8	8	10.67	10.71	10.71
	10	15	3	3	3	3	10	11.5	15	15	17.61	21.57	21.57
	15	22	2	2	2	2	10	15.5	22	22	25.68	35.19	35.19
	20	29	3	3	3	0	10	19	29	29	32.31	47.49	47.49
	25	36	3	3	3	2	9	23.5	36	36	45.49	60.78	60.78
	30	43	3	4	4	1	8	24.5	43	43	47.17	78.88	78.88
Column	–	13	5	5	5	9	10	13	13	13	35.44	35.69	35.69
Small Batch	–	10	8	8	8	10	10	10	10	10	4.23	4.23	4.23
Dynamic	–	12	4	4	4	9	10	12	12	12	12.22	12.25	12.25
		15	6	6	6	6	10	15	15	15	15.32	15.76	15.76
		20	10	10	10	3	10	18	20	20	20.02	20.76	20.76
		30	10	10	10	0	10	18	30	30	20.02	27.13	27.13
		37	10	10	10	0	2	18	34.5	37	20.02	28.01	30.05
		50	10	10	10	0	0	18	34.5	50	20.02	28.01	37.00

Target hits: runs that reach the reference target of Equation (8) of the main paper. Full budget: runs that use all B evaluations; RR uses the full budget in every run. Completion time: measured search time plus the recorded cost of every evaluated configuration; medians are over the ten runs. S: single descent from the best shared initial configuration; R: descents from the three shared initial configurations, best first, while budget remains; RR: R followed by descents from uniformly drawn unvisited configurations until the budget is exhausted. N: number of potential reactors of CSTR (–: not applicable); B: evaluation budget.

S14 D-SDA neighborhood

The process, Grid, and Lomustine D-SDA runs use the unit L_∞ neighborhood, whose $3^d - 1$ neighbors number 8 for $d = 2$ external variables and 26 for $d = 3$. The Catalytic Column, with $d = 6$ and 728 such neighbors, uses the separable neighborhood, which changes one external variable by ± 1 and has at most $2d$ neighbors. To test the effect of this choice, we repeated every D-SDA run on stored responses with the separable neighborhood and all other settings unchanged: the 1,050 Grid runs, the 140 single-descent process runs, and the 180 Lomustine runs in the six option orders of Section S5. Table S22 pairs each run with its L_∞ counterpart.

The separable neighborhood reaches more targets when the budget is small compared with the L_∞ neighborhood: on Grid at $B = 10$ (22 against 13 of 210 runs, with 9 and 0 discordant runs, exact McNemar $p = 0.004$), on Dynamic at $B = 12$ and 15 (16 against 10 of 20, $p = 0.031$), and on Lomustine over the six orders (99 against 90 of 180, $p = 0.022$). Its descents also stop earlier: from $B = 50$, the Grid runs end after a median of 25 evaluations in place of 37, and they reach 88 targets against 101 at $B = 70$ and 100 ($p = 0.011$). On CSTR, the separable neighborhood reaches 7 of the 60 targets against 16 ($p = 0.004$). Eighteen of its 53 misses stop on the edge $\xi_1 = \xi_2$ of the admissible region, where every one-coordinate move leaves the region or worsens the objective, whereas a diagonal move continues along the edge to the target at (N, N). Over the 140 process runs, the two neighborhoods reach 79 and 78 targets. Neither neighborhood dominates: the L_∞ neighborhood suits two external variables with budgets that cover its eight neighbors, and the separable neighborhood suits more external variables or smaller budgets.

Table S22: Grid-GDP, process GDP models, and Lomustine: target hits and median evaluations of D-SDA with the unit L_∞ neighborhood and with the separable neighborhood, on stored responses and with all other settings unchanged.

			Target hits				Median evaluations	
Benchmark	N	B	L_∞	Separable	Separable only	L_∞ only	L_∞	Separable
Grid-GDP	–	10	13	**22**	9	0	10	10
		30	78	**79**	14	13	30	25
		50	**91**	87	10	14	37	25
		70	**101**	88	5	18	37	25
		100	**101**	88	5	18	37	25
CSTR	5	8	2	2	0	0	8	8
	10	15	**3**	1	0	2	11.5	8
	15	22	**2**	1	0	1	15.5	10.5
	20	29	**3**	1	0	2	19	14
	25	36	**3**	1	0	2	23.5	15.5
	30	43	**3**	1	0	2	24.5	17
Column	–	13	5	**6**	3	2	13	12
Small Batch	–	10	8	**9**	2	1	10	10
Dynamic	–	12	4	**6**	2	0	12	12
		15	6	**10**	4	0	15	13
		20	10	10	0	0	18	13
		30	10	10	0	0	18	13
		37	10	10	0	0	18	13
		50	10	10	0	0	18	13
Lomustine	–	10	23	**27**	6	2	10	10
		15	32	**36**	4	0	15	11
		20	35	**36**	1	0	19	11

Target hits: runs that reach the reference target of Equation (8) of the main paper; Separable only and L_∞ only: paired runs in which only that neighborhood reaches the target. Paired runs: 210 per Grid-GDP row (21 functions and ten seeds), ten per process row, and 60 per Lomustine row (six option orders and ten seeds). Median evaluations: median number of evaluations per run. L_∞: unit neighborhood, which changes every external variable by at most one; Separable: neighborhood that changes one external variable by ± 1; process runs use the single descent. N: number of potential reactors of CSTR (–: not applicable); B: evaluation budget. Bold: larger number of target hits of the two neighborhoods in each row; equal counts are not bold.

S15 Joint comparison with conventional GDP methods

Figure S2 compares the eight process conditions not shown in the main joint completion-time figure. Each DFO condition has ten seeds, whereas each conventional GDP method has one run per problem. Timing and target criteria follow Figure 3 of the main paper. Table S23 reports all nine models, including final objectives and termination flags.

Table S23: Process GDP models: final objectives and full-run completion times of the conventional GDP methods, with one run per method and model.

Model	N		GLOA	LOA	RIC	Big-M	Hull
`CSTR`	5	Final objective (L)	3.13020†‡	3.06201†	3.06201†	**3.06198**	3.06201†
		Completion time (s)	**4.57**	24.58	15.38	53.68	15.43
	10	Final objective (L)	3.44796†‡	2.96605†‡	2.96605†‡	**2.88941**	2.88951†
		Completion time (s)	**3.10**	140.38	134.31	771.36	71.63
	15	Final objective (L)	3.56196†‡	2.82939†	2.82939†	–^{TL}	**2.82936**
		Completion time (s)	**6.17**	445.43	283.77	1800+	284.63
	20	Final objective (L)	3.61612†‡	2.79941†	2.79941†	2.80413^{TL}	**2.79936**
		Completion time (s)	**7.46**	733.53	607.72	1800+	1,150.67
	25	Final objective (L)	3.64786†‡	2.79125^{TL}	2.79125†‡	2.78771^{TL}	**2.78143**
		Completion time (s)	**5.20**	1800+	1,042.79	1800+	1,701.75
	30	Final objective (L)	3.66872†‡	2.87314^{TL}	2.77162^{TL}	**2.76957**^{TL}	**2.76957**^{TL}
		Completion time (s)	**5.52**	1800+	1800+	1800+	1800+
`Column`	–	Final objective	20,355.1†‡	**19,449.9**	**19,449.9**	–^{TL}	22,271.5$^{\mathrm{TL}\dagger}$
		Completion time (s)	**42.41**	199.22	124.03	1800+	1800+
`Small Batch`	–	Final objective (\$)	**167,428**	**167,428**	**167,428**	**167,428**	**167,428**
		Completion time (s)	10.32	5.61	13.15	0.51	**0.50**
`Dynamic`	–	Final objective	-197.140†‡	**-212.009**	**-212.009**	**-212.009**^{TL}	-196.931†‡
		Completion time (s)	238.96	550.49	**149.39**	1800+	921.28

Final objective: in the unit given in the row label, as in Table 1 of the main paper; the `Column` and `Dynamic` objectives have no physical unit. Completion time: measured full-run time, including model-template preparation. N: number of potential reactors of `CSTR`. GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; all three are `GDPopt` methods, which use `SCIP` for master problems and `IPOPT` for NLP subproblems. Big-M and Hull: the retained runs that solve these reformulations directly with `SCIP`. 1800+: stopped at the 1,800 s time limit. TL: stopped at the time limit, with reported `maxTimeLimit`; runs without TL report `optimal`. –: no accepted feasible solution in an objective cell; not applicable in the N column. †: reported lower bound exceeds the best checked saved numerically feasible objective f of the model by more than $10^{-5}\max\{1, |f|\}$ (Section S9). ‡: final objective above $f_{\mathrm{ref}} + 10^{-5}\max\{1, |f_{\mathrm{ref}}|\}$, the success threshold of the conventional GDP methods, despite reported `optimal` termination and a better checked numerically feasible solution. Feasibility: numerical feasibility under the original-GDP residual test at 10^{-5}, which does not establish exact feasibility. Bold: lowest final objective of each model, including ties within $10^{-5}\max\{1, |f_{\min}|\}$, where $f_{\min}$ is that lowest objective, and shortest completion time of each model, including ties.

Table S24: Process GDP models: NLP subproblems solved per run by the six DFO methods and uniform sampling, with ten runs per method and condition, and by the single run of each `GDPopt` method.

			DFO methods and uniform sampling (median of ten)							`GDPopt` (one run)		
Model	N	B	`D-SDA`	`NOMAD`	`RBFOpt`	`MISO`	`BAM`	`FMQA-SA`	Uniform	GLOA	LOA	RIC
`CSTR`	5	8	6	6	6	6	6	5.5	4.5	4	15	15
	10	15	10.5	10.5	12	11.5	8.5	9	7.5	2	55	55
	15	22	14	15	16.5	16.5	11	15	11	4	120	120
	20	29	18	20	20.5	19.5	17	17.5	15.5	4	210	210
	25	36	22	25	25.5	25	18.5	23	21	2	315	325
	30	43	23	25	31.5	29.5	21.5	30.5	21.5	2	243	455
`Column`	–	13	11	10.5	12	11	7.5	7	7	18	38	38
`Small Batch`	–	10	5	6	6	6	5	5	4	12	5	12
`Dynamic`	–	12	10	10	10	9	8	6.5	6	20	37	37
		15	12.5	12	13	12	9	8	6.5	20	37	37
		20	16.5	14.5	16	14	11	11	9.5	20	37	37
		30	16.5	14.5	21	21	17	14.5	15	20	37	37
		37	16.5	14.5	25	25	19	19	19	20	37	37
		50	16.5	14.5	31	29	26	24.5	26	20	37	37

DFO count: median over the ten runs of the number of evaluations that called `IPOPT`, whether or not its result was accepted; logical and preprocessing rejections are not counted. `GDPopt` count: number of `IPOPT` calls confirmed in the execution log of the run. The counts are numbers of subproblem solves, which differ in cost, and do not include solver iterations. Every `GDPopt` NLP call fixes a different configuration; `CSTR` with N potential reactors has $N(N+1)/2$ logically admissible configurations, and `Dynamic` has 37 (Section S11). Internal NLP counts of the direct Big-M and Hull solves are unavailable and are not treated as zero. N: number of potential reactors of `CSTR` (–: not applicable); B: evaluation budget of the DFO methods; the `GDPopt` methods have no budget, so the single `Dynamic` run of each appears in every `Dynamic` row. GLOA: global logic-based outer approximation; LOA: logic-based outer approximation; RIC: relaxation with integer cuts; the `GDPopt` methods use `SCIP` for master problems and `IPOPT` for NLP subproblems. Uniform: uniform sampling without replacement.

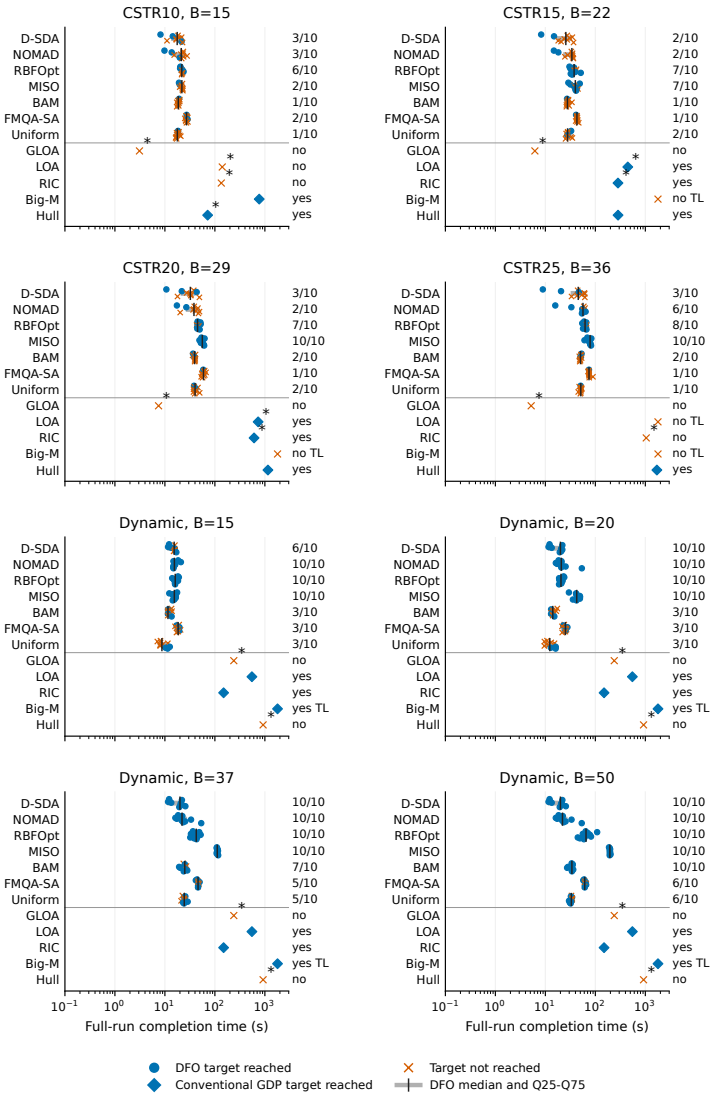


Figure S2: Full-run completion times on the eight process conditions not shown in Figure 3 of the main paper, by the six DFO methods and uniform sampling (ten seeds each) and by the single run of each conventional GDP method. Panels: `CSTR10` at $B = 15$, `CSTR15` at $B = 22$, `CSTR20` at $B = 29$, `CSTR25` at $B = 36$, and `Dynamic` at $B = 15$, 20, 37, and 50, where B is the evaluation budget of the DFO methods. Rows list `D-SDA`, `NOMAD`, `RBFOpt`, `MISO`, `BAM`, `FMQA-SA`, and Uniform, then the `GDPopt` methods GLOA (global logic-based outer approximation), LOA (logic-based outer approximation), and RIC (relaxation with integer cuts), and the Big-M and Hull reformulations solved directly by `SCIP`. DFO times add the search time to the measured cost of each evaluated configuration (Equation (10) of the main paper); conventional GDP times are measured and include model preparation; the time axis is logarithmic. Dots mark DFO runs that reach the reference target of Equation (8) of the main paper, an objective within $10^{-5}\max\{1, |f_{\text{ref}}|\}$ of the reference objective f_{ref}; diamonds mark conventional GDP runs whose feasible objective is at most $f_{\text{ref}} + 10^{-5}\max\{1, |f_{\text{ref}}|\}$; crosses mark the other runs; gray bars span the 25th to 75th percentiles of the ten DFO seeds, with a tick at the median. The right-hand labels give the target hits out of ten of each DFO method and the outcome, yes or no, of each conventional run, with TL for runs that report termination at the 1,800 s time limit. Asterisks mark conventional runs whose reported lower bound exceeds a checked feasible objective (Section S9). The conventional GDP methods have no evaluation budget, so the four `Dynamic` panels show the same conventional runs. Uniform: uniform sampling without replacement.